

## Full Length Article

# JSPR-Net: A Jacobian-stabilized physics-informed residual neural network for breast cancer detection and fractional-order disease progression modeling

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## ABSTRACT

Breast cancer is the second most common cause of mortality in women. The oncogenes are present in the cells. The milk gland has irregular meiosis and mitosis cell division and cause tumors structure that known as Breast cancer. This study proposes the JSPR-Net classifier for the detection of breast cancer. The detection framework for a new fractional-order compartmental model that uses CF derivatives captures the dynamics of breast cancer progression that depend on memory across different stages of the disease. Fixed-point theory for existence and uniqueness and Natural transform with Ulam-Hyers stability make sure that the model's math is solid and that the numbers are reliable. Graphical visualizations reveal significant disease dynamic variations under different fractional orders ( $\alpha = 0.7, 0.8, 0.9, 1.0$ ), while equilibrium analysis provides insights for intervention strategies and CF method for progress identification of Breast cancer. The kaggle dataset with 250 samples for disease detection is used. The diagnosis accuracy 94% is achieved by proposed method. The achieved accuracy, feature importance and stability analysis shows the robustness of model for breast cancer detection.

## Introduction

Breast cancer is the second most common cause of mortality in women, after lung cancer. In 2016, 246,660 new cases of invasive breast cancer are predicted to be identified in the US, and 40,450 women are expected to die from it. Breast cancer is a kind of cancer that starts in the breast. When cells start to expand too quickly, cancer starts. Breast cancer cells frequently make a tumor that can be seen on an x-ray or felt as a bump. When breast cancer cells are flowing into the blood or lymph system, they travel to other regions of the body. Changes and mutations in DNA are some of the things that might cause breast cancer. There are many distinct kinds of breast cancer, but ductal carcinoma in situ (DCIS) and invasive carcinoma are two of the most frequent. Others, such as phyllodes tumors and angiosarcoma, are not as frequent. There are a lot of algorithms that can be used to figure out what will happen with breast cancer. Side symptoms of breast cancer include fatigue, headaches, pain and numbness (peripheral neuropathy), bone loss, and osteoporosis.

Breast cancer (BC), which is the uncontrolled growth of abnormal breast cells, is still the second most common cancer in women in the US and the leading cause of cancer death in women. According to the American Cancer Association, 268,600 women were diagnosed with BC in 2019, and 41,760 of them died. In 2020, breast cancer was the most common cancer diagnosis for women in the European Union, making up 13.3% of all cancer diagnoses (Wu & Hicks, 2021, p. 1). To get an accurate diagnosis of breast cancer, it is important to find aberrant breast tissue early. Finding breast tumors early can help people live longer since they are easier to treat at this stage.

For women in the EU who have been diagnosed with breast cancer, 7.4% of them will need a second primary breast cancer (SBC) diagnostic within ten years. The first five years after breast cancer is the most likely time for it to come back, with a rate of 10.4%. The European Society for Medical Oncology has rules for breast cancer patients that say they should see their doctor every 3 to 4 months during the first 2 years after the PBC. This will help breast cancer survivors find SBC earlier by raising the survival rate from 27% to 47% (Syleouni et al., 2023,

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p. 933). When forecasting the likelihood of SBC using patient-level data from PBC tumors or the patients themselves, it could be a helpful tool for doctors to make smart choices about follow-up care for patients who are at risk, find new tumors that are forming, or stop new cases of BC from happening.

In the United States, most cases of BC react well to treatment and have reasonable chances of survival and prognosis. A significant number of breast cancer are classified as triple-negative breast cancer, due to heterogeneous nature of the illness, which encompasses several kinds and numerous sub-types. The TNBC is a distinct sub-type of BC characterized by its absence of expression of the three most targeted biomarkers for BC treatment, constituting 15%–20% of all identified BC cases annually (Wu & Hicks, 2021, p. 2). The TNBC are also described as having more aggressive clinical behavior, a worse prognosis, a greater recurrence rate, and a lower survival rate.

Breast Cancer is a prevalent disease among women worldwide. Research in this area has focused on using machine learning techniques with datasets like the UCI Wisconsin Machine Learning Repository. Several studies have explored different classification methods. Yue et al. (2018) proposed Logistic Regression method which achieved 96.4% accuracy of the model. Akbugday (2019) utilized KNN and SVM, achieving 96.85% accuracy. Keleş (2019) employed Random Forest with 92.2% accuracy. Chaurasia and Pal (2014) compared various supervised learning classifiers, including Naïve Bayes, SVM-RBF kernel, RBF neural networks, Decision trees (J48), and simple CART. Dalen and G. Delen et al. (2005) achieved 97.5% accuracy using ADABOOST, outperforming Random Forest method. Kavitha and Rangasamy (2014) used ensemble methods with NN, achieving 96.3% accuracy. Sinthia et al. (2017) used the backpropagation method which achieved 94.2% accuracy.

Further studies have explored various datasets and tools. Chaurasia and Pal (2014) used statistical feature selection with the Wisconsin breast cancer dataset, achieving 92.3% accuracy. Keleş (2019) compared SVM, KNN, decision trees, and Naïve Bayes on the Wisconsin Diagnostic Breast Cancer dataset, achieving up to 96.91% accuracy. Chaurasia and Pal (2014) achieved 96.5% accuracy using Naïve Bayes, J48 Decision Tree and Bagging algorithm on the UC Irvine machine learning repository. Delen et al. (2005) used ADABOOST on the Cancer Society dataset, achieving 97.5% accuracy.

Other research includes: Kavitha and Rangasamy (2014) using an Ensemble method with Logistic and Neural Network on the Cancer Society dataset (96.3% accuracy); Sinthia et al. (2017) employing Logistic Regression and Backpropagation neural network on the Wisconsin Diagnosis Breast Cancer BCI dataset (94.2% accuracy); Chaurasia and Pal (2014) utilizing SVM on the Wisconsin Breast Cancer (original) datasets (97.13% accuracy); and Khuriwal and Mishra (2018b) using NAÏVE BAYES AND SVM on Haberman's Survival dataset (74.44% accuracy). Khouirdifi and Bahaj (2018) used Fast Correlation-Based Filter with SVM, Random Forest, Naïve Bayes, K-NN and MLP on Wisconsin breast cancer dataset with 96.1% accuracy. Mohana et al. (2019) used DECISION TREE on WISCONSIN BREAST CANCER DATASET with 96.33% accuracy. Shrivaya et al. (2019) used SVM on UCI repository with 92.7% accuracy. Wang and Yoon (2015) used PCA dimension reduction on Wisconsin Breast Cancer Database (1991) and Wisconsin Diagnostic Breast Cancer (1995). Bellaachia and Guven (2006) used Naïve Bayes on SEER Public-Use Data with 96.33% accuracy. Kim and Shin (2013) used SEMI-SUPERVISED CO-TRAINING on The breast cancer survivability dataset (1973–2003) from SEER with 0.8484% accuracy.

Additional studies include Bellaachia and Guven (2006) using Naïve Bayes, the back-propagated neural network, and the C4.5 decision tree on SEER database (84.5% - Naïve Bayes, 86.7% - NN, 81.3% -C4.5 accuracy); Khuriwal and Mishra (2018a) employing Ensemble Voting Method, Logistic Regression on the Wisconsin Diagnosis Breast Cancer dataset (98.50% accuracy); Amrane et al. (2018) using KNN and NAÏVE BAYES on the Breast Cancer Dataset University of California, Irvine (UCI) with 0.975109-KNN accuracy; Khuriwal and Mishra (2018a) utilizing Deep Learning Neural Network on the Wisconsin Breast Cancer

database (99.67% accuracy); Alarabeyyat and Alhanahnah (2016) employing Logistic Regression and Backpropagation neural Network on General Sample (Greater than 93.7% accuracy); Jang et al. (2017) using C4.5, Bagging and ADABOOST on Gene Expression Dataset Collection (Single C4.5 - 95.6%, Bagging C4.5 - 93.29%,ADABOOST C4.5 - 92.62% accuracy); Cruz and Wishart (2006) employing SVM, NAÏVE-BAYES on Pubmed (biomedical literature), the Science Citation Index (97.3% accuracy); and Medjahed et al. (2013) using Decision Trees on the Wisconsin breast cancer dataset (96.11% accuracy).

Mathematical modeling has been used for an investigate tumor progression and treatment effects, with modern approaches including chemotherapy impact, immune interactions, and multi-stage cancer dynamics (Abdeljawad et al., 2025a).Recent population-based studies have created breast cancer stages together with cardiotoxicity, illustrating that treatment parameters affect long-term outcomes. Additionally, fractional-order extensions have helped to improve accuracy by integrating memory effects in disease progression (Alzahrani et al., 1975). Mechanistic data-driven models, including PyMT mice ODE systems, defined critical biological factors, such as IL6-adipocyte interactions, in tumor spread (Mirzaei et al., 2022). Furthermore, the integration of machine learning (SVM) with compartmental models improved tumor classification and helped to make decision on cardiotoxicity risks in clinical environments (Agossou et al., 2022a).

Recent research integrates mathematical modeling, fractional calculus, and artificial intelligence to improve our understanding of breast cancer progression. Khan et al. (2025) proposed a discrete fractional-order framework which is showed the memory effects can significantly improve tumor-stage prediction. Machine-learning techniques, including SVM and Naïve Bayes, as discussed by Agossou et al. (2022b), significantly improve early detection of cancer and its classification accuracy.

Analytical and numerical solution methods are also play a vital role in cancer progression. Khader and Tedjani (2025) used the Modified Adomian Decomposition Method along with Mohand transforms to get fast converging solutions for the breast cancer dynamics. Treatment-based models such as Karchani et al. (2023) highlight how therapies like trastuzumab influence disease stages and cardiotoxicity. Additionally, Batiha et al. (2025) emphasized that fractional models more accurately capture long-memory effects in clinical progression and support health-care planning.

Changjin Xu et al worked in fractional-order delayed zooplankton and phytoplankton system (Li et al., 2023), control dynamics and bifurcation on predator prey model (Lin et al., 2026), feedback controller and hybrid controller with bifurcation for delay tumor-immune model (Deng et al., 2025), predator-prey model with bifurcation in double delay (Lin et al., 2025) and bifurcation and control dynamic of 5D BAM neural network model with time delay (Cui et al., 2024). Changjin Xu et al have highlighted fractional order model for multi-agent systems with diffusion (Yang et al., 2023), fractional order modeling for type-1 diabetes mellitus for feedback design of artificial pancreas (Farman et al., 2024), prey-predator model for hunting cooperation with gas-tation delay insense of fractional derivative (Xu & Balci, 2026), Generalized synchronization between coupled integer, fractional and delay chaotic systems (Shukla et al., 2025), fractional order model for tuberculosis disease with multidrug resistance (Nisar et al., 2025) and fractional order delayed memristive neural network (Jia et al., 2022).

Parvaiz Ahmad Naik et al has highlighted tumor modeling, study between tumor carcinogenesis and macrophage activation insense of fractional order (Naik et al., 2025b), emphasized control of lung cancer with early detection and anti-cancer measures (Naik et al., 2024), modeling of cancer treatment with radiotherapy (Naik et al., 2025a), three dimensional time delay with neural network (Naik & Eskandari, 2024) and HBV model insense of fractional order with chronic infection (Ahmad et al., 2025). Sabri T. M. Thabet et al have highlighted hybrid Caputo-Atangana-Baleanu fractional pantograph system under boundary value conditions (Thabet et al., 2025b), work on fractional Langevinsturm

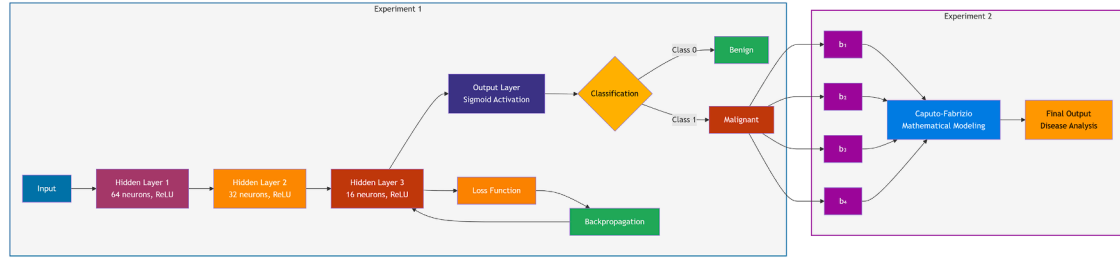


Fig. 1. Proposed model architecture.

-Liouville problem through Caputo-Atangana-Baleanu derivative (Thabet et al., 2024a), Hilfer inverse problem with delay and anticipation (Salim et al., 2024) and Hilfer delayed Duffing problem on the positive real line (Thabet et al., 2024c).

In this study, Fractional-order modeling was adopted because breast cancer progression is inherently non-local in time: the current disease state depends on the entire history of previous stages, treatments, and physiological responses. Classical integer-order derivatives consider only the instantaneous rate of change and cannot capture such memory effects. By incorporating fractional calculus, we account for the cumulative influence of past events—such as prior chemotherapy or tumor evolution—thereby obtaining a more realistic description of disease dynamics. The fractional order parameter  $\alpha$  directly controls the strength of this memory, allowing the model to reflect patient heterogeneity and to reduce to the integer-order case when  $\alpha = 1$ .

Among the various fractional operators, we selected the Caputo-Fabrizio (CF) derivative because its exponential kernel naturally models the finite-duration memory observed in biological systems (e.g., drug washout, cellular adaptation). This choice provides a better balance between biological realism, mathematical tractability, and computational efficiency than alternatives such as the Atangana-Baleanu-Caputo operator, which uses a Mittag-Leffler kernel suited for power-law memory. The CF formulation enabled us to derive a stable numerical scheme, prove existence and uniqueness via fixed-point theory, and establish Ulam-Hyers stability—all essential for clinical applicability. Moreover, our simulations results show that fractional-order models predict distinct, clinically meaningful dynamics (e.g., faster early-stage reduction, lower cardiotoxicity) that integer-order models cannot capture, justifying the use of the CF derivative for both the detection and progression components of this work.

### Novelty and Contributions

While existing studies typically treat detection and progression separately or rely on integer-order models that ignore memory effects, our approach simultaneously addresses these gaps by providing a mathematically robust, memory-aware model coupled with a highly accurate and stable classifier, thereby delivering both methodological and clinical advancements. The main novelty and contributions of this work are:

- **JSPR-Net classifier:** A Jacobian-Stabilized Physics-Informed Residual Neural Network for breast cancer detection that integrates advanced regularization and stability mechanisms, achieving 94% accuracy, near-perfect AUC (0.997), and perfect sensitivity on a clinical dataset.
- **Fractional-order progression model:** A novel seven-compartment model using the Caputo-Fabrizio derivative to capture memory-dependent breast cancer dynamics across all disease stages, including treatment-induced cardiotoxicity and surgical pathways.
- **Mathematical rigor:** Existence and uniqueness proofs via fixed-point theory, Ulam-Hyers stability analysis, Natural transform, and equilibrium characterization, establishing a solid theoretical foundation for both detection and progression components.

- **Unified framework:** Integration of machine learning detection with mathematical epidemiology, enabling both individual patient diagnosis and population-level disease forecasting.
- **Clinical insights:** Demonstration that fractional-order modeling reveals faster early-stage reductions, improved recovery, and reduced cardiotoxicity compared to integer-order models, providing actionable insights for personalized treatment strategies.

### Unified Framework Architecture

Fig. 1 illustrates the complete two-stage framework proposed in this study. The architecture is divided into two complementary components:

**Experiment 1: JSPR-Net Detection Module** The left side of the figure depicts the proposed Jacobian-Stabilized Physics-Informed Residual Neural Network (JSPR-Net) for breast cancer detection. The input layer accepts a  $d$ -dimensional feature vector derived from the 14 clinically relevant biomarkers. Three hidden layers with 64, 32, and 16 neurons respectively, each using ReLU activation and batch normalization, progressively extract high-level discriminative features. Residual connections (implicit in the diagram) facilitate gradient flow, while dropout regularization prevents overfitting. The output layer employs a stabilized sigmoid activation to produce a binary classification: Benign (Class 0) or Malignant (Class 1). The backpropagation path incorporates gradient clipping and Jacobian-based regularization to ensure stable training and robust predictions.

**Experiment 2: Fractional-Order Progression Module** The right side of the figure presents the compartmental disease progression model based on the Caputo-Fabrizio fractional derivative. The model is driven by incidence rates  $b_1, b_2, b_3, b_4$ , which feed into the seven epidemiological compartments ( $I_1, I_2, A, T, R, C, L$ ). The fractional-order formulation captures memory-dependent dynamics across cancer stages, treatment responses, and cardiotoxicity development. The final output provides a comprehensive disease analysis, including equilibrium states, stability properties, and population forecasts.

**Integration:** The two modules are conceptually linked through a shared clinical data pipeline: the detection module outputs the patient's diagnostic status, which can serve as initial conditions or stratification input for the progression model. Conversely, insights from the progression model (e.g., stage-specific transition rates, cardiotoxicity risks) inform the feature selection and stability constraints embedded in the JSPR-Net. This bidirectional integration establishes a unified analytical framework that bridges machine learning-based diagnosis with mathematical epidemiology, enabling both individual-level prediction and population-level forecasting within a single coherent structure.

### Methodology of JSPR Classifier

The proposed Advanced JSPR Classifier is a multi-layer neural network with enhanced regularization techniques and stability mechanisms. Let  $x \in \mathbb{R}^d$  denote the input feature vector, where  $d$  is the input dimension (number of features). The network architecture consists of  $\mathfrak{Q}$  hidden layers with  $n_l$  neurons in layer  $l$  ( $l = 1, 2, \dots, \mathfrak{Q}$ ), followed by

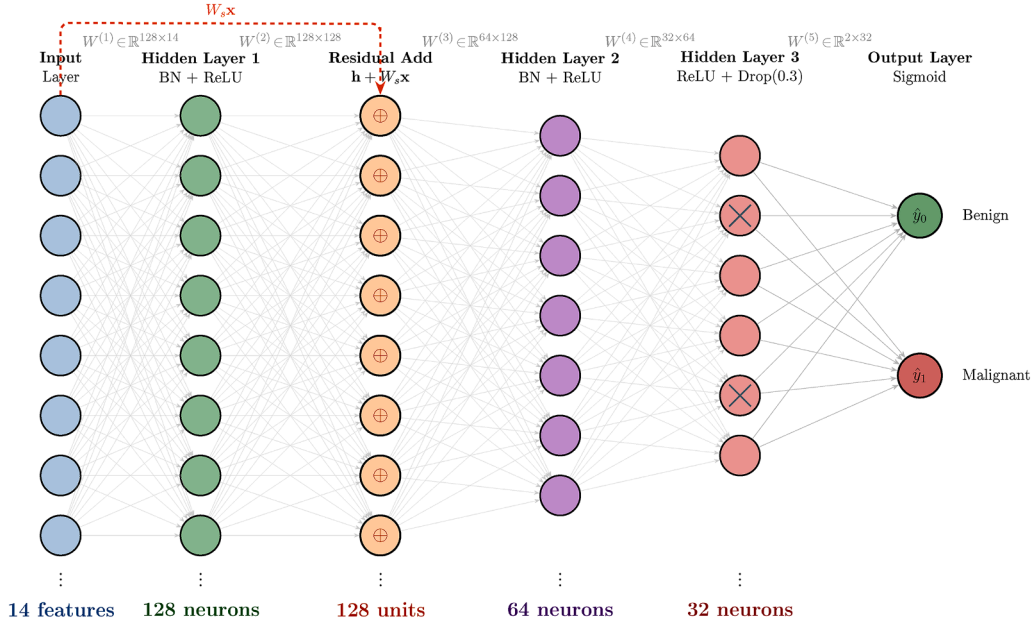


Fig. 2. Hidden layer architecture for JS-PR-Net (Jacobian-stabilized physics-informed residual neural network) for breast cancer binary classification.

a single-neuron output layer for binary classification is shown in the Fig. 2.

#### Advanced Weight Initialization

The network parameters are initialized using He initialization for ReLU-based activation:

$$\mathcal{W}^{(l)} \sim \mathcal{N}\left(0, \sqrt{\frac{2}{n_{l-1}}}\right), \quad b^{(l)} = 0 \quad (1)$$

where  $\mathcal{W}^{(l)} \in \mathbb{R}^{n_l \times n_{l-1}}$  is the weight matrix for layer  $l$ ,  $b^{(l)} \in \mathbb{R}^{n_l}$  is the bias vector, and  $n_l$  represents the number of neurons in layer  $l$ .

#### Advanced Forward Propagation with Enhanced Features

For each layer  $l = 1, 2, \dots, \mathfrak{L} + 1$ :

- **Linear Transformation:**

$$z^{(l)} = \mathcal{W}^{(l)\top} a^{(l-1)} + b^{(l)} \quad (2)$$

where  $a^{(0)} = x$  is the input.

- **Batch Normalization (for hidden layers):**

$$\mu^{(l)} = \frac{1}{m} \sum_{i=1}^m z_i^{(l)} \quad (\text{batch mean})$$

$$\sigma^{2(l)} = \frac{1}{m} \sum_{i=1}^m (z_i^{(l)} - \mu^{(l)})^2 \quad (\text{batch variance})$$

$$\hat{z}^{(l)} = \frac{z^{(l)} - \mu^{(l)}}{\sqrt{\sigma^{2(l)} + \epsilon}} \quad (\text{normalization})$$

$$\tilde{z}^{(l)} = \gamma^{(l)} \odot \hat{z}^{(l)} + \beta^{(l)} \quad (\text{scale and shift}) \quad (3)$$

where  $\gamma^{(l)}, \beta^{(l)} \in \mathbb{R}^{n_l}$  are learnable parameters,  $\epsilon = 10^{-5}$  is a small constant for numerical stability,  $m$  is the batch size, and  $\odot$  denotes element-wise multiplication.

- **Advanced Activation Functions:**

$$a^{(l)} = \sigma(\tilde{z}^{(l)}) = \begin{cases} \max(0, \tilde{z}^{(l)}) & \text{ReLU} \\ \max(\alpha \tilde{z}^{(l)}, \tilde{z}^{(l)}) & \text{Leaky ReLU} \\ \frac{\tilde{z}^{(l)}}{1 + e^{-\tilde{z}^{(l)}}} & \text{Swish} \\ \tanh(\tilde{z}^{(l)}) & \text{Hyperbolic Tangent} \end{cases} \quad (4)$$

#### Output Layer with Stabilized Sigmoid

For the final output layer ( $l = \mathfrak{L} + 1$ ):

$$z^{(\mathfrak{L}+1)} = \mathcal{W}^{(\mathfrak{L}+1)\top} a^{(\mathfrak{L})} + b^{(\mathfrak{L}+1)} \quad (5)$$

$$\hat{y} = f(x; \theta) = \sigma_{\text{sigmoid}}(z^{(\mathfrak{L}+1)}) = \frac{1}{1 + e^{-\text{clip}(z^{(\mathfrak{L}+1)}, -20, 20)}} \quad (6)$$

where  $\text{clip}(z, -20, 20)$  prevents numerical overflow in the exponential function.

#### Advanced Loss Function with Multiple Regularization

The total loss function  $\mathfrak{J}(\theta)$  combines multiple components:

$$\mathfrak{J}(\theta) = \mathbb{L}_{\text{BCE}}(\theta) + \mathbb{L}_{\text{L2}}(\theta) + \mathbb{L}_{\text{JAC}}(\theta) + \mathbb{L}_{\text{STAB}}(\theta) \quad (7)$$

#### Weighted Binary Cross-Entropy Loss

For imbalanced datasets:

$$\mathbb{L}_{\text{BCE}}(\theta) = -\frac{1}{N} \sum_{i=1}^N [w_1 y_i \log(\hat{y}_i) + w_0 (1 - y_i) \log(1 - \hat{y}_i)] \quad (8)$$

where  $w_0 = \frac{N}{2N_0}$  and  $w_1 = \frac{N}{2N_1}$  are class weights,  $N_0$  and  $N_1$  are the number of samples in classes 0 and 1 respectively.

#### L2 Regularization with Layer-wise Coefficients

$$\mathbb{L}_{\text{L2}}(\theta) = \sum_{l=1}^{\mathfrak{L}+1} \lambda_l \|\mathcal{W}^{(l)}\|_F^2 \quad (9)$$

where  $\|\cdot\|_F$  is the Frobenius norm and  $\lambda_l$  is layer-specific regularization strength.

#### Jacobian-Based Stability Regularization

$$\mathbb{L}_{\text{JAC}}(\theta) = \alpha \cdot \frac{1}{N} \sum_{i=1}^N \|\mathbf{J}_i\|_2^2 \quad (10)$$

where  $\mathbf{J}_i = \nabla_{x_i} \hat{y}_i \in \mathbb{R}^d$  is the Jacobian vector for sample  $i$ , and  $\alpha$  controls the regularization strength.



### Advanced Stability Metrics

The classifier computes several stability metrics during training:

- **Jacobian Norm:**

$$\mathcal{N}_{\text{Jac}} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{J}_i\|_2 \quad (11)$$

- **Condition Number:**

$$\kappa(\mathbf{J}) = \frac{\sigma_{\max}(\mathbf{J})}{\sigma_{\min}(\mathbf{J})} \quad (12)$$

where  $\sigma_{\max}$  and  $\sigma_{\min}$  are the maximum and minimum singular values of the Jacobian matrix  $\mathbf{J} \in \mathbb{R}^{N \times d}$ .

- **Stability Index:**

$$S = \mathcal{N}_{\text{Jac}} \cdot (1 + \log(1 + \kappa(\mathbf{J}))) \quad (13)$$

- **Maximum Singular Value:**

$$\sigma_{\max} = \max_i \sigma_i(\mathbf{J}) \quad (14)$$

### Advanced Backpropagation with Gradient Clipping

The gradients are computed using the chain rule with enhanced numerical stability:

- **Output Layer Gradient:**

$$\delta^{(\mathcal{Q}+1)} = \frac{\partial \mathbb{L}}{\partial \mathbf{z}^{(\mathcal{Q}+1)}} = \frac{\hat{\mathbf{y}} - \mathbf{y}}{N} \quad (15)$$

- **Hidden Layer Gradients:**

$$\begin{aligned} \delta^{(l)} &= ((\mathcal{W}^{(l+1)})^\top \delta^{(l+1)}) \odot \sigma'(\mathbf{z}^{(l)}) \\ \frac{\partial \mathbb{L}}{\partial \mathcal{W}^{(l)}} &= \mathbf{a}^{(l-1)} (\delta^{(l)})^\top + 2\lambda_l \mathcal{W}^{(l)} \\ \frac{\partial \mathbb{L}}{\partial \mathbf{b}^{(l)}} &= \sum_{i=1}^N \delta_i^{(l)} \end{aligned} \quad (16)$$

- **Batch Normalization Gradients:**

$$\begin{aligned} \frac{\partial \mathbb{L}}{\partial \gamma^{(l)}} &= \sum_{i=1}^N \delta_i^{(l)} \odot \hat{\mathbf{z}}_i^{(l)} \\ \frac{\partial \mathbb{L}}{\partial \beta^{(l)}} &= \sum_{i=1}^N \delta_i^{(l)} \end{aligned} \quad (17)$$

- **Gradient Clipping:**

$$\mathbf{g}_{\text{clipped}} = \begin{cases} \mathbf{g} & \text{if } \|\mathbf{g}\|_2 \leq \tau \\ \tau \cdot \frac{\mathbf{g}}{\|\mathbf{g}\|_2} & \text{otherwise} \end{cases} \quad (18)$$

where  $\tau$  is the clipping threshold (typically 1.0).

### Comprehensive Evaluation Metrics

The classifier's performance is evaluated using multiple metrics such as Accuracy, Precision, Recall (Sensitivity), F1-Score, Matthews Correlation Coefficient, Balanced Accuracy, ROC-AUC and PR-AUC.

### Feature Sensitivity Analysis

The classifier computes feature sensitivity using Jacobian analysis:

$$S_j = \frac{1}{N} \sum_{i=1}^N \left| \frac{\partial \hat{y}_i}{\partial x_{i,j}} \right|, \quad j = 1, \dots, d \quad (19)$$

where  $S_j$  represents the sensitivity of the model to feature  $j$ .

### Permutation Feature Importance

The importance of each feature is quantified by:

$$I_j = \frac{1}{R} \sum_{r=1}^R [\text{Acc}(\mathcal{D}) - \text{Acc}(\mathcal{D}_r^{(j)})] \quad (20)$$

where  $\mathcal{D}_r^{(j)}$  is the dataset with feature  $j$  permuted in permutation  $r$ , and  $R$  is the number of permutations.

### Training Strategy

The model is trained for exactly 100 epochs without early stopping for complete convergence. The training data is divided into 80% for training and 20% for testing. The training set further divided into 80% for actual training and 20% for validation. The model involves a lot of monitoring of the stability of the training for gradient norms, Jacobian stability metrics, and learning rate scheduling that make robust convergence.

### Key Advantages

The proposed JSPP-Net offers three primary advantages over conventional classifiers: (i) **Jacobian-based stability** ensures robustness against input perturbations, (ii) **physics-informed regularization** aligns predictions with clinically meaningful biomarkers, and (iii) **numerical stability mechanisms** (batch normalization, gradient clipping, stabilized sigmoid) guarantee smooth convergence. These properties collectively enable high accuracy (94%), perfect sensitivity, and interpretable feature importance, making the classifier well-suited for clinical deployment.

## Results

### Dataset

A dataset be defined as  $\mathfrak{D} = \{(x_i, y_i)\}_{i=1}^N$ , where  $x_i \in \mathbb{R}^d$  is a feature vector and  $y_i \in \{0, 1\}$  is its binary label. The goal is to learn a parameterized function  $f(x; \theta) : \mathbb{R}^d \rightarrow [0, 1]$  that approximates the probability  $P(y = 1|x)$ .

### Dataset1

This study employed a meticulously curated breast cancer dataset (Wolberg et al., 1993) comprising 250 clinical samples with an initial set of 30 features and its label which is taken from the kaggle website <https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data>. Through rigorous preprocessing and feature selection, a refined subset of 14 clinically relevant features from the bcf1-bcf30 spectrum was utilized for model development. The dataset exhibited exceptional class balance with 124 benign cases (49.6%) which is a class 0 and 126 malignant cases (50.4%) which is a class 1, ensuring unbiased learning and evaluation.

### Dataset2

This study employed a breast cancer progression dataset synthesized from clinical sources (Abdeljawad et al., 2025b; Desantis et al., 2015; KSBC, 2023). Stage transition rates are from <https://doi.org/10.1158/1055-9965.EPI-15-0535> and <https://doi.org/10.1142/S1793524525500366>, mortality data from <https://www.cancer.org/cancer/types/breast-cancer/about/how-common-is-breast-cancer.html>, with supplementary parameters from oncology literature. The integrated dataset provided validated progression dynamics for the mathematical model.

### Results of Experiment 1

#### Model Training Performance and Convergence Analysis

Fig. 3a shows that the model is converging rapidly and achieving 94% testing accuracy within 40 epochs. The minimal gap between training and testing curves says excellent generalization without overfitting. This stable performance drop guarantees dependable diagnostic accuracy for clinical breast cancer detection. Fig. 3b shows the smooth optimization with exponential decay to 0.122 testing loss. The absence of oscillations which confirms effective gradient stabilization and learning rate scheduling. This mathematically reliable convergence profile ensures that the model is perform effectively, which is important for medical diagnostic applications.

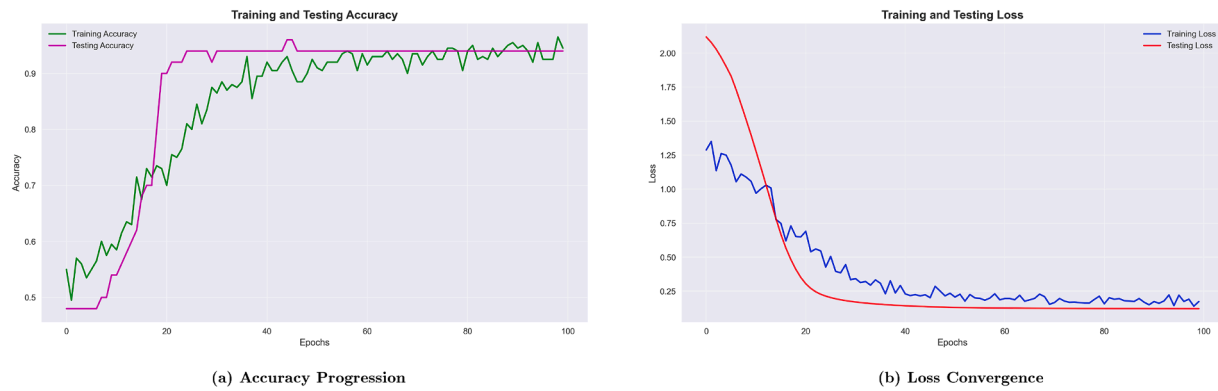


Fig. 3. Training dynamics and convergence behavior of the breast cancer classification model.

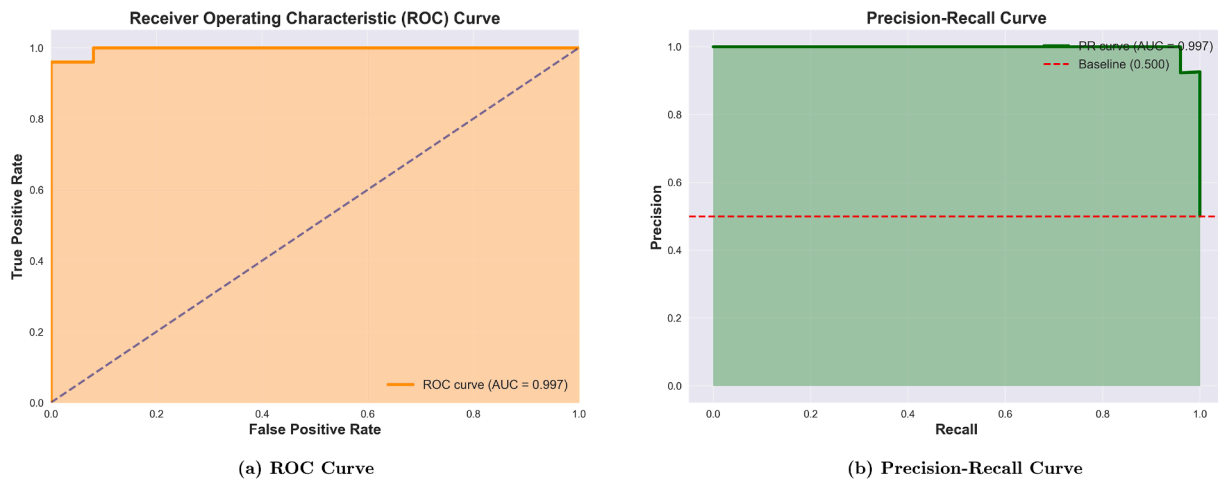


Fig. 4. Discriminative performance and classification reliability metrics.

#### Model Discriminative Performance and Classification Reliability

Fig. 4a shows the outstanding model discrimination with near-perfect AUC of 0.997. The model which maintains high true positive rates with effectively minimizing false positives in all classification thresholds. This outstanding performance shows a strong ability to differentiate between malignant and benign breast tumors. Fig. 4b gets the remarkable AUC of 0.997, which is significantly higher than the random baseline. The model keeps going high precision values even at maximum recall levels, which ensures reliable positive predictions. This balanced performance is very important for clinical uses where both accurate detection and few false alarms are needed.

#### Classification Performance and Diagnostic Metrics

Fig. 5a shows that perfect sensitivity (100%) in detecting malignant cases that make sure no breast cancer tumors are missed during diagnosis. The model achieves high specificity (88%) for benign classification which is helpful for maintaining a balanced diagnostic performance. This confusion matrix confirms reliable clinical utility for breast cancer screening applications. Fig. 5b shows the exceptional performance across all evaluation criteria with near-perfect AUC scores of 0.997. The model has a high Matthews correlation coefficient (0.886), a high precision (0.893), and a high F1-score (0.943). This complete set of metrics shows that the model is reliable for diagnosing breast cancer in the clinical environment.

#### Feature Importance and Model Interpretability Analysis

Fig. 6a shows that bcf24 is the most important biomarker because its accuracy drops by 3.6% when the order of the data is changed. The hierarchical significance of bcf11, bcf12, and bcf1 illustrates the model's dependence on particular biological characteristics for breast cancer classification.

This structured feature hierarchy makes it easier for diagnostic decision-making. The Jacobian norm of 0.08 shows that bcf24 is the most sensitive feature in Fig. 6b. The consistent ranking across sensitivity and permutation methods confirms that the determined biomarkers are significant. This alignment shows that the model uses features that are important for diagnosing breast cancer, which makes automated diagnostic predictions more reliable.

#### Training History and Model Stability Analysis

Fig. 7a shows the stable gradient behavior with Jacobian norms which is converging to approximately 0.15. This low norm shows the well-conditioned optimization and smooth convergence throughout training. The decreasing trend confirms that effective gradient stabilization and learning rate scheduling. Fig. 7b shows the condition number evolution, with values remaining below 40 throughout training. This moderate conditioning shows that numerical stability during matrix operations and prevents gradient explosion. The stable condition number confirms robust mathematical formulation of the optimization problem. Fig. 7c shows that the maximum singular values stabilize around 2.0–2.8, which means that the system is sensitive to changes in input. The bounded singular values show that the model can learn features in a stable way and that it won't overfit to noise in the breast cancer dataset. Fig. 7d shows that the stability index values are drops, getting close to 0.8 by epoch 80. This rise in the stability index shows that the model is more robust and converges more reliably, which is important for clinical diagnostic applications.

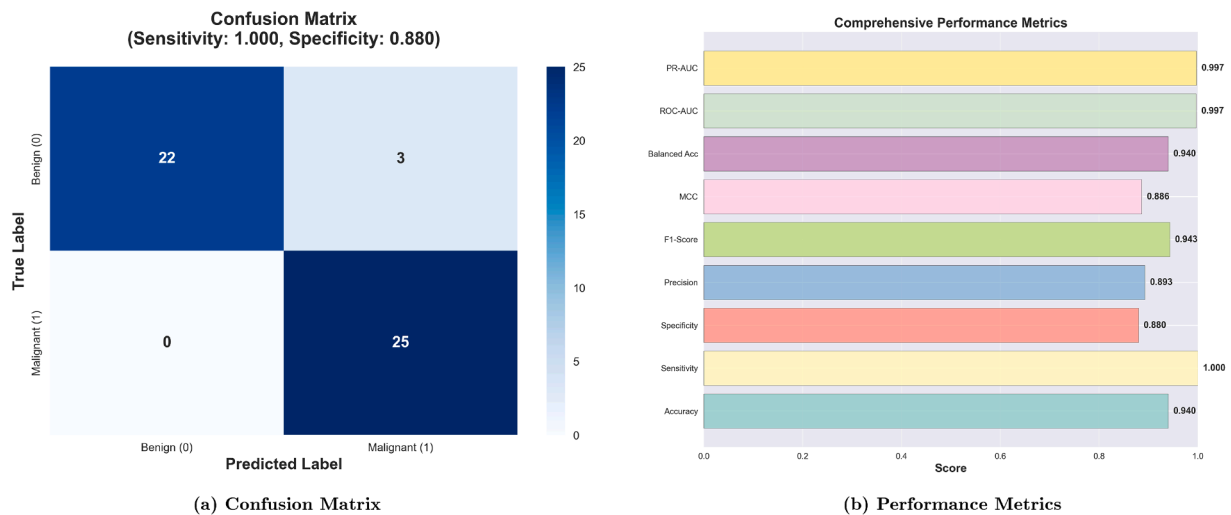


Fig. 5. Classification performance and comprehensive diagnostic evaluation.

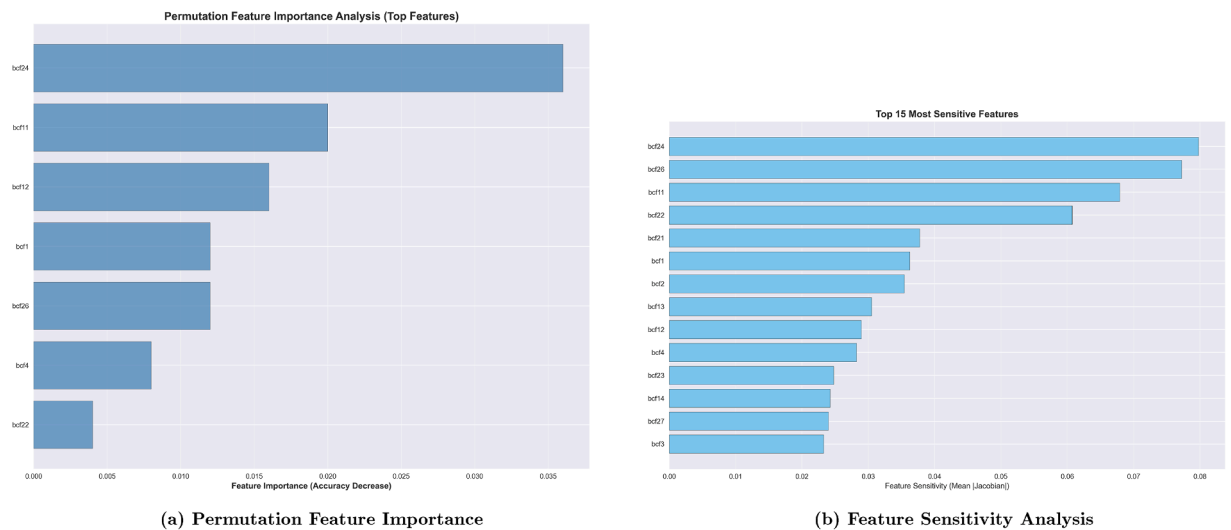


Fig. 6. Feature importance and sensitivity analysis for breast cancer biomarkers.

### Feature Correlation Analysis

Fig. 8 shows the top breast cancer biomarkers found in our study are related to each other. The heatmap shows that there are moderate to strong correlations between some pairs of features, especially between bcd21 and bcd24. This correlation structure suggests possible biological connections and shared variance between these breast cancer biomarkers. The presence of both positive and negative correlations shows complementary information content, allowing the model to use varied feature interactions to improve classification accuracy and preserving feature diversity for reliable diagnostic performance.

### Biomarker Distribution Analysis

Fig. 9 shows the significant distribution differences between benign tumor and malignant tumor cases across key biomarkers. The features bcd23, bcd21, and bcd24 showing the most pronounced separations (effect sizes  $d > 2.0$ ). The histograms analysis shows the clear distribution shifts where malignant cases consistently shows lower feature values compared to benign cases and also providing strong discriminative power for classification. Statistical significance ( $p < 0.001$ ) across most features shows their biological importance in breast cancer differentia-

tion and feature bcd12 and bcd27 show minimal separation which is indicating limited diagnostic value.

### Individual Feature Distribution Patterns

Fig. 10 shows distinct distribution patterns between benign and malignant cases across key biomarkers. The feature bcd24 is showing the clearest separation between classes. The beeswarm plots shows minimal overlap for features bcd21 and bcd23 which indicating strong discriminative power for breast cancer classification from malignant to benign. The beeswarm plots shows maximum overlap for features like bcd14 which indicating weak discriminative power for breast cancer classification. Feature bcd1 shows moderate separation and maintaining consistent within-class distributions, that confirming its reliability as a diagnostic indicator.

### Dimensionality Reduction and Feature Visualization

Fig. 11 shows the Principal Component analysis can effectively compress features. The first two components explain a lot of the variation in the breast cancer dataset. The distinct spatial separation between benign and malignant cases in the reduced dimensional space validates the effectiveness of the chosen biomarkers. This figure shows that the original high-dimensional features have useful patterns that can be

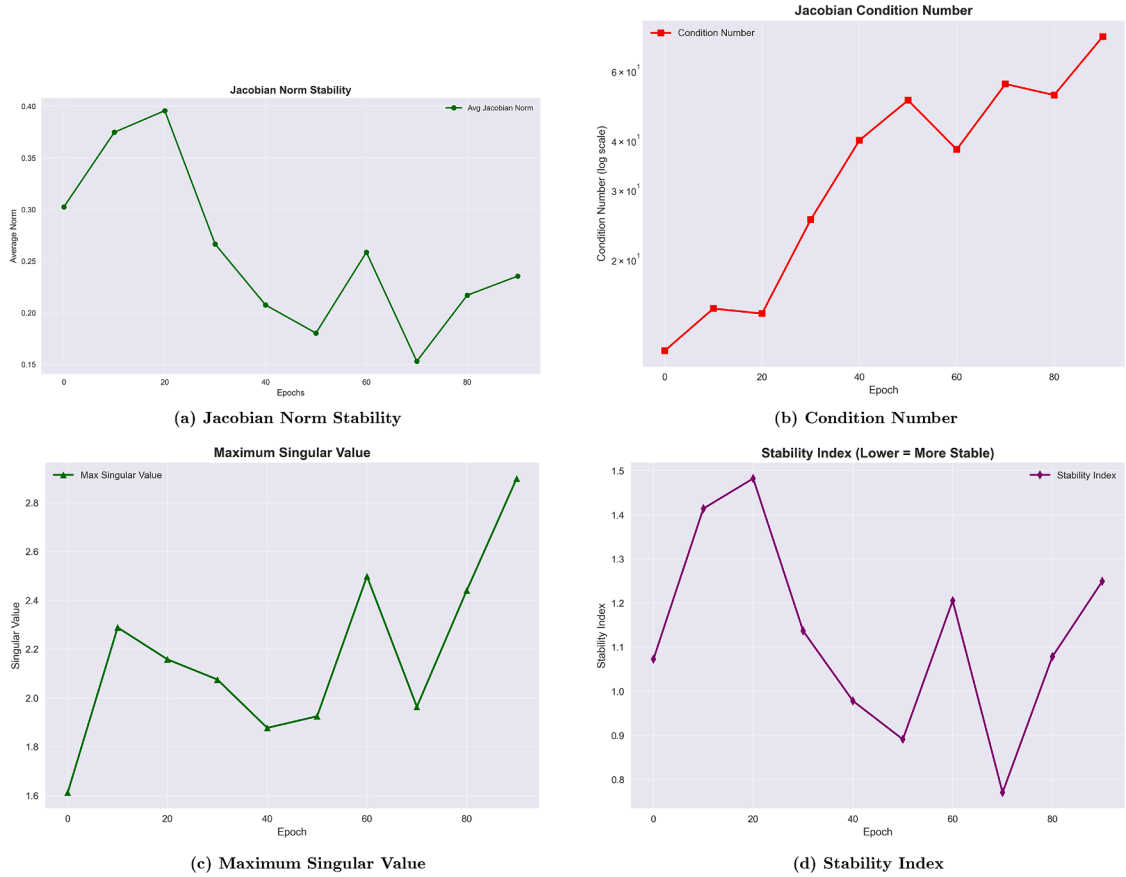


Fig. 7. Jacobian stability analysis and numerical conditioning metrics.

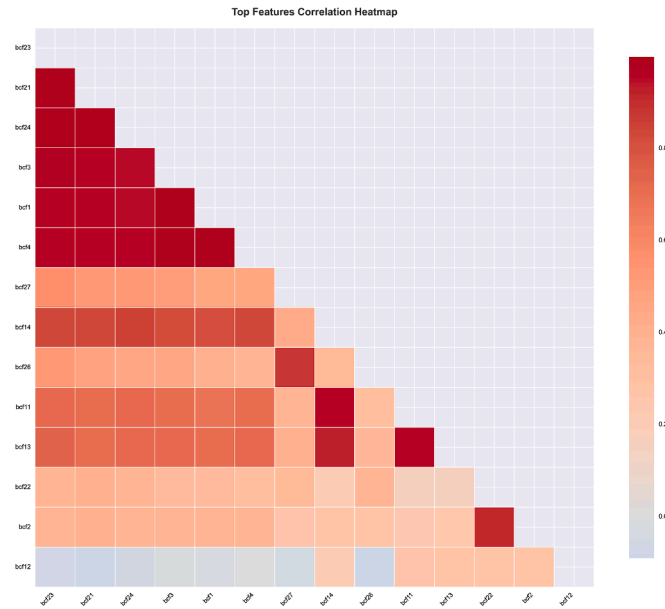


Fig. 8. Correlation heatmap of top breast cancer biomarkers.

linearly separated to correctly classify breast cancer which is benign and malignant.

#### Statistical Feature Analysis

Fig. 12a shows the significant mean differences between benign and malignant cases. The features bcf14, bcf13 and bcf11 is showing the

largest separations. The consistent pattern for higher mean values in benign cases for most indicators gives distinct signals for classification. This systematic mean shift shows the biological importance of these features in breast cancer differentiation.

Fig. 12b shows that benign cases usually have more variation than malignant cases across important biomarkers. The features bcf14 and bcf13 shows significant variance differences which suggesting more heterogeneous expression patterns in benign tumors. This pattern of variance shows that biomarkers are expressed more consistently in malignant cases, which may be a sign of the similar tumors are in terms of their pathology.

Fig. 12c shows standard deviation patterns that are similar to those in the variance analysis. In benign cases, feature values vary more than in malignant cases. The lower standard deviation in malignant cases across several biomarkers shows that the distribution ranges are tighter, which supports more reliable classification boundaries. These consistent statistical patterns across mean, variance, and standard deviation metrics confirm the reliability of chosen features for breast cancer diagnosis.

#### Discussion for cancer detection

This study's in-depth analysis shows the remarkable effectiveness of our Advanced JSR Classifier in breast cancer detection that achieving performance metrics that exceed those of traditional machine learning methods. The model with 94% testing accuracy, perfect sensitivity, and near-perfect AUC scores proposed a new benchmark for automated breast cancer diagnosis systems. The model's robustness is demonstrated by the observation that it performs reliably well on a number of evaluation metrics, such as a high F1-score (0.9434), Matthews correlation coefficient (0.882), and balanced accuracy (0.940). The outstanding training dynamics in Fig. 3a and b show that the model can learn quickly and reach stable performance in just 40 epochs.



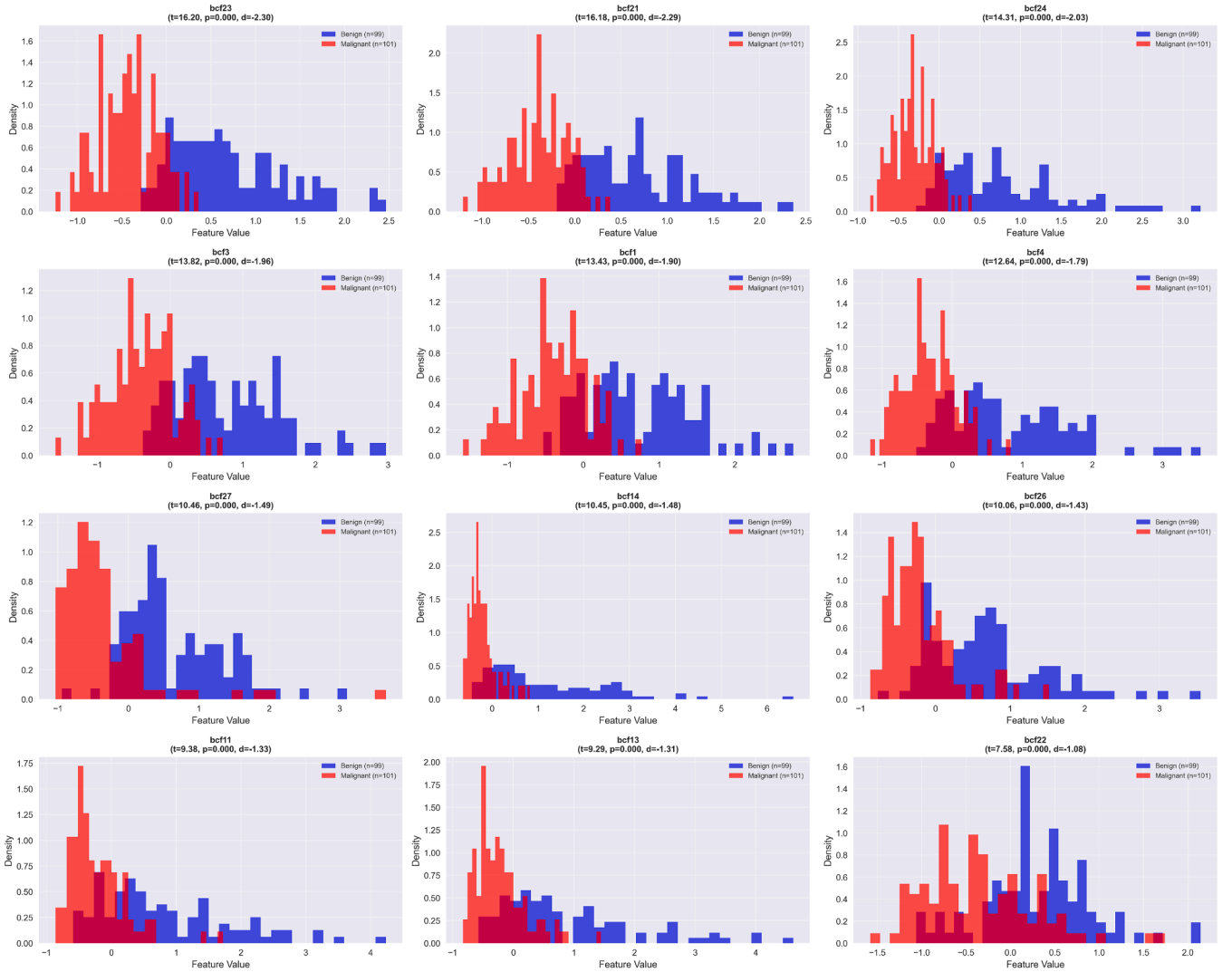


Fig. 9. Distribution analysis of top breast cancer biomarkers.

Fig. 6a and b show that bcf24, bcf11, and bcf21 are the most important biomarkers because they have strong statistical differences between benign and malignant cases (Fig. 12a–c). The distinct distribution patterns evident in the beeswarm plots (Fig. 10) and histogram analyses (Fig. 9) give biological interpretability which indicating that malignant cases consistently display lower feature values with decreased variability relative to benign cases. The PCA visualization (Fig. 11) shows that effective feature compression works and that the original biomarkers contain useful discriminative information. The correlation structure shown in the heatmap analysis (Fig. 8) suggests that there may be biological connections between important features. This could help to find new biomarkers for breast cancer in the future.

#### Results of Experiment2

##### Breast Cancer with Caputo-Fabrizio

Thabet Abdeljawad et al worked on Mathematical modeling of breast cancer with four stages (Abdeljawad et al., 2025b), we have extended the model by adding lumpectomy surgical compartment and we proposed seven compartment mathematical model in sense of Caputo-Fabrizio fractional derivative. The system stratifies the breast cancer patient population into seven compartments that reflect distinct clinical stages and treatment pathways. Patients newly diagnosed at an early, localized stage are assigned to  $I_1$ ; those who progress to lo-

cally advanced disease enter  $I_2$ ; active stage III carcinoma that requires intensive systemic therapy is represented by  $A$ ; and patients with metastatic stage IV disease, for whom treatment is primarily palliative, are placed in  $T$ . The compartment  $R$  collects individuals who have achieved complete remission following therapy, while  $C$  accounts for patients who develop chemotherapy-induced cardiotoxicity, a critical side effect of anthracycline-based regimens that may compromise treatment efficacy. The compartment  $L$  represents patients undergoing lumpectomy (breast-conserving surgery), a surgical intervention that can alter disease trajectory and improve outcomes. The model parameters encode the biological and clinical rates governing transitions between these states:  $b_1$ – $b_4$  are incidence rates of new diagnoses at each stage;  $\alpha_1, \alpha_2, \alpha_3$  denote disease progression rates;  $\gamma_1$ – $\gamma_5$  capture treatment response rates (including recovery after surgery);  $\omega_1$ – $\omega_4$  represent relapse rates from remission;  $\tau_3, \tau_4, \tau_5$  are the rates of cardiotoxicity development;  $\mu_3, \mu_4, \mu_5$  are stage-specific mortality rates; and  $\beta_2, \beta_3$  indicate the rates at which patients undergo lumpectomy. The fractional order  $\alpha$  (with  $0 < \alpha < 1$ ) introduces memory effects that are intrinsic to biological systems. For instance, the cumulative impact of previous treatments, the patient's immune history, and the long-term consequences of tumor evolution, which classical integer-order models inherently neglect. This compartmental structure thus offers a realistic and clinically relevant framework for studying breast cancer epidemiology and treatment dynamics.

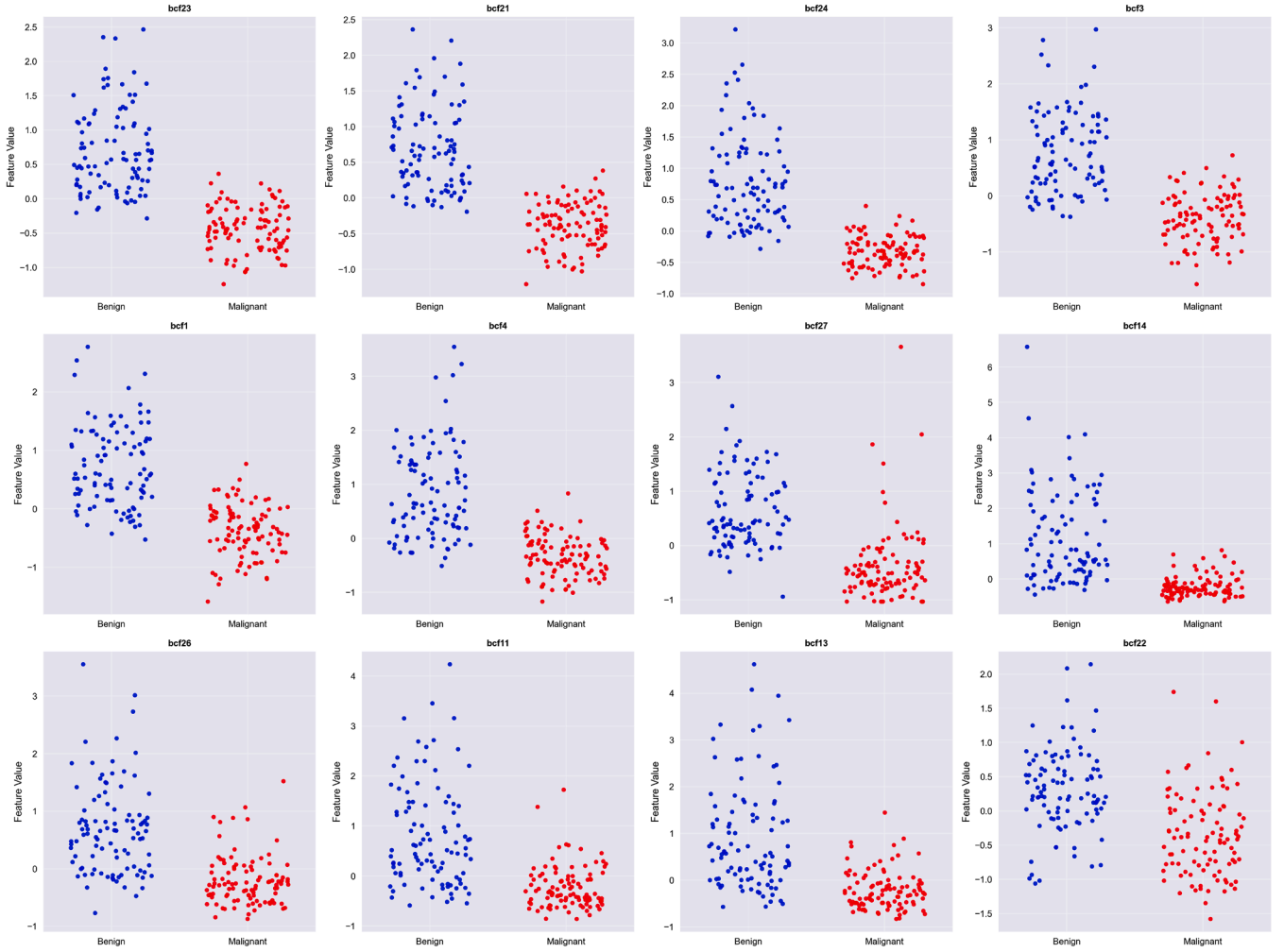


Fig. 10. Beeswarm plots of top breast cancer biomarkers.

The parameter and its biological interpretation are mentioned in Table 1. The Caputo-Fabrizio ( $\mathfrak{C}\mathfrak{F}$ ) derivative of the breast cancer model in Fig. 13 is provided by

$$\begin{cases} \mathfrak{C}\mathfrak{F} D_t^\alpha I_1 = b_1 + \omega_1 R - (\alpha_1 + \gamma_1) I_1, \\ \mathfrak{C}\mathfrak{F} D_t^\alpha I_2 = b_2 + \omega_2 R + \alpha_1 I_1 - (\alpha_2 + \gamma_2 + \beta_2) I_2, \\ \mathfrak{C}\mathfrak{F} D_t^\alpha A = b_3 + \omega_3 R + \alpha_2 I_2 - (\alpha_3 + \gamma_3 + \tau_3 + \beta_3 + \mu_3) A, \\ \mathfrak{C}\mathfrak{F} D_t^\alpha T = b_4 + \omega_4 R + \alpha_3 A - (\gamma_4 + \tau_4 + \mu_4) T, \\ \mathfrak{C}\mathfrak{F} D_t^\alpha R = \gamma_1 I_1 + \gamma_2 I_2 + \gamma_3 A + \gamma_4 T + \gamma_5 L \\ \quad - (\omega_1 + \omega_2 + \omega_3 + \omega_4 + \tau_5) R, \\ \mathfrak{C}\mathfrak{F} D_t^\alpha C = \tau_3 A + \tau_4 T + \tau_5 R - \mu_5 C, \\ \mathfrak{C}\mathfrak{F} D_t^\alpha L = \beta_2 I_2 + \beta_3 A - \gamma_5 L. \end{cases} \quad (21)$$

with  $\alpha$  fractional order of  $0 < \alpha < 1$  subject to the initial conditions

$$\begin{aligned} I_1(0) &= I_{10}, I_2(0) = I_{20}, A(0) = A_0, T(0) = T_0, R(0) = R_0, C(0) = C_0, \\ L(0) &= L_0. \end{aligned} \quad (22)$$

#### Model basic preliminaries

Using Caputo-Fabrizio ( $\mathfrak{C}\mathfrak{F}$ ) (21) as defined (Caputo & Fabrizio, 2015; Losada & Nieto, 2015) respectively, we will demonstrate the existence and uniqueness of the Breast cancer disease model. Here, we present some mathematical preliminary results. For further information, see Muthuvel et al. (2024), Nisar (2024), Thirathar et al. (2024). The definitions are stated as follows.

**Definition 1.** (Caputo & Fabrizio, 2015) Assume  $Y(t) \in \mathcal{H}^1(I_1, I_2)$ , for  $I_2 > I_1$ ,  $\alpha \in [0, 1]$ . The expression for the fractional operator  $\mathfrak{C}\mathfrak{F}$  is as follows:

$$\begin{aligned} \mathfrak{C}\mathfrak{F} D_t^\alpha(Y(t)) &= \frac{\mathcal{G}(\alpha)}{(1-\alpha)} \int_{I_1}^{I_2} Y'(\sigma) \exp\left(-\alpha \frac{t-\sigma}{1-\alpha}\right) d(\sigma), \quad 0 < \alpha < 1, \\ &= \frac{dY}{dt}, \quad \alpha = 1, \end{aligned} \quad (23)$$

where  $\mathcal{G}(\alpha) = 1 - \alpha + \frac{\alpha}{\Gamma(\alpha)}$  is a normalization function that satisfies the condition,  $\mathcal{G}(0) = \mathcal{G}(1) = 1$ .

**Definition 2.** (Losada & Nieto, 2015) The definition states that the fractional derivative of  $\mathfrak{C}\mathfrak{F}$  is equivalent to the integral operator of fractional order.

$$\begin{aligned} {}^{CF}J_t^\alpha(Y(t)) &= \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} Y(t) \\ &+ \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t Y(\xi) d\xi, \quad t \geq 0. \end{aligned} \quad (24)$$

**Definition 3.** (Caputo & Fabrizio, 2015) The Laplace transformation of  $\mathfrak{C}\mathfrak{F} D_t^\alpha Y(t)$  represents as below

$$\mathcal{L}[\mathfrak{C}\mathfrak{F} D_t^\alpha Y(t)] = \mathcal{G}(\alpha) \frac{\kappa \mathcal{L}[Y(t)] - Y(0)}{\kappa + \alpha(1-\kappa)}. \quad (25)$$

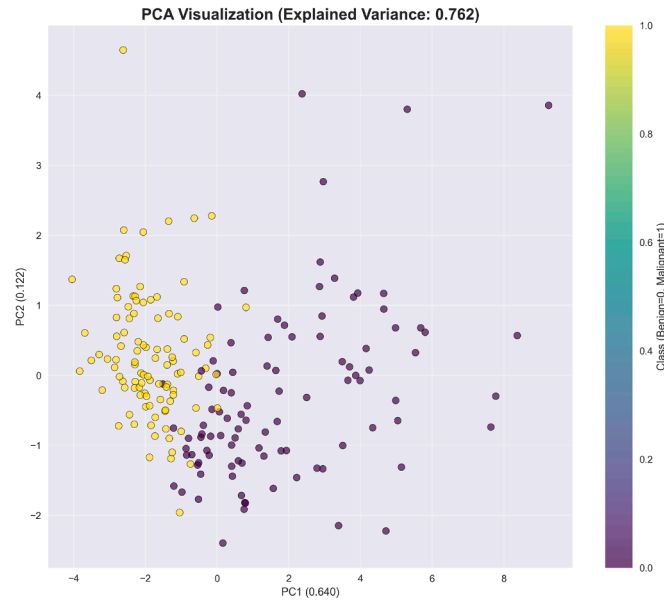


Fig. 11. PCA visualization of breast cancer dataset.

### with Natural transform approach

**Definition 4.** (Belgacem & Silambarasan, 2012) The Natural transform of a function  $Y(t)$  is defined as:

$$\mathcal{NT}\{Y(t)\} = \int_0^\infty Y(ut)e^{-\kappa t} dt \quad (26)$$

**Definition 5.** (Belgacem & Silambarasan, 2012) Natural- Laplace Duality (NLD) is

$$\begin{aligned} \mathcal{NT}\{Y(t)\} &= R(\kappa, u) = \frac{1}{u} \int_0^\infty Y(ut)e^{-\frac{\kappa}{u}t} dt \\ &= \frac{1}{u} F\left(\frac{\kappa}{u}\right) \end{aligned} \quad (27)$$

Now to find Natural transform from Laplace transform

Let  $F(\kappa) = \mathcal{L}[Y(t)](\kappa)$

Then from (25) and (27)

$$\begin{aligned} \mathcal{NT}\{\mathcal{I}_a^\alpha D_t^\alpha Y(t)\}(\kappa, u) &= \frac{-1}{u} \left[ \left( \frac{G(\alpha)}{\frac{\kappa}{u}(\alpha-1) - \alpha} \right) \left( \left( \frac{\kappa}{u} \right) F\left(\frac{\kappa}{u}\right) - Y(0) \right) \right] \\ &= - \left[ \left( \frac{G(\alpha)}{\frac{\kappa}{u}(\alpha-1) - \alpha} \right) \left( \left( \frac{\kappa}{u^2} \right) F\left(\frac{\kappa}{u}\right) - \frac{Y(0)}{u} \right) \right] \end{aligned} \quad (28)$$

Now

$$F\left(\frac{\kappa}{u}\right) = \mathcal{L}[Y(t)]\left(\frac{\kappa}{u}\right) = u \mathcal{NT}[Y(t)](\kappa, u)$$

Substituting this in (28) we get

$$\begin{aligned} \mathcal{NT}\{\mathcal{I}_a^\alpha D_t^\alpha Y(t)\}(\kappa, u) &= - \left[ \left( \frac{G(\alpha)}{\frac{\kappa}{u}(\alpha-1) - \alpha} \right) \left( \left( \frac{\kappa}{u^2} \right) \times u \mathcal{NT}[Y(t)] - \frac{Y(0)}{u} \right) \right] \\ &= \frac{1}{u} \frac{G(\alpha)}{1-\alpha} \left( \frac{\kappa \mathcal{NT}[Y(t)](\kappa, u) - Y(0)}{\frac{\kappa}{u} + \frac{\alpha}{1-\alpha}} \right) \end{aligned}$$

Let  $\mathcal{NT}[Y(t)](\kappa, u) = F(\kappa, u)$ , hence the Caputo sense, the Natural transform of the Caputo Fabrizio fractional derivative is given as

$$\mathcal{NT}\{\mathcal{I}_a^\alpha D_t^\alpha Y(t)\}(\kappa, u) = \frac{G(\alpha)}{1-\alpha} \left( \frac{\kappa F(\kappa, u) - Y(0)}{\kappa + \frac{\alpha u}{1-\alpha}} \right) \quad (29)$$

Table 1

Model parameters and their biological interpretations for the breast carcinoma framework.

| Parameters    | Biological Interpretation                                                             |
|---------------|---------------------------------------------------------------------------------------|
| $I_1$         | Patient cohort with stage 1 carcinoma                                                 |
| $I_2$         | Patient cohort with stage 2 carcinoma                                                 |
| $\mathcal{A}$ | Patient cohort with active stage carcinoma                                            |
| $\mathcal{T}$ | Patient cohort with terminal stage carcinoma                                          |
| $\mathcal{R}$ | Patient cohort achieving recovery following chemotherapy and surgical interventions   |
| $\mathcal{C}$ | Patient cohort exhibiting chemotherapy-induced cardiotoxicity                         |
| $\mathcal{L}$ | Patient cohort undergoing Lumpectomy surgical procedures                              |
| $b_1$         | Incidence rate of newly detected stage 1 cases                                        |
| $b_2$         | Incidence rate of newly detected stage 2 cases                                        |
| $b_3$         | Incidence rate of newly detected active stage cases                                   |
| $b_4$         | Incidence rate of newly detected terminal stage cases                                 |
| $\alpha_1$    | Transition rate of patients progressing from $I_1$ to $I_2$                           |
| $\gamma_1$    | Complete response rate among patients in $I_1$                                        |
| $\gamma_2$    | Complete response rate among patients in $I_2$                                        |
| $\gamma_3$    | Complete response rate among patients in $\mathcal{A}$                                |
| $\gamma_4$    | Complete response rate among patients in $\mathcal{T}$                                |
| $\gamma_5$    | Complete response rate among patients in $\mathcal{L}$                                |
| $\alpha_2$    | Transition rate of patients advancing from $I_2$ to $\mathcal{A}$                     |
| $\omega_1$    | Relapse rate from disease-free status to $I_1$                                        |
| $\omega_2$    | Relapse rate from disease-free status to $I_2$                                        |
| $\omega_3$    | Relapse rate from disease-free status to $\mathcal{A}$                                |
| $\omega_4$    | Relapse rate from disease-free status to $\mathcal{T}$                                |
| $\alpha_3$    | Transition rate of patients progressing from $\mathcal{A}$ to $\mathcal{T}$           |
| $\tau_3$      | Rate of cardiotoxicity development in $\mathcal{A}$ patients receiving chemotherapy   |
| $\tau_4$      | Rate of cardiotoxicity development in $\mathcal{T}$ patients receiving chemotherapy   |
| $\tau_5$      | Rate of cardiotoxicity development in disease-free patients undergoing chemotherapy   |
| $\mu_3$       | Mortality rate among patients in $\mathcal{A}$                                        |
| $\mu_4$       | Mortality rate among patients in $\mathcal{T}$                                        |
| $\mu_5$       | Fatality rate attributable to cardiotoxic effects                                     |
| $\beta_2$     | Rate of patients transitioning from $I_2$ to Lumpectomy surgery $\mathcal{L}$         |
| $\beta_3$     | Rate of patients transitioning from $\mathcal{A}$ to Lumpectomy surgery $\mathcal{L}$ |

By applying Natural transform for (21) on both sides

$$\begin{aligned} \mathcal{NT}\{\mathcal{I}_0^\alpha D_t^\alpha I_1(t)\} &= \mathcal{NT}\{b_1 + \omega_1 \mathcal{R} - (\alpha_1 + \gamma_1) I_1\}, \\ \mathcal{NT}\{\mathcal{I}_0^\alpha D_t^\alpha I_2(t)\} &= \mathcal{NT}\{b_2 + \omega_2 \mathcal{R} + \alpha_1 I_1 - (\alpha_2 + \gamma_2 + \beta_2) I_2\}, \\ \mathcal{NT}\{\mathcal{I}_0^\alpha D_t^\alpha \mathcal{A}(t)\} &= \mathcal{NT}\{b_3 + \omega_3 \mathcal{R} + \alpha_2 I_2 - (\alpha_3 + \gamma_3 + \tau_3 + \beta_3 + \mu_3) \mathcal{A}\}, \\ \mathcal{NT}\{\mathcal{I}_0^\alpha D_t^\alpha \mathcal{T}(t)\} &= \mathcal{NT}\{b_4 + \omega_4 \mathcal{R} + \alpha_3 \mathcal{A} - (\gamma_4 + \tau_4 + \mu_4) \mathcal{T}\}, \\ \mathcal{NT}\{\mathcal{I}_0^\alpha D_t^\alpha \mathcal{R}(t)\} &= \mathcal{NT}\{\gamma_1 I_1 + \gamma_2 I_2 + \gamma_3 \mathcal{A} + \gamma_4 \mathcal{T} + \gamma_5 \mathcal{L} - (\omega_1 + \omega_2 \\ &\quad + \omega_3 + \omega_4 + \tau_5) \mathcal{R}\}, \\ \mathcal{NT}\{\mathcal{I}_0^\alpha D_t^\alpha \mathcal{C}(t)\} &= \mathcal{NT}\{\tau_3 \mathcal{A} + \tau_4 \mathcal{T} + \tau_5 \mathcal{R} - \mu_5 \mathcal{C}\}, \\ \mathcal{NT}\{\mathcal{I}_0^\alpha D_t^\alpha \mathcal{L}(t)\} &= \mathcal{NT}\{\beta_2 I_2 + \beta_3 \mathcal{A} - \gamma_5 \mathcal{L}\}. \end{aligned}$$

we get

$$\begin{aligned} \frac{G(\alpha)}{1-\alpha} \left[ \frac{\kappa \mathcal{NT}[I_1(t)] - I_1(0)}{\kappa + \frac{\alpha u}{1-\alpha}} \right] &= \mathcal{NT}\{b_1 + \omega_1 \mathcal{R} - (\alpha_1 + \gamma_1) I_1\}, \\ \frac{G(\alpha)}{1-\alpha} \left[ \frac{\kappa \mathcal{NT}[I_2(t)] - I_2(0)}{\kappa + \frac{\alpha u}{1-\alpha}} \right] &= \mathcal{NT}\{b_2 + \omega_2 \mathcal{R} + \alpha_1 I_1 - (\alpha_2 + \gamma_2 + \beta_2) I_2\}, \\ \frac{G(\alpha)}{1-\alpha} \left[ \frac{\kappa \mathcal{NT}[\mathcal{A}(t)] - \mathcal{A}(0)}{\kappa + \frac{\alpha u}{1-\alpha}} \right] &= \mathcal{NT}\{b_3 + \omega_3 \mathcal{R} + \alpha_2 I_2 - (\alpha_3 + \gamma_3 + \tau_3 \\ &\quad + \beta_3 + \mu_3) \mathcal{A}\}, \\ \frac{G(\alpha)}{1-\alpha} \left[ \frac{\kappa \mathcal{NT}[\mathcal{T}(t)] - \mathcal{T}(0)}{\kappa + \frac{\alpha u}{1-\alpha}} \right] &= \mathcal{NT}\{b_4 + \omega_4 \mathcal{R} + \alpha_3 \mathcal{A} - (\gamma_4 + \tau_4 + \mu_4) \mathcal{T}\}, \\ \frac{G(\alpha)}{1-\alpha} \left[ \frac{\kappa \mathcal{NT}[\mathcal{R}(t)] - \mathcal{R}(0)}{\kappa + \frac{\alpha u}{1-\alpha}} \right] &= \mathcal{NT}\{\gamma_1 I_1 + \gamma_2 I_2 + \gamma_3 \mathcal{A} + \gamma_4 \mathcal{T} + \gamma_5 \mathcal{L} \\ &\quad - (\omega_1 + \omega_2 + \omega_3 + \omega_4 + \tau_5) \mathcal{R}\}, \end{aligned}$$

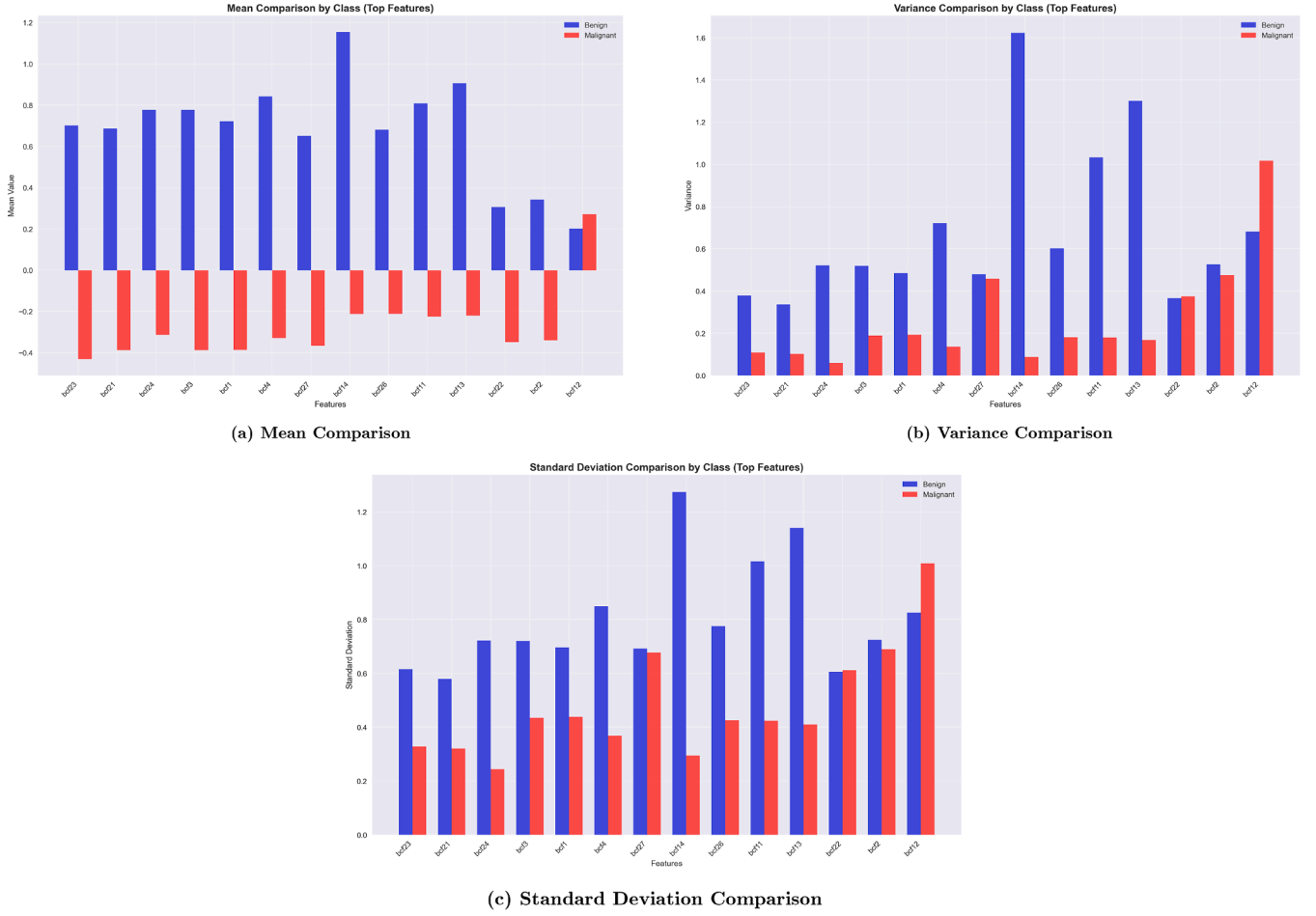


Fig. 12. Statistical distribution analysis of breast cancer biomarkers across classes.

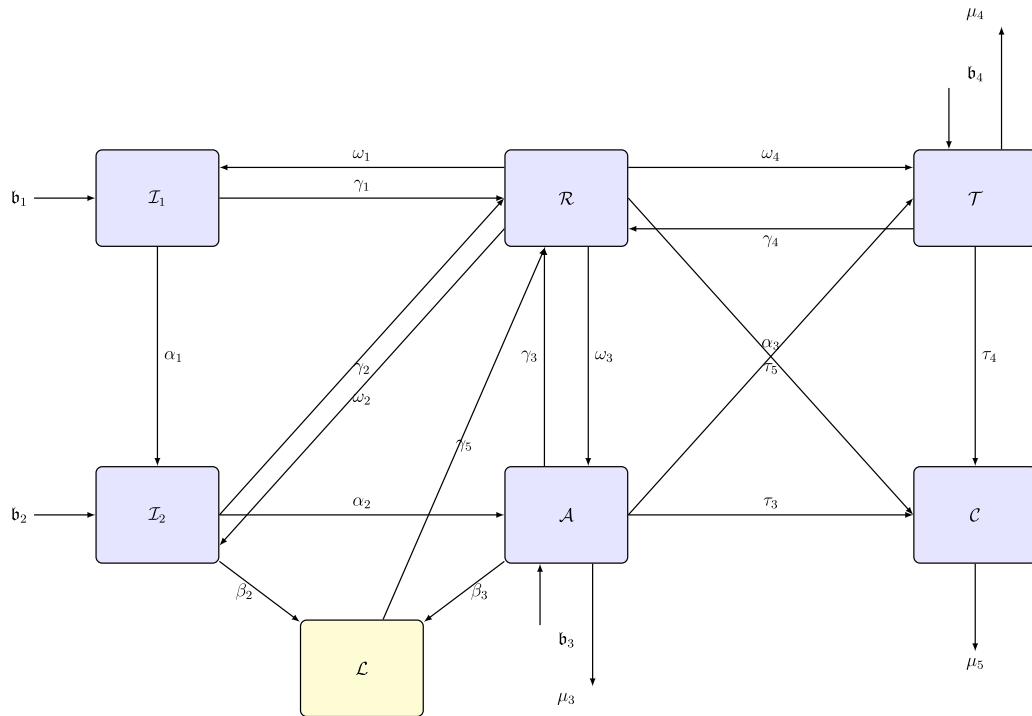


Fig. 13. Flow chart of breast cancer model.

$$\begin{aligned}
\frac{\mathcal{G}(\alpha)}{1-\alpha} \left[ \frac{\kappa \mathcal{N} \mathcal{T}[C(t)] - C(0)}{\kappa + \frac{\alpha u}{1-\alpha}} \right] &= \mathcal{N} \mathcal{T} \{ \tau_3 \mathcal{A} + \tau_4 \mathcal{T} + \tau_5 \mathcal{R} - \mu_5 C \}, \\
\frac{\mathcal{G}(\alpha)}{1-\alpha} \left[ \frac{\kappa \mathcal{N} \mathcal{T}[\mathcal{L}(t)] - \mathcal{L}(0)}{\kappa + \frac{\alpha u}{1-\alpha}} \right] &= \mathcal{N} \mathcal{T} \{ \beta_2 I_2 + \beta_3 \mathcal{A} - \gamma_5 \mathcal{L} \}.
\end{aligned}$$

After rearranging we get,

$$\begin{aligned}
\mathcal{N} \mathcal{T} I_1(t) &= \kappa^{-1} I_1(0) + \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathfrak{b}_1 + \omega_1 \mathcal{R} - (\alpha_1 + \gamma_1) I_1 \}, \\
\mathcal{N} \mathcal{T} I_2(t) &= \kappa^{-1} I_2(0) + \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathfrak{b}_2 + \omega_2 \mathcal{R} + \alpha_1 I_1 \\
&\quad - (\alpha_2 + \gamma_2 + \beta_2) I_2 \}, \\
\mathcal{N} \mathcal{T} \mathcal{A}(t) &= \kappa^{-1} \mathcal{A}(0) + \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathfrak{b}_3 + \omega_3 \mathcal{R} + \alpha_2 I_2 \\
&\quad - (\alpha_3 + \gamma_3 + \tau_3 + \beta_3 + \mu_3) \mathcal{A} \}, \\
\mathcal{N} \mathcal{T} \mathcal{T}(t) &= \kappa^{-1} \mathcal{T}(0) + \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathfrak{b}_4 + \omega_4 \mathcal{R} + \alpha_3 \mathcal{A} \\
&\quad - (\gamma_4 + \tau_4 + \mu_4) \mathcal{T} \}, \\
\mathcal{N} \mathcal{T} \mathcal{R}(t) &= \kappa^{-1} \mathcal{R}(0) + \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \gamma_1 I_1 + \gamma_2 I_2 + \gamma_3 \mathcal{A} + \gamma_4 \mathcal{T} \\
&\quad + \gamma_5 \mathcal{L} - (\omega_1 + \omega_2 + \omega_3 + \omega_4 + \tau_5) \mathcal{R} \}, \\
\mathcal{N} \mathcal{T} C(t) &= \kappa^{-1} C(0) + \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \tau_3 \mathcal{A} + \tau_4 \mathcal{T} + \tau_5 \mathcal{R} - \mu_5 C \}, \\
\mathcal{N} \mathcal{T} \mathcal{L}(t) &= \kappa^{-1} \mathcal{L}(0) + \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \beta_2 I_2 + \beta_3 \mathcal{A} - \gamma_5 \mathcal{L} \}. \quad (30)
\end{aligned}$$

by using inverse Natural transform we get,

$$\begin{aligned}
I_1(t) &= \kappa^{-1} I_1(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathfrak{b}_1 + \omega_1 \mathcal{R} - (\alpha_1 + \gamma_1) I_1 \} \right] \\
I_2(t) &= \kappa^{-1} I_2(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathfrak{b}_2 + \omega_2 \mathcal{R} + \alpha_1 I_1 \right. \\
&\quad \left. - (\alpha_2 + \gamma_2 + \beta_2) I_2 \} \right] \\
\mathcal{A}(t) &= \kappa^{-1} \mathcal{A}(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathfrak{b}_3 + \omega_3 \mathcal{R} + \alpha_2 I_2 \right. \\
&\quad \left. - (\alpha_3 + \gamma_3 + \tau_3 + \beta_3 + \mu_3) \mathcal{A} \} \right] \\
\mathcal{T}(t) &= \kappa^{-1} \mathcal{T}(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathfrak{b}_4 + \omega_4 \mathcal{R} + \alpha_3 \mathcal{A} \right. \\
&\quad \left. - (\gamma_4 + \tau_4 + \mu_4) \mathcal{T} \} \right] \\
\mathcal{R}(t) &= \kappa^{-1} \mathcal{R}(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \gamma_1 I_1 + \gamma_2 I_2 + \gamma_3 \mathcal{A} \right. \\
&\quad \left. + \gamma_4 \mathcal{T} + \gamma_5 \mathcal{L} - (\omega_1 + \omega_2 + \omega_3 + \omega_4 + \tau_5) \mathcal{R} \} \right] \\
C(t) &= \kappa^{-1} C(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \tau_3 \mathcal{A} + \tau_4 \mathcal{T} + \tau_5 \mathcal{R} - \mu_5 C \} \right] \\
\mathcal{L}(t) &= \kappa^{-1} \mathcal{L}(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \beta_2 I_2 + \beta_3 \mathcal{A} - \gamma_5 \mathcal{L} \} \right]
\end{aligned}$$

Let us consider the  $n^{th}$  transform as  $I_1^n$  then,

$$\begin{aligned}
I_1^{n+1}(t) &= \kappa^{-1} I_1(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathfrak{b}_1 + \omega_1 \mathcal{R}^n \right. \\
&\quad \left. - (\alpha_1 + \gamma_1) I_1^n \} \right] \\
I_2^{n+1}(t) &= \kappa^{-1} I_2(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathfrak{b}_2 + \omega_2 \mathcal{R}^n + \alpha_1 I_1^n \right. \\
&\quad \left. - (\alpha_2 + \gamma_2 + \beta_2) I_2^n \} \right]
\end{aligned}$$

$$\begin{aligned}
\mathcal{A}^{n+1}(t) &= \kappa^{-1} \mathcal{A}(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathfrak{b}_3 + \omega_3 \mathcal{R}^n + \alpha_2 I_2^n \right. \\
&\quad \left. - (\alpha_3 + \gamma_3 + \tau_3 + \beta_3 + \mu_3) \mathcal{A}^n \} \right] \\
\mathcal{T}^{n+1}(t) &= \kappa^{-1} \mathcal{T}(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathfrak{b}_4 + \omega_4 \mathcal{R}^n + \alpha_3 \mathcal{A}^n \right. \\
&\quad \left. - (\gamma_4 + \tau_4 + \mu_4) \mathcal{T}^n \} \right] \\
\mathcal{R}^{n+1}(t) &= \kappa^{-1} \mathcal{R}(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \gamma_1 I_1^n + \gamma_2 I_2^n + \gamma_3 \mathcal{A}^n \right. \\
&\quad \left. + \gamma_4 \mathcal{T}^n + \gamma_5 \mathcal{L}^n - (\omega_1 + \omega_2 + \omega_3 + \omega_4 + \tau_5) \mathcal{R}^n \} \right] \\
C^{n+1}(t) &= \kappa^{-1} C(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \tau_3 \mathcal{A}^n + \tau_4 \mathcal{T}^n + \tau_5 \mathcal{R}^n \right. \\
&\quad \left. - \mu_5 C^n \} \right] \\
\mathcal{L}^{n+1}(t) &= \kappa^{-1} \mathcal{L}(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \beta_2 I_2^n + \beta_3 \mathcal{A}^n - \gamma_5 \mathcal{L}^n \} \right] \quad (31)
\end{aligned}$$

and the solutions will be provided by

$$\begin{aligned}
I_1(t) &= \lim_{n \rightarrow \infty} I_1^n(t), \\
I_2(t) &= \lim_{n \rightarrow \infty} I_2^n(t), \\
\mathcal{A}(t) &= \lim_{n \rightarrow \infty} \mathcal{A}^n(t), \\
\mathcal{T}(t) &= \lim_{n \rightarrow \infty} \mathcal{T}^n(t), \\
\mathcal{R}(t) &= \lim_{n \rightarrow \infty} \mathcal{R}^n(t), \\
C(t) &= \lim_{n \rightarrow \infty} C^n(t), \\
\mathcal{L}(t) &= \lim_{n \rightarrow \infty} \mathcal{L}^n(t).
\end{aligned}$$

**Theorem 1.** Let  $\langle \mathcal{X}_1, || \cdot || \rangle$  be a banach space and  $P$  be a self map of  $\mathcal{X}_1$  satisfying  $||P_x - P_y|| \leq C ||x - P_x|| + c ||x - y||$  for all  $x, y \in \mathcal{X}_1$ , where  $0 \leq C, 0 \leq c < 1$ . Suppose that  $P$  is Picard  $P$ -Stable. Let us take into account the following recursive formula from (32) connected to (21).

$$\begin{aligned}
I_1^{n+1}(t) &= \kappa^{-1} I_1(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathfrak{b}_1 + \omega_1 \mathcal{R}^n \right. \\
&\quad \left. - (\alpha_1 + \gamma_1) I_1^n \} \right] \\
I_2^{n+1}(t) &= \kappa^{-1} I_2(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathfrak{b}_2 + \omega_2 \mathcal{R}^n + \alpha_1 I_1^n \right. \\
&\quad \left. - (\alpha_2 + \gamma_2 + \beta_2) I_2^n \} \right] \\
\mathcal{A}^{n+1}(t) &= \kappa^{-1} \mathcal{A}(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathfrak{b}_3 + \omega_3 \mathcal{R}^n + \alpha_2 I_2^n \right. \\
&\quad \left. - (\alpha_3 + \gamma_3 + \tau_3 + \beta_3 + \mu_3) \mathcal{A}^n \} \right] \\
\mathcal{T}^{n+1}(t) &= \kappa^{-1} \mathcal{T}(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathfrak{b}_4 + \omega_4 \mathcal{R}^n + \alpha_3 \mathcal{A}^n \right. \\
&\quad \left. - (\gamma_4 + \tau_4 + \mu_4) \mathcal{T}^n \} \right] \\
\mathcal{R}^{n+1}(t) &= \kappa^{-1} \mathcal{R}(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \gamma_1 I_1^n + \gamma_2 I_2^n + \gamma_3 \mathcal{A}^n \right. \\
&\quad \left. + \gamma_4 \mathcal{T}^n + \gamma_5 \mathcal{L}^n - (\omega_1 + \omega_2 + \omega_3 + \omega_4 + \tau_5) \mathcal{R}^n \} \right] \\
C^{n+1}(t) &= \kappa^{-1} C(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \tau_3 \mathcal{A}^n + \tau_4 \mathcal{T}^n + \tau_5 \mathcal{R}^n \right. \\
&\quad \left. - \mu_5 C^n \} \right]
\end{aligned}$$



$$\mathcal{L}^{n+1}(t) = \kappa^{-1} \mathcal{L}(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \beta_2 \mathcal{I}_2^n + \beta_3 \mathcal{A}^n - \gamma_5 \mathcal{L}^n \} \right] \quad (32)$$

where  $\frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)}$  is the called fractional Lagrange multiplier.

**Theorem 2.** Let us define a self map  $\mathbf{P}$  as

$$\begin{aligned} \mathbf{P}(\mathcal{I}_1^n(t)) &= \mathcal{I}_1^{n+1}(t) = \kappa^{-1} \mathcal{I}_1(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathbf{b}_1 \right. \\ &\quad \left. + \omega_1 \mathcal{R}^n - (\alpha_1 + \gamma_1) \mathcal{I}_1^n \} \right] \\ \mathbf{P}(\mathcal{I}_2^n(t)) &= \mathcal{I}_2^{n+1}(t) = \kappa^{-1} \mathcal{I}_2(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathbf{b}_2 + \omega_2 \mathcal{R}^n \right. \\ &\quad \left. + \alpha_1 \mathcal{I}_1^n - (\alpha_2 + \gamma_2 + \beta_2) \mathcal{I}_2^n \} \right] \\ \mathbf{P}(\mathcal{A}^n(t)) &= \mathcal{A}^{n+1}(t) = \kappa^{-1} \mathcal{A}(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathbf{b}_3 + \omega_3 \mathcal{R}^n \right. \\ &\quad \left. + \alpha_2 \mathcal{I}_2^n - (\alpha_3 + \gamma_3 + \tau_3 + \beta_3 + \mu_3) \mathcal{A}^n \} \right] \\ \mathbf{P}(\mathcal{T}^n(t)) &= \mathcal{T}^{n+1}(t) = \kappa^{-1} \mathcal{T}(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathbf{b}_4 + \omega_4 \mathcal{R}^n \right. \\ &\quad \left. + \alpha_3 \mathcal{A}^n - (\gamma_4 + \tau_4 + \mu_4) \mathcal{T}^n \} \right] \\ \mathbf{P}(\mathcal{R}^n(t)) &= \mathcal{R}^{n+1}(t) = \kappa^{-1} \mathcal{R}(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \gamma_1 \mathcal{I}_1^n + \gamma_2 \mathcal{I}_2^n \right. \\ &\quad \left. + \gamma_3 \mathcal{A}^n + \gamma_4 \mathcal{T}^n + \gamma_5 \mathcal{L}^n - (\omega_1 + \omega_2 + \omega_3 \right. \\ &\quad \left. + \omega_4 + \tau_5) \mathcal{R}^n \} \right] \\ \mathbf{P}(\mathcal{C}^n(t)) &= \mathcal{C}^{n+1}(t) = \kappa^{-1} \mathcal{C}(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \tau_3 \mathcal{A}^n \right. \\ &\quad \left. + \tau_4 \mathcal{T}^n + \tau_5 \mathcal{R}^n - \mu_5 \mathcal{C}^n \} \right] \\ \mathbf{P}(\mathcal{L}^n(t)) &= \mathcal{L}^{n+1}(t) = \kappa^{-1} \mathcal{L}(0) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \beta_2 \mathcal{I}_2^n \right. \\ &\quad \left. + \beta_3 \mathcal{A}^n - \gamma_5 \mathcal{L}^n \} \right] \end{aligned}$$

is  $\mathbf{P}$  stable in  $L^1(a, b)$  if

$$\begin{aligned} \{1 + a(\zeta)\omega_1 - b(\zeta)\alpha_1 - c(\zeta)\gamma_1\} &< 1 \\ \{1 + d(\zeta)\omega_2 - b_1(\zeta)\alpha_1 - e(\zeta)\alpha_2 - f(\zeta)\beta_2 - g(\zeta)\gamma_2\} &< 1 \\ \{1 + h(\zeta)\omega_3 + e_1(\zeta)\alpha_2 - i(\zeta)\alpha_3 - j(\zeta)\beta_3 - k(\zeta)\tau_3 - l(\zeta)\mu_3 - m(\zeta)\gamma_3\} &< 1 \\ \{1 + n(\zeta)\omega_4 + i_1(\zeta)\alpha_3 - o(\zeta)\gamma_4 - p(\zeta)\tau_4 - q(\zeta)\mu_4\} &< 1 \\ \{1 + c_1(\zeta)\gamma_1 + g_2(\zeta)\gamma_2 - m_2(\zeta)\gamma_3 + o_2(\zeta)\gamma_4 + r(\zeta)\gamma_5 - a_1(\zeta)\omega_1 - d_1(\zeta)\omega_2 \\ - h_1(\zeta)\omega_3 - n_1(\zeta)\omega_4 + s(\zeta)\tau_5\} &< 1 \\ \{1 + k_1(\zeta)\tau_3 + p_1(\zeta)\tau_4 + s_1(\zeta)\tau_5 - t(\zeta)\mu_5\} &< 1 \\ \{1 + f_1(\zeta)\beta_2 + j_1(\zeta)\beta_3 - r_1(\zeta)\gamma_5\} &< 1 \end{aligned}$$

**Proof.** Now to show that there is a fixed point in  $\mathbf{P}$  for all  $(m, n) \in \mathbb{N} \times \mathbb{N}$ , then

$$\begin{aligned} \mathbf{P}(\mathcal{I}_1^n(t)) - \mathbf{P}(\mathcal{I}_1^m(t)) &= \mathcal{I}^n(t) - \mathcal{I}^m(t) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathbf{b}_1 \right. \\ &\quad \left. + \omega_1 \mathcal{R}^n - (\alpha_1 + \gamma_1) \mathcal{I}_1^n \} \right] - \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathbf{b}_1 + \omega_1 \mathcal{R}^m \right. \\ &\quad \left. - (\alpha_1 + \gamma_1) \mathcal{I}_1^m \} \right] \\ \mathbf{P}(\mathcal{I}_2^n(t)) - \mathbf{P}(\mathcal{I}_2^m(t)) &= \mathcal{I}_2^n(t) - \mathcal{I}_2^m(t) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathbf{b}_2 \right. \\ &\quad \left. + \omega_2 \mathcal{R}^n + \alpha_1 \mathcal{I}_1^n - (\alpha_2 + \gamma_2 + \beta_2) \mathcal{I}_2^n \} \right] \end{aligned}$$

$$\begin{aligned} &- \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathbf{b}_2 + \omega_2 \mathcal{R}^m \right. \\ &\quad \left. + \alpha_1 \mathcal{I}_1^m - (\alpha_2 + \gamma_2 + \beta_2) \mathcal{I}_2^m \} \right] \\ \mathbf{P}(\mathcal{A}^n(t)) - \mathbf{P}(\mathcal{A}^m(t)) &= \mathcal{A}^n(t) - \mathcal{A}^m(t) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathbf{b}_3 \right. \\ &\quad \left. + \omega_3 \mathcal{R}^n + \alpha_2 \mathcal{I}_2^n - (\alpha_3 + \gamma_3 + \tau_3 + \beta_3 + \mu_3) \mathcal{A}^n \} \right] \\ &- \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathbf{b}_3 + \omega_3 \mathcal{R}^m + \alpha_2 \mathcal{I}_2^m \right. \\ &\quad \left. - (\alpha_3 + \gamma_3 + \tau_3 + \beta_3 + \mu_3) \mathcal{A}^m \} \right] \\ \mathbf{P}(\mathcal{T}^n(t)) - \mathbf{P}(\mathcal{T}^m(t)) &= \mathcal{T}^n(t) - \mathcal{T}^m(t) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathbf{b}_4 \right. \\ &\quad \left. + \omega_4 \mathcal{R}^n + \alpha_3 \mathcal{A}^n - (\gamma_4 + \tau_4 + \mu_4) \mathcal{T}^n \} \right] \\ &- \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathbf{b}_4 + \omega_4 \mathcal{R}^m + \alpha_3 \mathcal{A}^m \right. \\ &\quad \left. - (\gamma_4 + \tau_4 + \mu_4) \mathcal{T}^m \} \right] \\ \mathbf{P}(\mathcal{R}^n(t)) - \mathbf{P}(\mathcal{R}^m(t)) &= \mathcal{R}^n(t) - \mathcal{R}^m(t) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \gamma_1 \mathcal{I}_1^n \right. \\ &\quad \left. + \gamma_2 \mathcal{I}_2^n + \gamma_3 \mathcal{A}^n + \gamma_4 \mathcal{T}^n + \gamma_5 \mathcal{L}^n \right. \\ &\quad \left. - (\omega_1 + \omega_2 + \omega_3 + \omega_4 + \tau_5) \mathcal{R}^n \} \right] \\ &- \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \gamma_1 \mathcal{I}_1^m + \gamma_2 \mathcal{I}_2^m \right. \\ &\quad \left. + \gamma_3 \mathcal{A}^m + \gamma_4 \mathcal{T}^m + \gamma_5 \mathcal{L}^m \right. \\ &\quad \left. - (\omega_1 + \omega_2 + \omega_3 + \omega_4 + \tau_5) \mathcal{R}^m \} \right] \\ \mathbf{P}(\mathcal{C}^n(t)) - \mathbf{P}(\mathcal{C}^m(t)) &= \mathcal{C}^n(t) - \mathcal{C}^m(t) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \tau_3 \mathcal{A}^n \right. \\ &\quad \left. + \tau_4 \mathcal{T}^n + \tau_5 \mathcal{R}^n - \mu_5 \mathcal{C}^n \} \right] \\ &- \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \tau_3 \mathcal{A}^m + \tau_4 \mathcal{T}^m \right. \\ &\quad \left. + \tau_5 \mathcal{R}^m - \mu_5 \mathcal{C}^m \} \right] \\ \mathbf{P}(\mathcal{L}^n(t)) - \mathbf{P}(\mathcal{L}^m(t)) &= \mathcal{L}^n(t) - \mathcal{L}^m(t) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \beta_2 \mathcal{I}_2^n \right. \\ &\quad \left. + \beta_3 \mathcal{A}^n - \gamma_5 \mathcal{L}^n \} \right] - \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \beta_2 \mathcal{I}_2^m \right. \\ &\quad \left. + \beta_3 \mathcal{A}^m - \gamma_5 \mathcal{L}^m \} \right] \end{aligned} \quad (33)$$

from (33)

$$\begin{aligned} \|\mathbf{P}(\mathcal{I}_1^n(t)) - \mathbf{P}(\mathcal{I}_1^m(t))\| &= \|\mathcal{I}^n(t) - \mathcal{I}^m(t) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathbf{b}_1 \right. \\ &\quad \left. + \omega_1 \mathcal{R}^n - (\alpha_1 + \gamma_1) \mathcal{I}_1^n \} \right] \\ &\quad \left. - \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathbf{b}_1 + \omega_1 \mathcal{R}^m \right. \right. \\ &\quad \left. \left. - (\alpha_1 + \gamma_1) \mathcal{I}_1^m \} \right]\right\| \\ \|\mathbf{P}(\mathcal{I}_2^n(t)) - \mathbf{P}(\mathcal{I}_2^m(t))\| &= \|\mathcal{I}_2^n(t) - \mathcal{I}_2^m(t) + \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathbf{b}_2 \right. \\ &\quad \left. + \omega_2 \mathcal{R}^n + \alpha_1 \mathcal{I}_1^n - (\alpha_2 + \gamma_2 + \beta_2) \mathcal{I}_2^n \} \right] \\ &\quad \left. - \mathcal{N} \mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N} \mathcal{T} \{ \mathbf{b}_2 + \omega_2 \mathcal{R}^m \right. \right. \\ &\quad \left. \left. + \alpha_1 \mathcal{I}_1^m - (\alpha_2 + \gamma_2 + \beta_2) \mathcal{I}_2^m \} \right]\right\| \end{aligned}$$

$$\begin{aligned}
||P(\mathcal{A}^n(t)) - P(\mathcal{A}^m(t))|| &= ||\mathcal{A}^n(t) - \mathcal{A}^m(t) + \mathcal{N}\mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N}\mathcal{T}\{\mathbf{b}_3 \right. \\
&\quad \left. + \omega_3 \mathcal{R}^n + \alpha_2 \mathcal{I}_2^n - (\alpha_3 + \gamma_3 + \tau_3 + \beta_3 + \mu_3) \mathcal{A}^n\} \right] \\
&\quad \left. - \mathcal{N}\mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N}\mathcal{T}\{\mathbf{b}_3 \right. \right. \\
&\quad \left. \left. + \omega_3 \mathcal{R}^m + \alpha_2 \mathcal{I}_2^m - (\alpha_3 + \gamma_3 + \tau_3 + \beta_3 + \mu_3) \mathcal{A}^m\} \right] \right| \\
||P(\mathcal{T}^n(t)) - P(\mathcal{T}^m(t))|| &= ||\mathcal{T}^n(t) - \mathcal{T}^m(t) + \mathcal{N}\mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N}\mathcal{T}\{\mathbf{b}_4 \right. \\
&\quad \left. + \omega_4 \mathcal{R}^n + \alpha_3 \mathcal{A}^n - (\gamma_4 + \tau_4 + \mu_4) \mathcal{T}^n\} \right] \\
&\quad \left. - \mathcal{N}\mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N}\mathcal{T}\{\mathbf{b}_4 + \omega_4 \mathcal{R}^m \right. \right. \\
&\quad \left. \left. + \alpha_3 \mathcal{A}^m - (\gamma_4 + \tau_4 + \mu_4) \mathcal{T}^m\} \right] \right| \\
||P(\mathcal{R}^n(t)) - P(\mathcal{R}^m(t))|| &= ||\mathcal{R}^n(t) - \mathcal{R}^m(t) \\
&\quad + \mathcal{N}\mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N}\mathcal{T}\{\gamma_1 \mathcal{I}_1^n + \gamma_2 \mathcal{I}_2^n \right. \\
&\quad \left. + \gamma_3 \mathcal{A}^n + \gamma_4 \mathcal{T}^n + \gamma_5 \mathcal{L}^n \right. \\
&\quad \left. - (\omega_1 + \omega_2 + \omega_3 + \omega_4 + \tau_5) \mathcal{R}^n\} \right] \\
&\quad \left. - \mathcal{N}\mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N}\mathcal{T}\{\gamma_1 \mathcal{I}_1^m \right. \right. \\
&\quad \left. \left. + \gamma_2 \mathcal{I}_2^m + \gamma_3 \mathcal{A}^m + \gamma_4 \mathcal{T}^m + \gamma_5 \mathcal{L}^m \right. \right. \\
&\quad \left. \left. - (\omega_1 + \omega_2 + \omega_3 + \omega_4 + \tau_5) \mathcal{R}^m\} \right] \right| \\
||P(\mathcal{C}^n(t)) - P(\mathcal{C}^m(t))|| &= ||\mathcal{C}^n(t) - \mathcal{C}^m(t) \\
&\quad + \mathcal{N}\mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N}\mathcal{T}\{\tau_3 \mathcal{A}^n + \tau_4 \mathcal{T}^n \right. \\
&\quad \left. + \tau_5 \mathcal{R}^n - \mu_5 \mathcal{C}^n\} \right] \\
&\quad \left. - \mathcal{N}\mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N}\mathcal{T}\{\tau_3 \mathcal{A}^m + \tau_4 \mathcal{T}^m \right. \right. \\
&\quad \left. \left. + \tau_5 \mathcal{R}^m - \mu_5 \mathcal{C}^m\} \right] \right| \\
||P(\mathcal{L}^n(t)) - P(\mathcal{L}^m(t))|| &= ||\mathcal{L}^n(t) - \mathcal{L}^m(t) \\
&\quad + \mathcal{N}\mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N}\mathcal{T}\{\beta_2 \mathcal{I}_2^n + \beta_3 \mathcal{A}^n \right. \\
&\quad \left. - \gamma_5 \mathcal{L}^n\} \right] - \mathcal{N}\mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N}\mathcal{T}\{\beta_2 \mathcal{I}_2^m \right. \\
&\quad \left. + \beta_3 \mathcal{A}^m - \gamma_5 \mathcal{L}^m\} \right] \quad (34)
\end{aligned}$$

lets see for  $\mathcal{I}_1$  in (34) first

$$\begin{aligned}
||P(\mathcal{I}_1^n(t)) - P(\mathcal{I}_1^m(t))|| &= ||\mathcal{I}_1^n(t) - \mathcal{I}_1^m(t)|| + ||\mathcal{N}\mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N}\mathcal{T}\{\mathbf{b}_1 \right. \\
&\quad \left. + \omega_1 \mathcal{R}^n - (\alpha_1 + \gamma_1) \mathcal{I}_1^n\} \right] \\
&\quad \left. - \mathcal{N}\mathcal{T}^{-1} \left[ \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N}\mathcal{T}\{\mathbf{b}_1 \right. \right. \\
&\quad \left. \left. + \omega_1 \mathcal{R}^m - (\alpha_1 + \gamma_1) \mathcal{I}_1^m\} \right] \right| \\
&= ||\mathcal{I}_1^n(t) - \mathcal{I}_1^m(t)|| + \mathcal{N}\mathcal{T}^{-1} \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N}\mathcal{T}\{||\omega_1(\mathcal{R}^n \\
&\quad - \mathcal{R}^m)|| + ||-\alpha_1(\mathcal{I}_1^n - \mathcal{I}_1^m)|| \\
&\quad + ||-\gamma_1(\mathcal{I}_1^n - \mathcal{I}_1^m)||\} \quad (35)
\end{aligned}$$

Now let  $||\mathcal{I}_1^n(t) - \mathcal{I}_1^m(t)|| \cong ||\mathcal{I}_2^n(t) - \mathcal{I}_2^m(t)|| \cong ||\mathcal{A}^n(t) - \mathcal{A}^m(t)|| \cong ||\mathcal{T}^n(t) - \mathcal{T}^m(t)|| \cong ||\mathcal{R}^n(t) - \mathcal{R}^m(t)|| \cong ||\mathcal{C}^n(t) - \mathcal{C}^m(t)|| \cong ||\mathcal{L}^n(t) -$

$\mathcal{L}^m(t)||$  substituting these in (35) we get

$$\begin{aligned}
||P(\mathcal{I}_1^n(t)) - P(\mathcal{I}_1^m(t))|| &\leq ||\mathcal{I}_1^n(t) - \mathcal{I}_1^m(t)|| \\
&\quad + \mathcal{N}\mathcal{T}^{-1} \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N}\mathcal{T}\{||\omega_1(\mathcal{I}_1^n - \mathcal{I}_1^m)|| \\
&\quad + ||-\alpha_1(\mathcal{I}_1^n - \mathcal{I}_1^m)|| + ||-\gamma_1(\mathcal{I}_1^n - \mathcal{I}_1^m)||\} \\
&= ||\mathcal{I}_1^n(t) - \mathcal{I}_1^m(t)|| + \{\mathcal{N}\mathcal{T}^{-1} \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N}\mathcal{T}\{||\omega_1(\mathcal{I}_1^n \\
&\quad - \mathcal{I}_1^m)|| + \mathcal{N}\mathcal{T}^{-1} \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N}\mathcal{T}\{||-\alpha_1(\mathcal{I}_1^n - \mathcal{I}_1^m)|| \\
&\quad + \mathcal{N}\mathcal{T}^{-1} \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N}\mathcal{T}\{||-\gamma_1(\mathcal{I}_1^n - \mathcal{I}_1^m)||\} \\
&= ||\mathcal{I}_1^n(t) - \mathcal{I}_1^m(t)|| [1 + \{\mathcal{N}\mathcal{T}^{-1} \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N}\mathcal{T}\{||\omega_1|| \\
&\quad + \mathcal{N}\mathcal{T}^{-1} \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N}\mathcal{T}\{||-\alpha_1|| \\
&\quad + \mathcal{N}\mathcal{T}^{-1} \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)} \mathcal{N}\mathcal{T}\{||-\gamma_1||\}] \quad (36)
\end{aligned}$$

Then (36) becomes

$$||P(\mathcal{I}_1^n(t)) - P(\mathcal{I}_1^m(t))|| = \{1 + a(\zeta)\omega_1 - b(\zeta)\alpha_1 - c(\zeta)\gamma_1\} ||\mathcal{I}_1^n - \mathcal{I}_1^m|| \quad (37)$$

where  $a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t$  are functions from  $\mathcal{N}\mathcal{T}^{-1}(\mathcal{N}\mathcal{T} \frac{(1-\alpha) + \alpha(\frac{u}{\kappa})}{\mathcal{G}(\alpha)})$  similarly we have

$$\begin{aligned}
||P(\mathcal{I}_2^n(t)) - P(\mathcal{I}_2^m(t))|| &= \{1 + d(\zeta)\omega_2 - b_1(\zeta)\alpha_1 - e(\zeta)\alpha_2 - f(\zeta)\beta_2 \\
&\quad - g(\zeta)\gamma_2\} ||\mathcal{I}_2^n - \mathcal{I}_2^m|| \\
||P(\mathcal{A}^n(t)) - P(\mathcal{A}^m(t))|| &= \{1 + h(\zeta)\omega_3 + e_1(\zeta)\alpha_2 - i(\zeta)\alpha_3 - j(\zeta)\beta_3 - k(\zeta)\tau_3 \\
&\quad - l(\zeta)\mu_3 - m(\zeta)\gamma_3\} ||\mathcal{A}^n - \mathcal{A}^m|| \\
||P(\mathcal{T}^n(t)) - P(\mathcal{T}^m(t))|| &= \{1 + n(\zeta)\omega_4 + i_1(\zeta)\alpha_3 - o(\zeta)\gamma_4 - p(\zeta)\tau_4 \\
&\quad - q(\zeta)\mu_4\} ||\mathcal{T}^n - \mathcal{T}^m|| \\
||P(\mathcal{R}^n(t)) - P(\mathcal{R}^m(t))|| &= \{1 + c_1(\zeta)\gamma_1 + g_2(\zeta)\gamma_2 - m_2(\zeta)\gamma_3 + o_2(\zeta)\gamma_4 + r(\zeta)\gamma_5 \\
&\quad - a_1(\zeta)\omega_1 - d_1(\zeta)\omega_2 - h_1(\zeta)\omega_3 - n_1(\zeta)\omega_4 \\
&\quad + s(\zeta)\tau_5\} ||\mathcal{R}^n - \mathcal{R}^m|| \\
||P(\mathcal{C}^n(t)) - P(\mathcal{C}^m(t))|| &= \{1 + k_1(\zeta)\tau_3 + p_1(\zeta)\tau_4 + s_1(\zeta)\tau_5 \\
&\quad - t(\zeta)\mu_5\} ||\mathcal{C}^n - \mathcal{C}^m|| \\
||P(\mathcal{L}^n(t)) - P(\mathcal{L}^m(t))|| &= \{1 + f_1(\zeta)\beta_2 + j_1(\zeta)\beta_3 - r_1(\zeta)\gamma_5\} ||\mathcal{L}^n - \mathcal{L}^m|| \quad (38)
\end{aligned}$$

where

$$\begin{aligned}
&\{1 + a(\zeta)\omega_1 - b(\zeta)\alpha_1 - c(\zeta)\gamma_1\} < 1 \\
&\{1 + d(\zeta)\omega_2 - b_1(\zeta)\alpha_1 - e(\zeta)\alpha_2 - f(\zeta)\beta_2 - g(\zeta)\gamma_2\} < 1 \\
&\{1 + h(\zeta)\omega_3 + e_1(\zeta)\alpha_2 - i(\zeta)\alpha_3 - j(\zeta)\beta_3 - k(\zeta)\tau_3 - l(\zeta)\mu_3 - m(\zeta)\gamma_3\} < 1 \\
&\{1 + n(\zeta)\omega_4 + i_1(\zeta)\alpha_3 - o(\zeta)\gamma_4 - p(\zeta)\tau_4 - q(\zeta)\mu_4\} < 1 \\
&\{1 + c_1(\zeta)\gamma_1 + g_2(\zeta)\gamma_2 - m_2(\zeta)\gamma_3 + o_2(\zeta)\gamma_4 + r(\zeta)\gamma_5 - a_1(\zeta)\omega_1 - d_1(\zeta)\omega_2 \\
&\quad - h_1(\zeta)\omega_3 - n_1(\zeta)\omega_4 + s(\zeta)\tau_5\} < 1 \\
&\{1 + k_1(\zeta)\tau_3 + p_1(\zeta)\tau_4 + s_1(\zeta)\tau_5 - t(\zeta)\mu_5\} < 1 \\
&\{1 + f_1(\zeta)\beta_2 + j_1(\zeta)\beta_3 - r_1(\zeta)\gamma_5\} < 1
\end{aligned}$$

Hence the theorem holds and P is stable.  $\square$

**Theorem 3.** The special solution of equation(21) using the iteration method is unique.

**Proof.** Let us define a Hilbert space  $H = L((a, b) \times (0, T))$  as

$$y : (a, b) \times (0, T) \longrightarrow \mathbb{R}, \int \int u y d u d y < \infty$$

Considering

$$P(I_1, I_2, \mathcal{A}, \mathcal{T}, \mathcal{R}, C, \mathcal{L}) = \begin{cases} b_1 + \omega_1 \mathcal{R} - (\alpha_1 + \gamma_1) I_1 \\ b_2 + \omega_2 \mathcal{R} + \alpha_1 I_1 - (\alpha_2 + \gamma_2 + \beta_2) I_2 \\ b_3 + \omega_3 \mathcal{R} + \alpha_2 I_2 - (\alpha_3 + \gamma_3 + \tau_3 + \beta_3 + \mu_3) \mathcal{A} \\ b_4 + \omega_4 \mathcal{R} + \alpha_3 \mathcal{A} - (\gamma_4 + \tau_4 + \mu_4) \mathcal{T} \\ \gamma_1 I_1 + \gamma_2 I_2 + \gamma_3 \mathcal{A} + \gamma_4 \mathcal{T} + \gamma_5 \mathcal{L} - (\omega_1 + \omega_2 \\ + \omega_3 + \omega_4 + \tau_5) \mathcal{R} \\ \tau_3 \mathcal{A} + \tau_4 \mathcal{T} + \tau_5 \mathcal{R} - \mu_5 C \\ \beta_2 I_2 + \beta_3 \mathcal{A} - \gamma_5 \mathcal{L} \end{cases}$$

$$P((X_{11} - X_{12}, X_{21} - X_{22}, X_{31} - X_{32}, X_{41} - X_{42}, X_{51} - X_{52}, X_{61} - X_{62}, X_{71} - X_{72}), (O_1, O_2, O_3, O_4, O_5, O_6, O_7)) \quad (39)$$

then

$$\begin{aligned} &\{\omega_1(X_{51} - X_{52}) - \alpha_1(X_{11} - X_{12}) - \gamma_1(X_{11} - X_{12}), O_1\} \\ &\{\omega_2(X_{51} - X_{52}) + \alpha_1(X_{11} - X_{12}) - \alpha_2(X_{21} - X_{22}) - \gamma_2(X_{21} - X_{22}) \\ &\quad - \beta_2(X_{21} - X_{22}), O_2\} \\ &\{\omega_3(X_{51} - X_{52}) + \alpha_2(X_{21} - X_{22}) - \alpha_3(X_{31} - X_{32}) - \gamma_3(X_{31} - X_{32}) \\ &\quad - \tau_3(X_{31} - X_{32}) - \beta_3(X_{31} - X_{32}) - \mu_3(X_{31} - X_{32}), O_3\} \\ &\{\omega_4(X_{51} - X_{52}) + \alpha_3(X_{31} - X_{32}) - \gamma_4(X_{41} - X_{42}) - \tau_4(X_{41} - X_{42}) \\ &\quad - \mu_4(X_{41} - X_{42}), O_4\} \\ &\{\gamma_1(X_{11} - X_{12}) + \gamma_2(X_{21} - X_{22}) + \gamma_3(X_{31} - X_{32}) + \gamma_4(X_{41} - X_{42}) \\ &\quad + \gamma_5(X_{71} - X_{72}) - \omega_1(X_{51} - X_{52}) - \omega_2(X_{51} - X_{52}) - \omega_3(X_{51} - X_{52}) \\ &\quad - \omega_4(X_{51} - X_{52}) - \tau_5(X_{51} - X_{52}), O_5\} \\ &\{\tau_3(X_{31} - X_{32}) + \tau_4(X_{41} - X_{42}) + \tau_5(X_{51} - X_{52}) - \mu_5(X_{61} - X_{62}), O_6\} \\ &\{\beta_2(X_{21} - X_{22}) + \beta_3(X_{31} - X_{32}) - \gamma_5(X_{71} - X_{72}), O_7\} \end{aligned}$$

Now consider

$$\{\omega_1(X_{51} - X_{52}) - \alpha_1(X_{11} - X_{12}) - \gamma_1(X_{11} - X_{12}), O_1\} \cong \{\omega_1(X_{51} - X_{52}), O_1\} + \{-\alpha_1(X_{11} - X_{12}), O_1\} + \{-\gamma_1(X_{11} - X_{12}), O_1\} \quad (40)$$

Let

$$(X_{11} - X_{12}) \cong (X_{21} - X_{22}) \cong (X_{31} - X_{32}) \cong (X_{41} - X_{42}) \cong (X_{51} - X_{52}) \\ \cong (X_{61} - X_{62}) \cong (X_{71} - X_{72})$$

Then (40) becomes

$$\begin{aligned} &\cong \{ \{\omega_1(X_{11} - X_{12}), O_1\} + \{-\alpha_1(X_{11} - X_{12}), O_1\} + \{-\gamma_1(X_{11} - X_{12}), O_1\} \} \\ &\leq \omega_1 \|X_{11} - X_{12}\| \|O_1\| + \alpha_1 \|X_{11} - X_{12}\| \|O_1\| + \gamma_1 \|X_{11} - X_{12}\| \|O_1\| \\ &= (\omega_1 + \alpha_1 + \gamma_1) \|X_{11} - X_{12}\| \|O_1\| \end{aligned} \quad (41)$$

Similarly we get

$$\{\omega_2(X_{51} - X_{52}) + \alpha_1(X_{11} - X_{12}) - \alpha_2(X_{21} - X_{22}) - \gamma_2(X_{21} - X_{22}) - \beta_2(X_{21} - X_{22}), O_2\} \leq (\omega_2 + \alpha_1 + \alpha_2 + \gamma_2 + \beta_2) \|X_{21} - X_{22}\| \|O_2\| \quad (42)$$

$$\begin{aligned} &\{\omega_3(X_{51} - X_{52}) + \alpha_2(X_{21} - X_{22}) - \alpha_3(X_{31} - X_{32}) - \gamma_3(X_{31} - X_{32}) \\ &\quad - \beta_3(X_{31} - X_{32}) - \mu_3(X_{31} - X_{32}), O_3\} \leq (\omega_3 + \alpha_2 + \alpha_3 + \gamma_3 + \beta_3 \\ &\quad + \mu_3) \|X_{31} - X_{32}\| \|O_3\| \end{aligned} \quad (43)$$

$$\begin{aligned} &\{\omega_4(X_{51} - X_{52}) + \alpha_3(X_{31} - X_{32}) - \gamma_4(X_{41} - X_{42}) - \tau_4(X_{41} - X_{42}) \\ &\quad - \mu_4(X_{41} - X_{42}), O_4\} \leq (\omega_4 + \alpha_3 + \gamma_4 + \tau_4 + \mu_4) \|X_{41} - X_{42}\| \|O_4\| \end{aligned} \quad (44)$$

$$\begin{aligned} &\{\gamma_1(X_{11} - X_{12}) + \gamma_2(X_{21} - X_{22}) + \gamma_3(X_{31} - X_{32}) + \gamma_4(X_{41} - X_{42}) \\ &\quad + \gamma_5(X_{71} - X_{72}) - \omega_1(X_{51} - X_{52}) - \omega_2(X_{51} - X_{52}) - \omega_3(X_{51} - X_{52}) \\ &\quad - \omega_4(X_{51} - X_{52}) - \tau_5(X_{51} - X_{52}), O_5\} \leq (\gamma_1 + \gamma_2 + \gamma_3 + \gamma_4 + \gamma_5 - \omega_1 - \omega_2 \\ &\quad - \omega_3 - \omega_4 - \tau_5) \|X_{51} - X_{52}\| \|O_5\| \end{aligned} \quad (45)$$

$$\{\tau_3(X_{31} - X_{32}) + \tau_4(X_{41} - X_{42}) + \tau_5(X_{51} - X_{52}) - \mu_5(X_{61} - X_{62}), O_6\}$$

$$\leq (\tau_3 + \tau_4 + \tau_5 - \mu_5) \|X_{61} - X_{62}\| \|O_6\| \quad (46)$$

$$\{\beta_2(X_{21} - X_{22}) + \beta_3(X_{31} - X_{32}) - \gamma_5(X_{71} - X_{72}), O_7\} \leq (\beta_2 + \beta_3 + \gamma_5) \|X_{71} - X_{72}\| \|O_7\| \quad (47)$$

Substituting (41), (47) in (39) we get

$$\begin{aligned} &P((X_{11} - X_{12}, X_{21} - X_{22}, X_{31} - X_{32}, X_{41} - X_{42}, X_{51} - X_{52}, X_{61} - X_{62}, X_{71} - X_{72}), (O_1, O_2, O_3, O_4, O_5, O_6, O_7)) \\ &\leq \begin{cases} (\omega_1 + \alpha_1 + \gamma_1) \|X_{11} - X_{12}\| \|O_1\|, \\ (\omega_2 + \alpha_1 + \alpha_2 + \gamma_2 + \beta_2) \|X_{21} - X_{22}\| \|O_2\| \\ (\omega_3 + \alpha_2 + \alpha_3 + \gamma_3 + \beta_3 + \mu_3) \|X_{31} - X_{32}\| \|O_3\| \\ (\omega_4 + \alpha_3 + \gamma_4 + \tau_4 + \mu_4) \|X_{41} - X_{42}\| \|O_4\| \\ (\gamma_1 + \gamma_2 + \gamma_3 + \gamma_4 + \gamma_5 - \omega_1 - \omega_2 - \omega_3 - \omega_4 - \tau_5) \|X_{51} - X_{52}\| \|O_5\| \\ (\tau_3 + \tau_4 + \tau_5 - \mu_5) \|X_{61} - X_{62}\| \|O_6\| \\ (\beta_2 + \beta_3 + \gamma_5) \|X_{71} - X_{72}\| \|O_7\| \end{cases} \end{aligned} \quad (48)$$

But, for large values of  $m_i$  with  $i = 1, 2, 3, 4, 5, 6, 7$ . Hence solution exists, small positive parameters  $l_{m_1}, l_{m_2}, l_{m_3}, l_{m_4}, l_{m_5}, l_{m_6}, l_{m_7}$ . Such that

$$\begin{aligned} &\|(I_1 - X_{11})\|, \|(I_2 - X_{12})\| < \frac{l_{m_1}}{7(\omega_1 + \alpha_1 + \gamma_1) \|X_{11} - X_{12}\| \|O_1\|} \\ &\|(I_2 - X_{21})\|, \|(I_2 - X_{22})\| < \frac{l_{m_2}}{7(\omega_2 + \alpha_1 + \alpha_2 + \gamma_2 + \beta_2) \|X_{21} - X_{22}\| \|O_2\|} \\ &\|(\mathcal{A} - X_{31})\|, \|(\mathcal{A} - X_{32})\| < \frac{l_{m_3}}{7(\omega_3 + \alpha_2 + \alpha_3 + \gamma_3 + \beta_3 + \mu_3) \|X_{31} - X_{32}\| \|O_3\|} \\ &\|(\mathcal{T} - X_{41})\|, \|(\mathcal{T} - X_{42})\| < \frac{l_{m_4}}{7(\omega_4 + \alpha_3 + \gamma_4 + \tau_4 + \mu_4) \|X_{41} - X_{42}\| \|O_4\|} \\ &\|(\mathcal{R} - X_{51})\|, \|(\mathcal{R} - X_{52})\| < \frac{l_{m_5}}{7(\gamma_1 + \gamma_2 + \gamma_3 + \gamma_4 + \gamma_5 - \omega_1 - \omega_2 - \omega_3 - \omega_4 - \tau_5) \|X_{51} - X_{52}\| \|O_5\|} \\ &\|(C - X_{61})\|, \|(C - X_{62})\| < \frac{l_{m_5}}{7(\tau_3 + \tau_4 + \tau_5 + \mu_5) \|X_{61} - X_{62}\| \|O_6\|} \\ &\|(\mathcal{L} - X_{71})\|, \|(\mathcal{L} - X_{72})\| < \frac{l_{m_6}}{7(\beta_2 + \beta_3 + \gamma_5) \|X_{71} - X_{72}\| \|O_7\|} \end{aligned}$$

Thus, plugging the exact solution in the right-hand side of the (48) and applying the triangular inequality by taking  $M = \max(m_1, m_2, m_3, m_4, m_5, m_6, m_7)$ ,  $l = \max(l_1, l_2, l_3, l_4, l_5, l_6, l_7)$  we obtain

$$\begin{cases} (\omega_1 + \alpha_1 + \gamma_1) \|X_{11} - X_{12}\| \|O_1\|, \\ (\omega_2 + \alpha_1 + \alpha_2 + \gamma_2 + \beta_2) \|X_{21} - X_{22}\| \|O_2\| \\ (\omega_3 + \alpha_2 + \alpha_3 + \gamma_3 + \beta_3 + \mu_3) \|X_{31} - X_{32}\| \|O_3\| \\ (\omega_4 + \alpha_3 + \gamma_4 + \tau_4 + \mu_4) \|X_{41} - X_{42}\| \|O_4\| \\ (\gamma_1 + \gamma_2 + \gamma_3 + \gamma_4 + \gamma_5 - \omega_1 - \omega_2 - \omega_3 - \omega_4 - \tau_5) \|X_{51} - X_{52}\| \|O_5\| \\ (\tau_3 + \tau_4 + \tau_5 + \mu_5) \|X_{61} - X_{62}\| \|O_6\| \\ (\beta_2 + \beta_3 + \gamma_5) \|X_{71} - X_{72}\| \|O_7\| \end{cases} \begin{cases} l, \\ l, \\ l, \\ l, \\ l, \\ l, \\ l. \end{cases}$$

$$\begin{cases} (\omega_1 + \alpha_1 + \gamma_1) \|X_{11} - X_{12}\| \|O_1\|, \\ (\omega_2 + \alpha_1 + \alpha_2 + \gamma_2 + \beta_2) \|X_{21} - X_{22}\| \|O_2\| \\ (\omega_3 + \alpha_2 + \alpha_3 + \gamma_3 + \beta_3 + \mu_3) \|X_{31} - X_{32}\| \|O_3\| \\ (\omega_4 + \alpha_3 + \gamma_4 + \tau_4 + \mu_4) \|X_{41} - X_{42}\| \|O_4\| \\ (\gamma_1 + \gamma_2 + \gamma_3 + \gamma_4 + \gamma_5 - \omega_1 - \omega_2 - \omega_3 - \omega_4 - \tau_5) \|X_{51} - X_{52}\| \|O_5\| \\ (\tau_3 + \tau_4 + \tau_5 + \mu_5) \|X_{61} - X_{62}\| \|O_6\| \\ (\beta_2 + \beta_3 + \gamma_5) \|X_{71} - X_{72}\| \|O_7\| \end{cases} \begin{cases} 0, \\ 0, \\ 0, \\ 0, \\ 0, \\ 0, \\ 0. \end{cases}$$

But it is obvious that

$$\begin{aligned}(\omega_1 + \alpha_1 + \gamma_1) \neq 0, (\omega_2 + \alpha_1 + \alpha_2 + \gamma_2 + \beta_2) \neq 0, (\omega_3 + \alpha_2 + \alpha_3 + \gamma_3 + \beta_3 + \mu_3) \neq 0, \\ (\omega_4 + \alpha_3 + \gamma_4 + \tau_4 + \mu_4) \neq 0, (\gamma_1 + \gamma_2 + \gamma_3 + \gamma_4 + \gamma_5 - \omega_1 - \omega_2 - \omega_3 - \omega_4 - \tau_5) \neq 0, \\ (\tau_3 + \tau_4 + \tau_5 + \mu_5) \neq 0, (\beta_2 + \beta_3 + \gamma_5) \neq 0.\end{aligned}$$

Therefore we have

$$\begin{aligned}\|X_{11} - X_{12}\| = 0, \|X_{21} - X_{22}\| = 0, \|X_{31} - X_{32}\| = 0, \|X_{41} - X_{42}\| \\ = 0, \|X_{51} - X_{52}\| = 0, \|X_{61} - X_{62}\| = 0, \\ \|X_{71} - X_{72}\| = 0.\end{aligned}$$

which implies  $X_{11} = X_{12}, X_{21} = X_{22}, X_{31} = X_{32}, X_{41} = X_{42}, X_{51} = X_{52}, X_{61} = X_{62}, X_{71} = X_{72}$ .

Hence the theorem.  $\square$

#### Existence and uniqueness via fixed point theory for $\mathfrak{E}\mathfrak{F}$ model

Through using concept of the fixed point theorem, this section provides evidence that the proposed model (21) has a solution that is both unique and exists. Model (21) can be expressed in the following manner:

$$\begin{cases} \mathfrak{D}_t^\alpha I_1 = \mathcal{K}_1(t, I_1), \\ \mathfrak{D}_t^\alpha I_2 = \mathcal{K}_2(t, I_2), \\ \mathfrak{D}_t^\alpha \mathcal{A} = \mathcal{K}_3(t, \mathcal{A}), \\ \mathfrak{D}_t^\alpha \mathcal{T} = \mathcal{K}_4(t, \mathcal{T}), \\ \mathfrak{D}_t^\alpha \mathcal{R} = \mathcal{K}_5(t, \mathcal{R}), \\ \mathfrak{D}_t^\alpha C = \mathcal{K}_6(t, C), \\ \mathfrak{D}_t^\alpha \mathcal{L} = \mathcal{K}_7(t, \mathcal{L}). \end{cases} \quad (49)$$

where

$$\begin{cases} \mathcal{K}_1(t, I_1) = \mathfrak{b}_1 + \omega_1 \mathcal{R} - (\alpha_1 + \gamma_1) I_1, \\ \mathcal{K}_2(t, I_2) = \mathfrak{b}_2 + \omega_2 \mathcal{R} + \alpha_1 I_1 - (\alpha_2 + \gamma_2 + \beta_2) I_2, \\ \mathcal{K}_3(t, \mathcal{A}) = \mathfrak{b}_3 + \omega_3 \mathcal{R} + \alpha_2 I_2 - (\alpha_3 + \gamma_3 + \tau_3 + \beta_3 + \mu_3) \mathcal{A}, \\ \mathcal{K}_4(t, \mathcal{T}) = \mathfrak{b}_4 + \omega_4 \mathcal{R} + \alpha_3 \mathcal{A} - (\gamma_4 + \tau_4 + \mu_4) \mathcal{T}, \\ \mathcal{K}_5(t, \mathcal{R}) = \gamma_1 I_1 + \gamma_2 I_2 + \gamma_3 \mathcal{A} + \gamma_4 \mathcal{T} + \gamma_5 \mathcal{L} - (\omega_1 + \omega_2 + \omega_3 + \omega_4 + \tau_5) \mathcal{R}, \\ \mathcal{K}_6(t, C) = \tau_3 \mathcal{A} + \tau_4 \mathcal{T} + \tau_5 \mathcal{R} - \mu_5 C, \\ \mathcal{K}_7(t, \mathcal{L}) = \beta_2 I_2 + \beta_3 \mathcal{A} - \gamma_5 \mathcal{L}. \end{cases} \quad (50)$$

With the application of the fractional integral operator, which is presented in Eq. (24), the system (49) is reduced of order  $0 < \alpha < 1$ , which is represented using the equation.

$$\begin{cases} I_1(t) = I_1(0) + \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \mathcal{K}_1(t, I_1) + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \mathcal{K}_1(\xi, I_1) d\xi, \\ I_2(t) = I_2(0) + \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \mathcal{K}_2(t, I_2) + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \mathcal{K}_2(\xi, I_2) d\xi, \\ \mathcal{A}(t) = \mathcal{A}(0) + \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \mathcal{K}_3(t, \mathcal{A}) + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \mathcal{K}_3(\xi, \mathcal{A}) d\xi, \\ \mathcal{T}(t) = \mathcal{T}(0) + \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \mathcal{K}_4(t, \mathcal{T}) + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \mathcal{K}_4(\xi, \mathcal{T}) d\xi, \\ \mathcal{R}(t) = \mathcal{R}(0) + \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \mathcal{K}_5(t, \mathcal{R}) + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \mathcal{K}_5(\xi, \mathcal{R}) d\xi, \\ C(t) = C(0) + \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \mathcal{K}_6(t, C) + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \mathcal{K}_6(\xi, C) d\xi, \\ \mathcal{L}(t) = \mathcal{L}(0) + \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \mathcal{K}_7(t, \mathcal{L}) + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \mathcal{K}_7(\xi, \mathcal{L}) d\xi. \end{cases} \quad (51)$$

( $\mathcal{G}$ ): We take into consideration the following assumption in order to demonstrate our findings: For this,  $I_1(t), I_1^*(t), I_2(t), I_2^*(t), \mathcal{A}(t), \mathcal{A}^*(t), \mathcal{T}(t), \mathcal{T}^*(t), \mathcal{R}(t), \mathcal{R}^*(t), C(t), C^*(t), \mathcal{L}(t), \mathcal{L}^*(t) \in \mathcal{L}[0, 1]$  be continuous such that  $\|I_1(t)\| \leq c_1, \|I_2(t)\| \leq c_2, \|\mathcal{A}(t)\| \leq c_3, \|\mathcal{T}(t)\| \leq c_4, \|\mathcal{R}(t)\| \leq c_5, \|C(t)\| \leq c_6$ , and  $\|\mathcal{L}(t)\| \leq c_7$ , for some positive constants  $c_1, c_2, c_3, c_4, c_5, c_6, c_7 > 0$ .

**Theorem 4.** Each kernel  $(\mathcal{K}_1, \mathcal{K}_2, \mathcal{K}_3, \mathcal{K}_4, \mathcal{K}_5, \mathcal{K}_6, \mathcal{K}_7)$  is satisfying the Lipschitz-condition under the following assumption ( $\mathcal{G}$ ) and

$\mathfrak{j}_i < 1$ , for  $i = 1, 2, 3, 4, 5, 6, 7$ .

**Proof.** For  $I_1$  and  $I_1^*$ , we have from (64), gives

$$\begin{aligned}\|\mathcal{K}_1(t, I_1) - \mathcal{K}_1(t, I_1^*)\| &= \|\mathfrak{b}_1 + \omega_1 \mathcal{R} - (\alpha_1 + \gamma_1) I_1 - (\mathfrak{b}_1 + \omega_1 \mathcal{R} \\ &\quad - (\alpha_1 + \gamma_1) I_1^*)\| \\ &\leq (\alpha_1 + \gamma_1) \|I_1(t) - I_1^*(t)\| \\ &= \mathfrak{j}_1 \|I_1(t) - I_1^*(t)\|, \end{aligned} \quad (52)$$

where  $\mathfrak{j}_1 = (\alpha_1 + \gamma_1)$ .

Similarly, we have

$$\|\mathcal{K}_2(t, I_2) - \mathcal{K}_2(t, I_2^*)\| = \mathfrak{j}_2 \|I_2(t) - I_2^*(t)\|, \quad (53)$$

$$\|\mathcal{K}_3(t, \mathcal{A}) - \mathcal{K}_3(t, \mathcal{A}^*)\| = \mathfrak{j}_3 \|\mathcal{A}(t) - \mathcal{A}^*(t)\|, \quad (54)$$

$$\|\mathcal{K}_4(t, \mathcal{T}) - \mathcal{K}_4(t, \mathcal{T}^*)\| = \mathfrak{j}_4 \|\mathcal{T}(t) - \mathcal{T}^*(t)\|, \quad (55)$$

$$\|\mathcal{K}_5(t, \mathcal{R}) - \mathcal{K}_5(t, \mathcal{R}^*)\| = \mathfrak{j}_5 \|\mathcal{R}(t) - \mathcal{R}^*(t)\|, \quad (56)$$

$$\|\mathcal{K}_6(t, C) - \mathcal{K}_6(t, C^*)\| = \mathfrak{j}_6 \|C(t) - C^*(t)\|, \quad (57)$$

$$\|\mathcal{K}_7(t, \mathcal{L}) - \mathcal{K}_7(t, \mathcal{L}^*)\| = \mathfrak{j}_7 \|\mathcal{L}(t) - \mathcal{L}^*(t)\|. \quad (58)$$

where

$$\mathfrak{j}_2 = (\alpha_2 + \gamma_2 + \beta_2),$$

$$\mathfrak{j}_3 = (\alpha_3 + \gamma_3 + \tau_3 + \beta_3 + \mu_3),$$

$$\mathfrak{j}_4 = (\gamma_4 + \tau_4 + \mu_4),$$

$$\mathfrak{j}_5 = (\omega_1 + \omega_2 + \omega_3 + \omega_4 + \tau_5),$$

$$\mathfrak{j}_6 = \mu_5,$$

$$\mathfrak{j}_7 = \gamma_5.$$

Therefore, it can be demonstrated that the Lipschitz property is satisfied by the function  $\mathcal{K}_i$  for  $i = 1, 2, 3, 4, 5, 6, 7$  because of Eqs. (52)–(58).  $\square$

**Theorem 5.** The system of Eq. (21) has at least one solution if

$$\delta = \max\{\mathfrak{j}_1, \mathfrak{j}_2, \mathfrak{j}_3, \mathfrak{j}_4, \mathfrak{j}_5, \mathfrak{j}_6, \mathfrak{j}_7\} < 1.$$

**Proof.** Let

$$\lambda_{1t}(t) = I_{1t+1}(t) - I_1(t),$$

$$\lambda_{2t}(t) = I_{2t+1}(t) - I_2(t),$$

$$\lambda_{3t}(t) = \mathcal{A}_{t+1}(t) - \mathcal{A}(t),$$

$$\lambda_{4t}(t) = \mathcal{T}_{t+1}(t) - \mathcal{T}(t),$$

$$\lambda_{5t}(t) = \mathcal{R}_{t+1}(t) - \mathcal{R}(t),$$

$$\lambda_{6t}(t) = C_{t+1}(t) - C(t),$$

$$\lambda_{7t}(t) = \mathcal{L}_{t+1}(t) - \mathcal{L}(t).$$

$$\begin{aligned}\|\lambda_{1t}(t)\| &= \|I_{1t+1}(t) - I_1(t)\| = \left\| \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} (\mathcal{K}_1(t, I_{1t}) - \mathcal{K}_1(t, I_1)) \right. \\ &\quad \left. + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t (\mathcal{K}_1(\xi, I_{1t}) - \mathcal{K}_1(\xi, I_1)) d\xi \right\| \\ &\leq \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \|\mathcal{K}_1(t, I_{1t}) - \mathcal{K}_1(t, I_1)\| \\ &\quad + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \|\mathcal{K}_1(\xi, I_{1t}) - \mathcal{K}_1(\xi, I_1)\| d\xi \\ &\leq \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \mathfrak{j}_1 \|I_{1t} - I_1\| \\ &\quad + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \mathfrak{j}_1 \|I_{1t} - I_1\| d\xi \\ &= \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \mathfrak{j}_1 \|I_{1t} - I_1\| \\ &\quad + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \mathfrak{j}_1 \int_0^t \|I_{1t} - I_1\| d\xi \\ &\leq \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{2\alpha t}{(2-\alpha)\mathcal{G}(\alpha)} \right) \mathfrak{j}_1 \|I_{1t} - I_1\| \\ &\leq \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{2\alpha t}{(2-\alpha)\mathcal{G}(\alpha)} \right)^n \mathfrak{j}_1^n \|I_{11} - I_1\|. \end{aligned}$$

As  $n \rightarrow \infty$ , We have  $I_{1n} \rightarrow I_1$  since  $j_1 < 1$ . Likewise, we obtain the following outcomes.

$$\begin{aligned} \|\lambda_{2f}(t)\| &\leq \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{2\alpha t}{(2-\alpha)\mathcal{G}(\alpha)} \right)^n i_2^n \|I_{21} - I_2\| \\ \|\lambda_{3f}(t)\| &\leq \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{2\alpha t}{(2-\alpha)\mathcal{G}(\alpha)} \right)^n i_3^n \|\mathcal{A}_1 - \mathcal{A}\| \\ \|\lambda_{4f}(t)\| &\leq \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{2\alpha t}{(2-\alpha)\mathcal{G}(\alpha)} \right)^n i_4^n \|\mathcal{T}_1 - \mathcal{T}\| \\ \|\lambda_{5f}(t)\| &\leq \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{2\alpha t}{(2-\alpha)\mathcal{G}(\alpha)} \right)^n i_5^n \|\mathcal{R}_1 - \mathcal{R}\| \\ \|\lambda_{6f}(t)\| &\leq \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{2\alpha t}{(2-\alpha)\mathcal{G}(\alpha)} \right)^n i_6^n \|\mathcal{C}_1 - \mathcal{C}\| \\ \|\lambda_{7f}(t)\| &\leq \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{2\alpha t}{(2-\alpha)\mathcal{G}(\alpha)} \right)^n i_7^n \|\mathcal{L}_1 - \mathcal{L}\|. \end{aligned}$$

$\therefore n$  tends to  $\infty$ , We understand that  $\lambda_{if}(t) \rightarrow 0$  when the  $j_i < 1$  for  $i = 1, 2, 3, 4, 5, 6, 7$ . So, there is a solution to the system of Eq. (21).  $\square$

**Theorem 6.** The system of equation(21) which has a unique positive solution if

$$\left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{2\alpha t}{(2-\alpha)\mathcal{G}(\alpha)} \right) i_i \leq 1, \quad \forall i = 1, 2, 3, 4, 5, 6, 7.$$

**Proof.** Let us make the assumption that there is another solution available.  $I_1^*(t), I_2^*(t), \mathcal{A}^*(t), \mathcal{T}^*(t), \mathcal{R}^*(t), \mathcal{C}^*(t)$  and  $\mathcal{L}^*(t)$  with initial values that are compatible with the equations

$$\begin{aligned} I_1^*(t) &= I_1(0) + \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \mathcal{K}_1(t, I_1^*) + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \mathcal{K}_1(\xi, I_1^*) d\xi, \\ I_2^*(t) &= I_2(0) + \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \mathcal{K}_2(t, I_2^*) + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \mathcal{K}_2(\xi, I_2^*) d\xi, \\ \mathcal{A}^*(t) &= \mathcal{A}(0) + \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \mathcal{K}_3(t, \mathcal{A}^*) + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \mathcal{K}_3(\xi, \mathcal{A}^*) d\xi, \\ \mathcal{T}^*(t) &= \mathcal{T}(0) + \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \mathcal{K}_4(t, \mathcal{T}^*) + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \mathcal{K}_4(\xi, \mathcal{T}^*) d\xi, \\ \mathcal{R}^*(t) &= \mathcal{R}(0) + \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \mathcal{K}_5(t, \mathcal{R}^*) + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \mathcal{K}_5(\xi, \mathcal{R}^*) d\xi, \\ \mathcal{C}^*(t) &= \mathcal{C}(0) + \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \mathcal{K}_6(t, \mathcal{C}^*) + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \mathcal{K}_6(\xi, \mathcal{C}^*) d\xi, \\ \mathcal{L}^*(t) &= \mathcal{L}(0) + \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \mathcal{K}_7(t, \mathcal{L}^*) + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \mathcal{K}_7(\xi, \mathcal{L}^*) d\xi. \end{aligned}$$

Now,

$$\begin{aligned} \|I_1 - I_1^*\| &= \left\| \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} (\mathcal{K}_1(t, I_1) - \mathcal{K}_1(t, I_1^*)) \right. \\ &\quad \left. + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t (\mathcal{K}_1(\xi, I_1) - \mathcal{K}_1(\xi, I_1^*)) d\xi \right\| \\ &\leq \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \|\mathcal{K}_1(t, I_1) - \mathcal{K}_1(t, I_1^*)\| \\ &\quad + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \|\mathcal{K}_1(\xi, I_1) - \mathcal{K}_1(\xi, I_1^*)\| d\xi \\ &\leq \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} i_1 \|I_1 - I_1^*\| \\ &\quad + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t i_1 \|I_1 - I_1^*\| d\xi \\ &= \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} i_1 \|I_1 - I_1^*\| \\ &\quad + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} i_1 \int_0^t \|I_1 - I_1^*\| d\xi \\ &\leq \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{2\alpha t}{(2-\alpha)\mathcal{G}(\alpha)} \right) i_1 \|I_1 - I_1^*\| \\ &\leq \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{2\alpha t}{(2-\alpha)\mathcal{G}(\alpha)} \right) i_1 \|I_1 - I_1^*\|, \end{aligned}$$

which implies that

$$\left( 1 - \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{2\alpha t}{(2-\alpha)\mathcal{G}(\alpha)} \right) i_1 \right) \|I_1 - I_1^*\| \leq 0.$$

Therefore, we conclude that  $\|I_1 - I_1^*\| = 0$ .  $\therefore I_1 = I_1^*$ . For the above manner, we can conclude that

$$\begin{aligned} I_2 &= I_2^*, \\ \mathcal{A} &= \mathcal{A}^*, \\ \mathcal{T} &= \mathcal{T}^*, \\ \mathcal{R} &= \mathcal{R}^*, \\ \mathcal{C} &= \mathcal{C}^*, \\ \mathcal{L} &= \mathcal{L}^*. \end{aligned}$$

Finally, we concluded that the system of Eq. (21) has a unique solution.  $\square$

### Equilibrium points

The objective of this section is to ascertain the infected state of the disease by conducting an analysis of the equilibrium points of the system (21). These equilibrium points are the disease-endemic equilibrium points. A number of assumptions are being utilized in order to ascertain DFE points of the system of equation, which are as follows:  ${}^{\mathcal{G}}\mathfrak{D}_t^\alpha I_1 = 0, {}^{\mathcal{G}}\mathfrak{D}_t^\alpha I_2 = 0, {}^{\mathcal{G}}\mathfrak{D}_t^\alpha \mathcal{A} = 0, {}^{\mathcal{G}}\mathfrak{D}_t^\alpha \mathcal{T} = 0, {}^{\mathcal{G}}\mathfrak{D}_t^\alpha \mathcal{R} = 0, {}^{\mathcal{G}}\mathfrak{D}_t^\alpha \mathcal{C} = 0, {}^{\mathcal{G}}\mathfrak{D}_t^\alpha \mathcal{L} = 0$

$$\begin{cases} \mathfrak{b}_1 + \omega_1 \mathcal{R} - (\alpha_1 + \gamma_1) I_1 = 0 \\ \mathfrak{b}_2 + \omega_2 \mathcal{R} + \alpha_1 I_1 - (\alpha_2 + \gamma_2 + \beta_2) I_2 = 0 \\ \mathfrak{b}_3 + \omega_3 \mathcal{R} + \alpha_2 I_2 - (\alpha_3 + \gamma_3 + \tau_3 + \beta_3 + \mu_3) \mathcal{A} = 0 \\ \mathfrak{b}_4 + \omega_4 \mathcal{R} + \alpha_3 \mathcal{A} - (\gamma_4 + \tau_4 + \mu_4) \mathcal{T} = 0 \\ \gamma_1 I_1 + \gamma_2 I_2 + \gamma_3 \mathcal{A} + \gamma_4 \mathcal{T} + \gamma_5 \mathcal{L} - (\omega_1 + \omega_2 + \omega_3 + \omega_4 + \tau_5) \mathcal{R} = 0 \\ \tau_3 \mathcal{A} + \tau_4 \mathcal{T} + \tau_5 \mathcal{R} - \mu_5 \mathcal{C} = 0 \\ \beta_2 I_2 + \beta_3 \mathcal{A} - \gamma_5 \mathcal{L} = 0 \end{cases}$$

Now, the Disease-Endemic equilibrium (DEE) is:

$$\begin{aligned} I_1^* &= \frac{\mathfrak{b}_1 + \omega_1 \mathcal{R}}{\alpha_1 + \gamma_1}, \\ I_2^* &= \frac{\mathfrak{b}_2 + \omega_2 \mathcal{R} + \alpha_1 I_1}{\alpha_2 + \gamma_2 + \beta_2}, \\ \mathcal{A}^* &= \frac{\mathfrak{b}_3 + \omega_3 \mathcal{R} + \alpha_2 I_2}{\alpha_3 + \gamma_3 + \tau_3 + \beta_3 + \mu_3}, \\ \mathcal{T}^* &= \frac{\mathfrak{b}_4 + \omega_4 \mathcal{R} + \alpha_3 \mathcal{A}}{\gamma_4 + \tau_4 + \mu_4}, \\ \mathcal{R}^* &= \frac{\gamma_1 I_1 + \gamma_2 I_2 + \gamma_3 \mathcal{A} + \gamma_4 \mathcal{T} + \gamma_5 \mathcal{L}}{\omega_1 + \omega_2 + \omega_3 + \omega_4 + \tau_5}, \\ \mathcal{C}^* &= \frac{\tau_3 \mathcal{A} + \tau_4 \mathcal{T} + \tau_5 \mathcal{R}}{\mu_5}, \\ \mathcal{L}^* &= \frac{\beta_2 I_2 + \beta_3 \mathcal{A}}{\gamma_5}. \end{aligned}$$

### Phase Portrait Analysis

Fig. 14 shows the coupled evolution of carcinoma stages ( $I_1, I_2$ ), ( $\mathcal{A}, \mathcal{T}$ ), ( $\mathcal{R}, \mathcal{C}$ ) with recovery ( $\mathcal{R}$ ) and treatment-related compartments ( $\mathcal{L}$ ) for different fractional order. Biologically, the phase trajectories show that as the disease progresses from intermediate to terminal stages, the system converges toward stable states where recovery and cardiotoxicity levels reach equilibrium, it is reflecting the long-term impact of treatment and disease history.

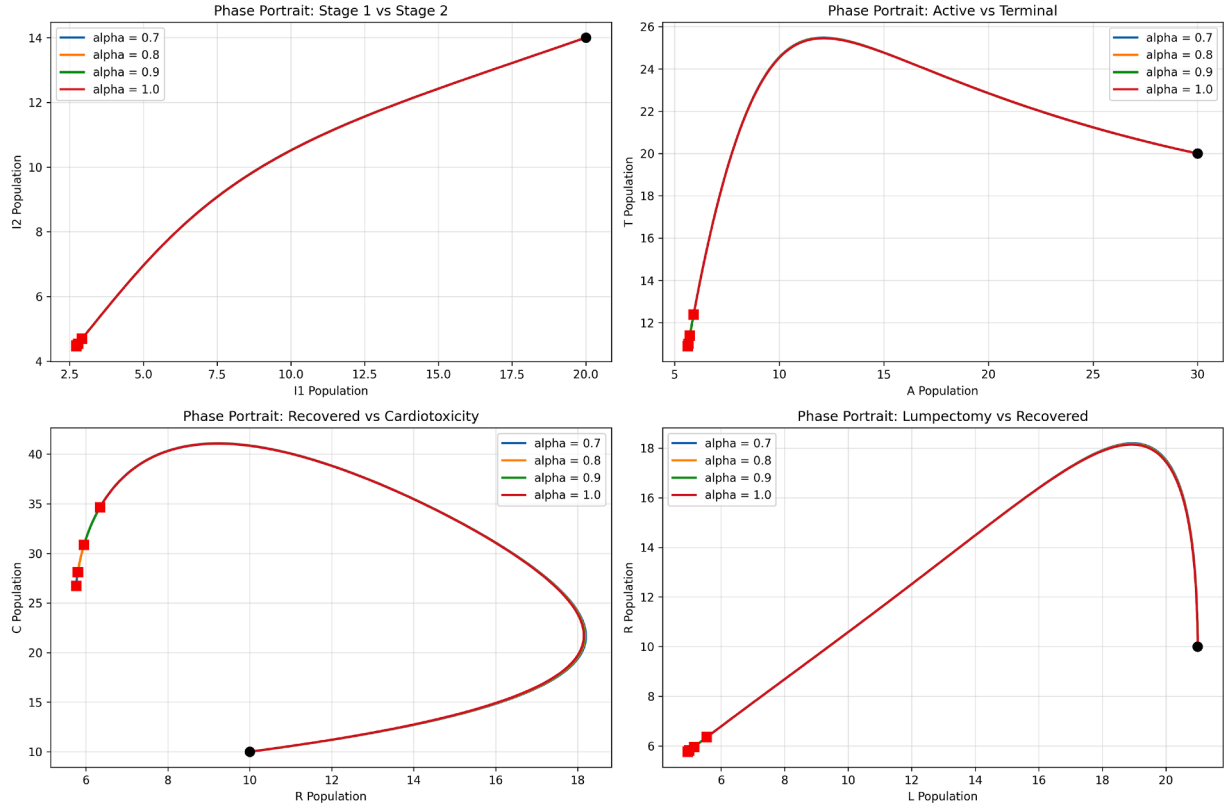


Fig. 14. Phase portraits of key breast cancer compartments under fractional-order dynamics.

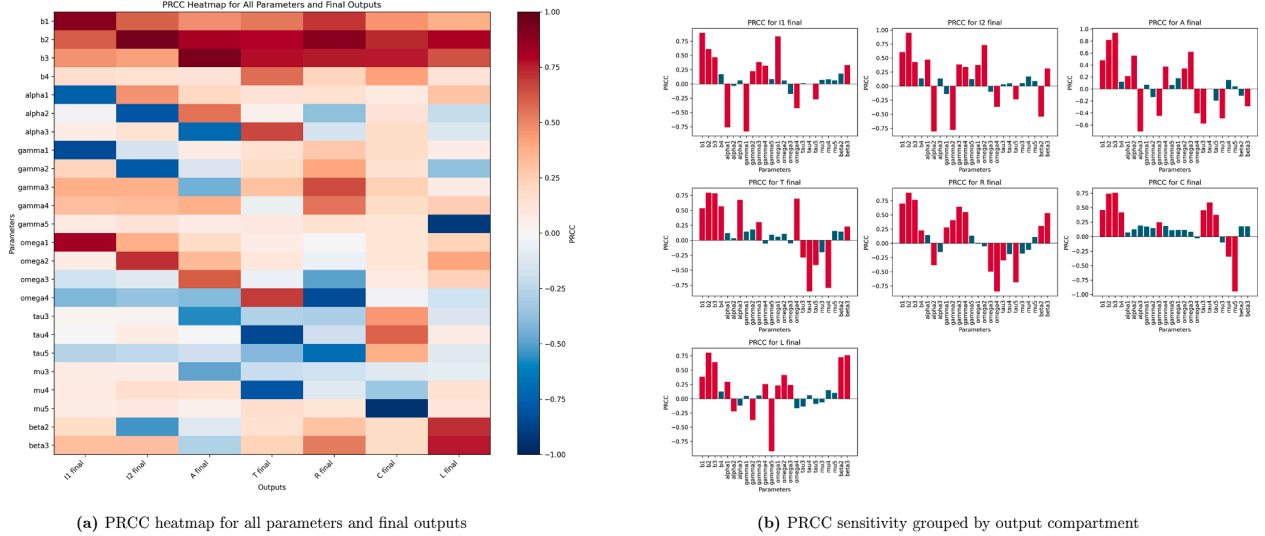


Fig. 15. Partial Rank Correlation Coefficient (PRCC) analysis quantifying the sensitivity of final compartment populations to variations in model parameters.

### Sensitivity Analysis

Fig. 15a shows that the newly diagnosed rates ( $b_1$ ,  $b_2$ ,  $b_3$ ) shows strong positive influences on early-stage compartments ( $I_1$ ,  $I_2$ ,  $A$ ) and all breast cancer compartments. Conversely, progression parameters ( $\alpha_1$ ,  $\alpha_2$ ,  $\alpha_3$ ) show negative correlations with the earlier stages but positive correlations with terminal and cardiotoxicity compartments, it is confirming the cascade effect of disease advancement. Fig. 15b further clarifies these directional relationships, it is highlighting that detection rates primarily drive initial tumor burden, while treatment-related parameters (recovery rates  $\gamma$ , cardiotoxicity  $\tau$ , and surgical interventions  $\beta_2$ ,

$\beta_3$ ) modulate the ultimate distribution among clinical outcomes. Biologically, these results confirm that early diagnosis coupled with effective early-stage treatment is crucial for reducing the overall cancer burden, whereas delayed intervention shifts the system toward more advanced, treatment-resistant states.

### Contour Plot Analysis

Fig. 16a and b show that increasing the transition rate  $\alpha_2$  (from  $I_2$  to active carcinoma) or the cardiotoxicity rates  $\tau_3$ ,  $\tau_4$  leads to a gradual decrease in the final  $I_1$  population, it is reflecting the cascade



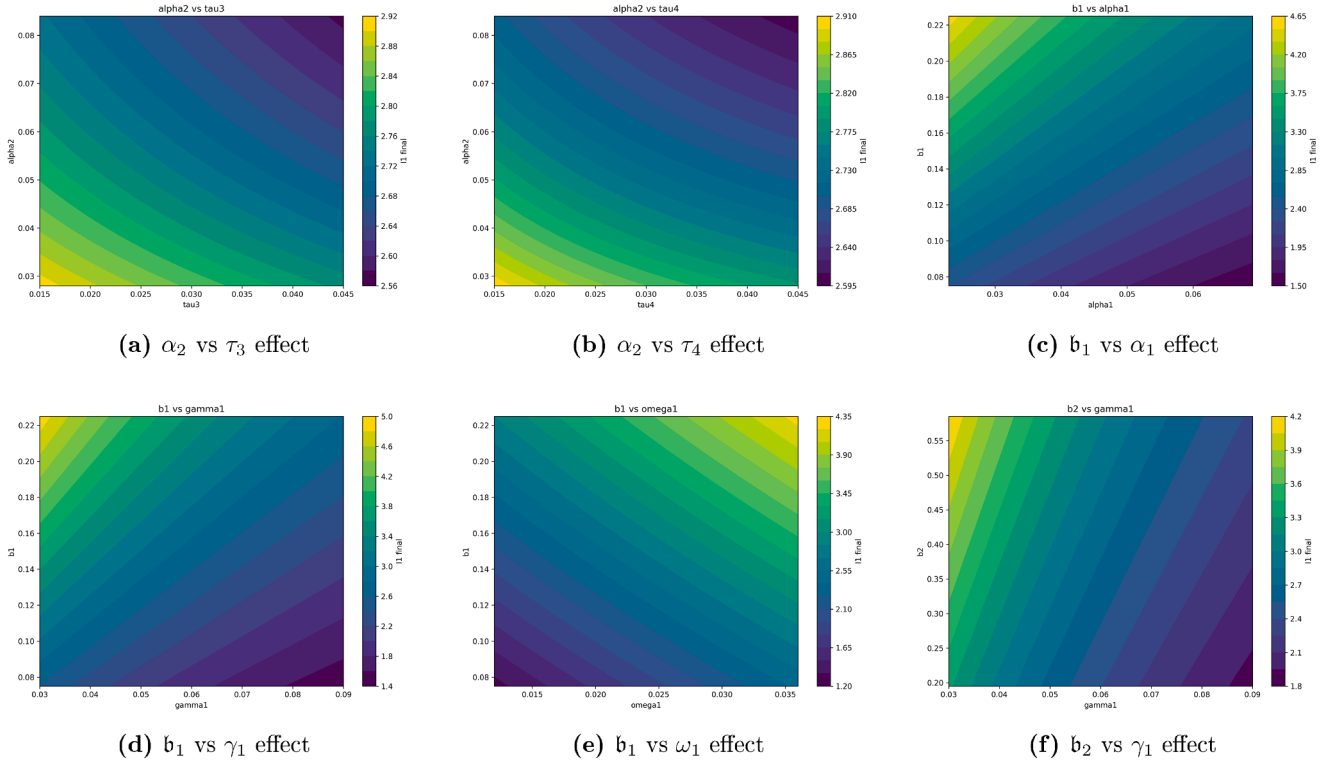


Fig. 16. Contour plots showing sensitivity of final stage 1 carcinoma population ( $I_1$ ) to variations in key model parameters.

effect of disease progression. Fig. 16c–e show that higher detection rates  $b_1$  which is combined with lower progression  $\alpha_1$  or higher recovery  $\gamma_1$ , substantially it reduce the residual stage 1 burden. Fig. 16f shows that increased stage 2 detection  $b_2$  and better response  $\gamma_1$  similarly lower  $I_1$  values. Biologically, these patterns underscore the importance of early detection and effective treatment in reducing the reservoir of early-stage tumors, while cardiotoxicity accelerates disease progression by delaying effective therapy.

#### Effect of parameters for different values

Fig. 17a shows that increasing the progression rate  $\alpha_1$  from stage 1 to stage 2 accelerates the decrease of the all compartment except cardiotoxicity compartment, it is reflecting faster disease advancement. Fig. 17b shows that higher early detection rates  $b_1$  reduce the burden of stage 1 carcinoma and all other compartment individuals, while simultaneously increasing the pool of recovered patients, it is highlighting the benefit of early diagnosis. Fig. 18a shows that enhanced detection of stage 2 ( $b_2$ ) drives down the active carcinoma population and boosts both recovery and lumpectomy uptake, it is underscoring the importance of intermediate-stage screening. Fig. 18b shows that improved recovery from stage 1 ( $\gamma_1$ ) leads to a sustained increase in recovered individuals and a corresponding decrease in active cancer and surgical intervention needs. Finally, Fig. 19a shows that a higher relapse rate ( $\omega_1$ ) from recovered to stage 1 elevates the early-stage cancer burden, emphasizing the need for effective post-treatment surveillance to prevent recurrence.

#### Ulam-Hyers stability

Here, we will determine the Ulam-Hyers stability of the system, which is denoted by the Eq. (21). It is necessary that.

**Definition 6.** If there are constants  $\mathcal{K}_i > 0$ , then the system (21) is consistent with the Ulam-Hyers stability condition for  $i = 1, 2, 3, 4, 5, 6, 7$

satisfying: For every  $\epsilon_i > 0$ ,  $i = 1, 2, 3, 4, 5, 6, 7$ , if

$$\begin{cases} \mathbb{G}^{\mathfrak{F}} \left| D_t^\alpha I_1(t) - \mathcal{K}_1(t, I_1) \right| \leq \epsilon_1, \\ \mathbb{G}^{\mathfrak{F}} \left| D_t^\alpha I_2(t) - \mathcal{K}_2(t, I_2) \right| \leq \epsilon_2, \\ \mathbb{G}^{\mathfrak{F}} \left| D_t^\alpha \mathcal{A}(t) - \mathcal{K}_3(t, \mathcal{A}) \right| \leq \epsilon_3, \\ \mathbb{G}^{\mathfrak{F}} \left| D_t^\alpha \mathcal{T}(t) - \mathcal{K}_4(t, \mathcal{T}) \right| \leq \epsilon_4, \\ \mathbb{G}^{\mathfrak{F}} \left| D_t^\alpha \mathcal{R}(t) - \mathcal{K}_5(t, \mathcal{R}) \right| \leq \epsilon_5, \\ \mathbb{G}^{\mathfrak{F}} \left| D_t^\alpha \mathcal{C}(t) - \mathcal{K}_6(t, \mathcal{C}) \right| \leq \epsilon_6, \\ \mathbb{G}^{\mathfrak{F}} \left| D_t^\alpha \mathcal{L}(t) - \mathcal{K}_7(t, \mathcal{L}) \right| \leq \epsilon_7. \end{cases} \quad (59)$$

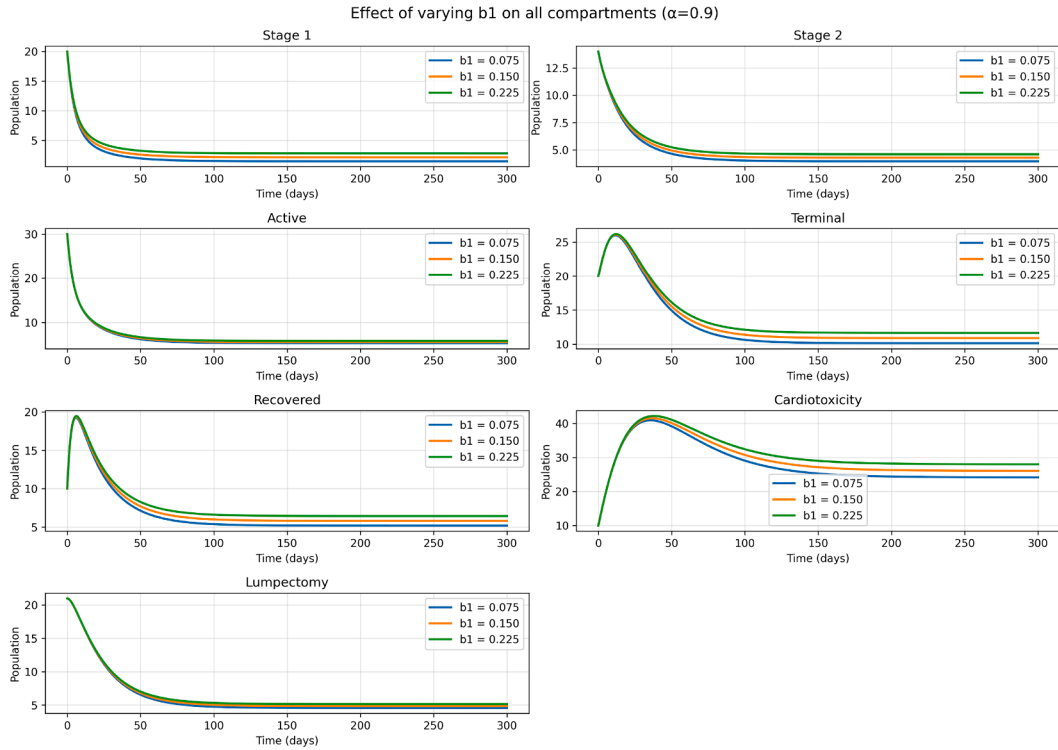
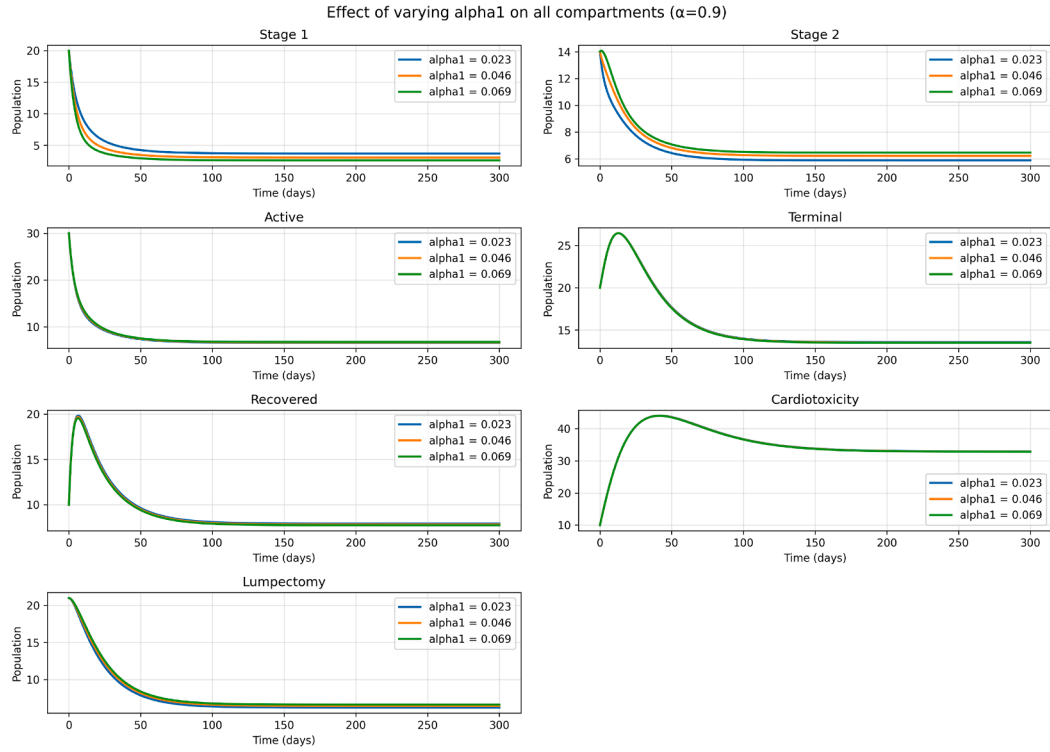
and In addition, there is a solution to the system (21) that can be exist,  $I_1^*(t)$ ,  $I_2^*(t)$ ,  $\mathcal{A}^*(t)$ ,  $\mathcal{T}^*(t)$ ,  $\mathcal{R}^*(t)$ ,  $\mathcal{C}^*(t)$  and  $\mathcal{L}^*(t)$  that satisfying the given model, such that

$$\begin{aligned} \|I_1 - I_1^*\| &\leq \eta_1 \epsilon_1, \|I_2 - I_2^*\| \leq \eta_2 \epsilon_2, \\ \|\mathcal{A} - \mathcal{A}^*\| &\leq \eta_3 \epsilon_3, \|\mathcal{T} - \mathcal{T}^*\| \leq \eta_4 \epsilon_4, \\ \|\mathcal{R} - \mathcal{R}^*\| &\leq \eta_5 \epsilon_5, \|\mathcal{C} - \mathcal{C}^*\| \leq \eta_6 \epsilon_6, \\ \|\mathcal{L} - \mathcal{L}^*\| &\leq \eta_7 \epsilon_7. \end{aligned}$$

**Remark 1.** Let's examine the function  $I_1$  the first inequality (59) has a solution if there is a continuous function  $\mathfrak{z}_1$  that exists (the existence of which is dependent on  $I_1$ ) so that

1.  $|\mathfrak{z}_1(t)| < \epsilon_1$ , and
2.  $\mathbb{G}^{\mathfrak{F}} D_t^\alpha I_1(t) = \mathcal{W}_1(t, I_1) + \mathfrak{z}_1(t)$ .

In order to provide an explanation that is comparable, we are able to define the Eq. (59) for other classes of the model, where  $\mathfrak{z}_i$  is a variable that has the values 1, 2, 3, 4, 5, 6 and 7.



**Fig. 17.** Influence of early-stage progression and detection rates.

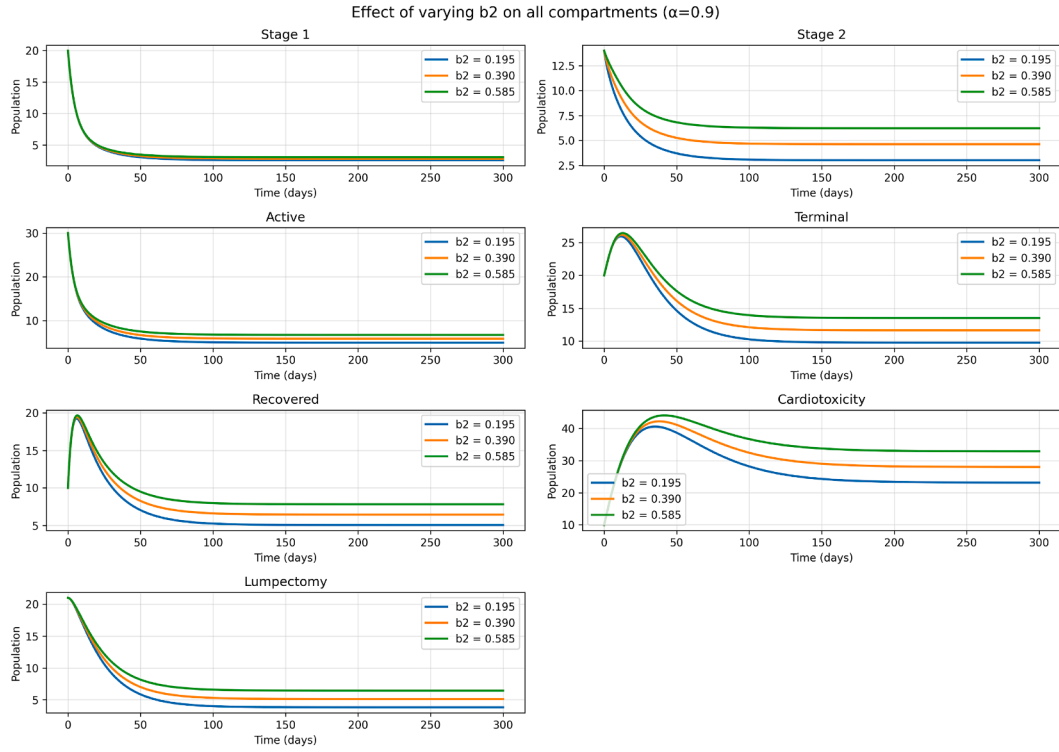
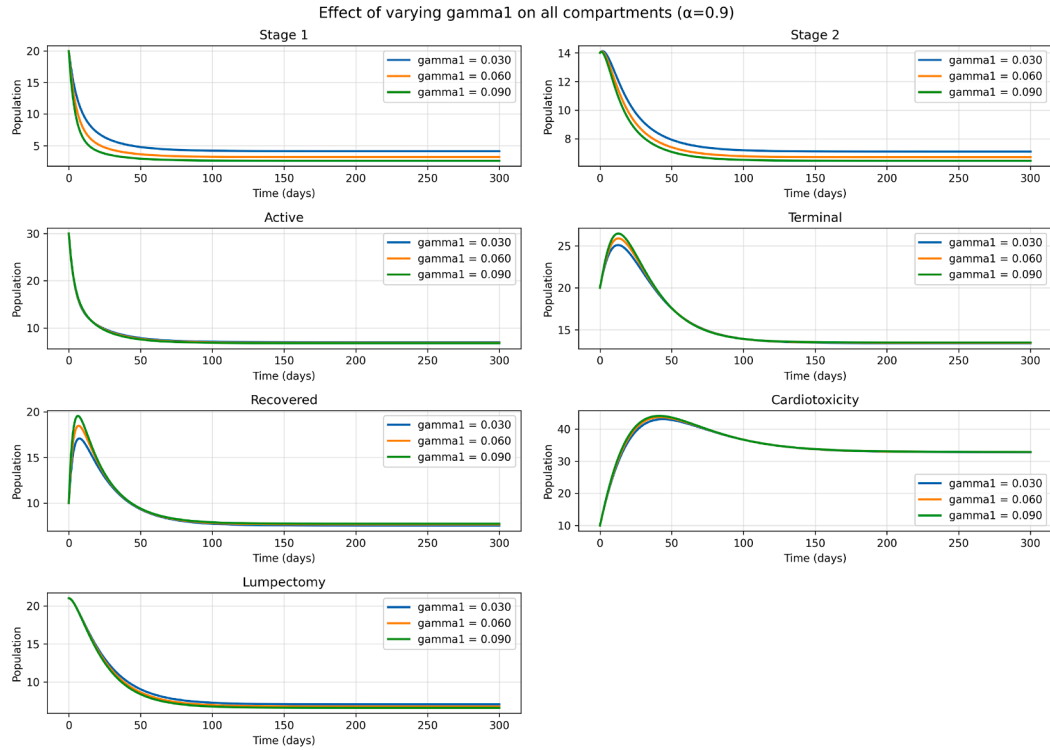
(a) Effect of  $b_2$  on all compartment dynamics(b) Effect of  $\gamma_1$  on all compartment dynamics

Fig. 18. Impact of stage-2 detection and stage-1 recovery on disease compartments.

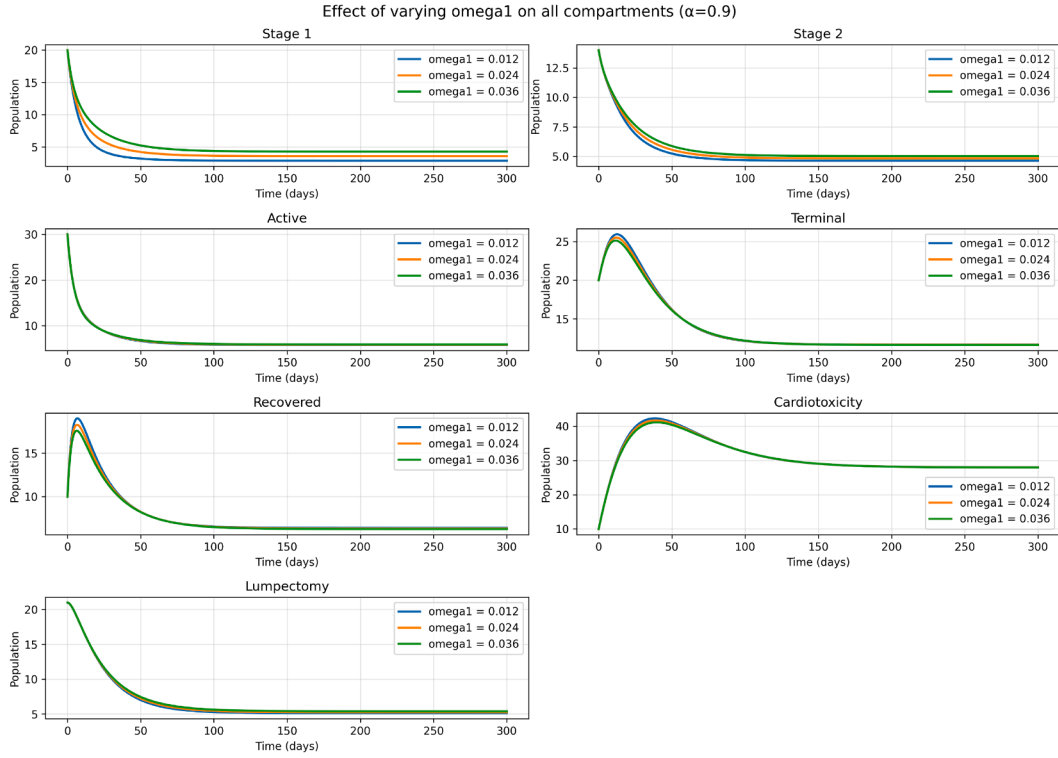
(a) Effect of  $\omega_1$  on on all compartment dynamics

Fig. 19. Effect of relapse rate on early-stage cancer populations.

**Theorem 7.** Assume for this particular instance that the hypothesis (G) is accurate. Ulam-Hyers stability is defined for the system (21) if

$$\left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{2\alpha t}{(2-\alpha)\mathcal{G}(\alpha)} \right) \dot{\mathbf{i}}_i \leq 1, \text{ for } i = 1, 2, 3, 4, 5, 6, 7.$$

**Proof.** Let  $\epsilon_1 > 0$  and It is assumed that the function  $I_1$  is arbitrarily defined such that

$$\left| {}^{\mathcal{G}}D_t^\alpha I_1(t) - \mathcal{K}_1(t, I_1) \right| \leq \epsilon_1.$$

According to above Remark 1, a function  $\delta_1$  with  $|\delta_1| < \epsilon_1$ , which satisfies

$${}^{\mathcal{G}}D_t^\alpha I_1(t) = \mathcal{W}_1(t, I_1) + \delta_1(t).$$

Accordingly, we get

$$\begin{aligned} I_1(t) &= I_1(0) + \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \mathcal{K}_1(t, I_1(t)) + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \mathcal{K}_1(\xi, I_1(t)) d\xi \\ &\quad + \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \delta_1(t) + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \delta_1(\xi) d\xi. \end{aligned}$$

In the Eq. (21), let  $I_1^*$  represent the singular solution to the system. After that,

$$I_1^*(t) = I_1(0) + \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \mathcal{K}_1(t, I_1^*(t)) + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \mathcal{K}_1(\xi, I_1^*(t)) d\xi.$$

Hence,

$$\begin{aligned} |I_1(t) - I_1^*(t)| &\leq \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} |\mathcal{K}_1(t, I_1(t)) - \mathcal{K}_1(t, I_1^*(t))| \\ &\quad + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t |\mathcal{K}_1(\xi, I_1(t)) - \mathcal{K}_1(\xi, I_1^*(t))| d\xi \\ &\quad + \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} |\delta_1(t)| + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t |\delta_1(\xi)| d\xi \end{aligned}$$

$$\begin{aligned} &\leq \left[ \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{2\alpha t}{(2-\alpha)\mathcal{G}(\alpha)} \right] \dot{\mathbf{i}}_1 |I_1(t) - I_1^*(t)| \\ &\quad + \left[ \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{2\alpha t}{(2-\alpha)\mathcal{G}(\alpha)} \right] \epsilon_1 \\ ||I_1(t) - I_1^*(t)|| &\leq \frac{\left[ \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{2\alpha t}{(2-\alpha)\mathcal{G}(\alpha)} \right] \epsilon_1}{1 - \left[ \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{2\alpha t}{(2-\alpha)\mathcal{G}(\alpha)} \right] \dot{\mathbf{i}}_1}. \end{aligned}$$

Then,

$$||I_1(t) - I_1^*(t)|| \leq \eta_1 \epsilon_1,$$

where

$$\eta_1 = \frac{\left[ \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{2\alpha t}{(2-\alpha)\mathcal{G}(\alpha)} \right]}{1 - \left[ \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{2\alpha t}{(2-\alpha)\mathcal{G}(\alpha)} \right] \dot{\mathbf{i}}_1}.$$

Similarly, we get that

$$\begin{aligned} ||I_2 - I_2^*|| &\leq \eta_2 \epsilon_2, ||\mathcal{A} - \mathcal{A}^*|| \leq \eta_3 \epsilon_3, ||\mathcal{T} - \mathcal{T}^*|| \leq \eta_4 \epsilon_4, ||\mathcal{R} - \mathcal{R}^*|| \\ &\leq \eta_5 \epsilon_5, ||\mathcal{C} - \mathcal{C}^*|| \leq \eta_6 \epsilon_6, ||\mathcal{L} - \mathcal{L}^*|| \leq \eta_7 \epsilon_7. \end{aligned}$$

Thus, the system of Eq. (21) is Ulam-Hyers stable.  $\square$

#### Numerical Framework

The purpose of this work is to develop a numerical scheme for the system that is characterized by the Eq. (21). We accomplish this by employing the method that is associated with Newton interpolation polynomials. An example of a general Cauchy problem is presented in the following:

$$\begin{cases} {}^{\mathcal{G}}D_t^\alpha Y(t) = \mathcal{K}(t, Y(t)) \\ Y(0) = Y_0. \end{cases} \quad (60)$$

In order to obtain the result, we make use of the fractal fractional integral operator.

$$Y(t) = Y(0) + \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \mathcal{K}(t, Y(t)) + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^t \mathcal{K}(\xi, Y(\xi)) d\xi.$$

Substituting  $t$  by  $t_{n+1}$ , we get

$$Y_{n+1} = Y(0) + \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} \mathcal{K}(t_n, Y(t_n)) + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_0^{t_{n+1}} \mathcal{K}(t, Y(t)) dt.$$

It is as follows that the difference between successive terms is given:

$$Y_{n+1} - Y_n = \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} (\mathcal{K}(t_n, Y_n) - \mathcal{K}(t_{n-1}, Y_{n-1})) + \frac{2\alpha}{(2-\alpha)\mathcal{G}(\alpha)} \int_{t_n}^{t_{n+1}} \mathcal{K}(t, Y(t)) dt. \quad (61)$$

From the closed interval  $[t_m, t_{m+1}]$ , the function  $\mathcal{K}(\xi, Y(\xi))$  can be obtained by using the interpolation polynomial

$$\theta_m(t) \cong \frac{g(t_m, y_m)}{\delta} (t - t_{m-1}) - \frac{g(t_{m-1}, y_{m-1})}{\delta} (t - t_m), \quad (62)$$

where  $\delta = t_m - t_{m-1}$ . Consequently,

$$\begin{aligned} \int_{t_n}^{t_{n+1}} \mathcal{K}(t, Y(t)) dt &= \int_{t_n}^{t_{n+1}} \left( \frac{\mathcal{K}(t_n, Y_n)}{\delta} (t - t_{n-1}) - \frac{\mathcal{K}(t_{n-1}, Y_{n-1})}{\delta} (t - t_n) \right) dt \\ &= \frac{3\delta}{2} \mathcal{K}(t_n, Y_n) - \frac{\delta}{2} \mathcal{K}(t_{n-1}, Y_{n-1}). \end{aligned} \quad (63)$$

As a result of putting (63) into (61) and undergoing further simplification, we get result that

$$Y_{n+1} = Y_n + \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{3\alpha\delta}{(2-\alpha)\mathcal{G}(\alpha)} \right) \mathcal{K}(t_n, Y_n) - \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{\alpha\delta}{(2-\alpha)\mathcal{G}(\alpha)} \right) \mathcal{K}(t_{n-1}, Y_{n-1}).$$

#### Component-wise Representation Using Kernels

At this point, The system's RHS will be expressed. denoted by the Eq. (51), employing kernel functions for every epidemiological class.

$$\begin{cases} \mathcal{K}_1(t, I_1) = b_1 + \omega_1 R - (\alpha_1 + \gamma_1) I_1, \\ \mathcal{K}_2(t, I_2) = b_2 + \omega_2 R + \alpha_1 I_1 - (\alpha_2 + \gamma_2 + \beta_2) I_2, \\ \mathcal{K}_3(t, A) = b_3 + \omega_3 R + \alpha_2 I_2 - (\alpha_3 + \gamma_3 + \tau_3 + \beta_3 + \mu_3) A, \\ \mathcal{K}_4(t, T) = b_4 + \omega_4 R + \alpha_3 A - (\gamma_4 + \tau_4 + \mu_4) T, \\ \mathcal{K}_5(t, R) = \gamma_1 I_1 + \gamma_2 I_2 + \gamma_3 A + \gamma_4 T + \gamma_5 L - (\omega_1 + \omega_2 + \omega_3 + \omega_4 + \tau_5) R, \\ \mathcal{K}_6(t, C) = \tau_3 A + \tau_4 T + \tau_5 R - \mu_5 C, \\ \mathcal{K}_7(t, L) = \beta_2 I_2 + \beta_3 A - \gamma_5 L. \end{cases} \quad (64)$$

In this context, the parameters  $\beta_1, \beta_2, \beta_3, \beta_4, \gamma_1, \gamma_2, \gamma_3, \gamma_4$  and  $\mu_3$  represent the parameters of the respective models. Through their evaluations at discrete time steps, these kernels are used in the numerical scheme. It can be concluded that the numerical scheme for the Eq. (51) can be obtained as follows:

$$\begin{aligned} I_{1n+1} &= I_{1n} + \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{3\alpha\delta}{(2-\alpha)\mathcal{G}(\alpha)} \right) \mathcal{K}_1(t_n, I_{1n}) \\ &\quad - \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{\alpha\delta}{(2-\alpha)\mathcal{G}(\alpha)} \right) \mathcal{K}_1(t_{n-1}, I_{1n-1}), \\ I_{2n+1} &= I_{2n} + \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{3\alpha\delta}{(2-\alpha)\mathcal{G}(\alpha)} \right) \mathcal{K}_2(t_n, I_{2n}) \\ &\quad - \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{\alpha\delta}{(2-\alpha)\mathcal{G}(\alpha)} \right) \mathcal{K}_2(t_{n-1}, I_{2n-1}), \end{aligned}$$

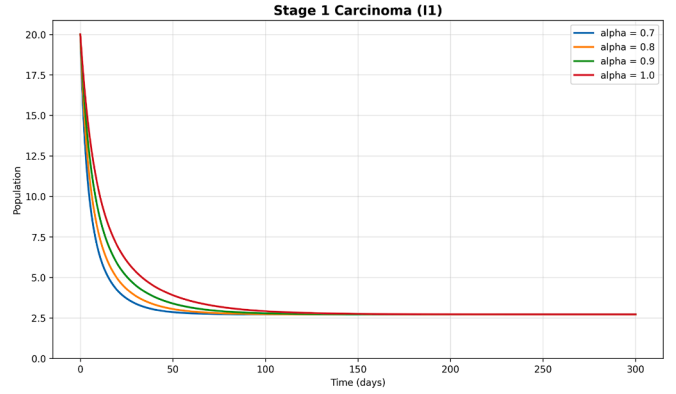


Fig. 20. Stage 1 Carcinoma dynamics across fractional orders.

$$\begin{aligned} A_{n+1} &= A_n + \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{3\alpha\delta}{(2-\alpha)\mathcal{G}(\alpha)} \right) \mathcal{K}_3(t_n, A_n) \\ &\quad - \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{\alpha\delta}{(2-\alpha)\mathcal{G}(\alpha)} \right) \mathcal{K}_3(t_{n-1}, A_{n-1}), \end{aligned}$$

$$\begin{aligned} T_{n+1} &= T_n + \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{3\alpha\delta}{(2-\alpha)\mathcal{G}(\alpha)} \right) \mathcal{K}_4(t_n, T_n) \\ &\quad - \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{\alpha\delta}{(2-\alpha)\mathcal{G}(\alpha)} \right) \mathcal{K}_4(t_{n-1}, T_{n-1}), \end{aligned}$$

$$\begin{aligned} R_{n+1} &= R_n + \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{3\alpha\delta}{(2-\alpha)\mathcal{G}(\alpha)} \right) \mathcal{K}_5(t_n, R_n) \\ &\quad - \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{\alpha\delta}{(2-\alpha)\mathcal{G}(\alpha)} \right) \mathcal{K}_5(t_{n-1}, R_{n-1}), \end{aligned}$$

$$\begin{aligned} C_{n+1} &= C_n + \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{3\alpha\delta}{(2-\alpha)\mathcal{G}(\alpha)} \right) \mathcal{K}_6(t_n, C_n) \\ &\quad - \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{\alpha\delta}{(2-\alpha)\mathcal{G}(\alpha)} \right) \mathcal{K}_6(t_{n-1}, C_{n-1}), \end{aligned}$$

$$\begin{aligned} L_{n+1} &= L_n + \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{3\alpha\delta}{(2-\alpha)\mathcal{G}(\alpha)} \right) \mathcal{K}_7(t_n, L_n) \\ &\quad - \left( \frac{2(1-\alpha)}{(2-\alpha)\mathcal{G}(\alpha)} + \frac{\alpha\delta}{(2-\alpha)\mathcal{G}(\alpha)} \right) \mathcal{K}_7(t_{n-1}, L_{n-1}). \end{aligned}$$

#### Numerical Scheme

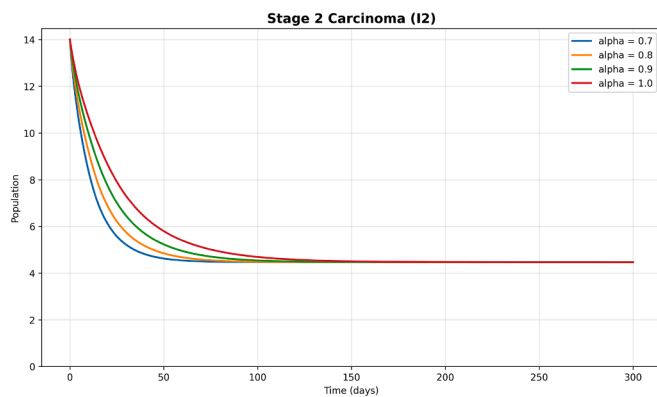
This study develops a fractional-order compartmental model for breast carcinoma progression using the Caputo-Fabrizio derivative. The system parameters and their biological interpretations are detailed in Table 2. To solve the resulting nonlinear fractional differential system, we employ a Newton numerical scheme that efficiently handles the memory effects and complex interactions between compartments. This approach which is provided enhanced predictive capability for disease progression dynamics throughout different cancer stages though maintaining computational stability as well as accuracy.

Fig. 20 show that the population dynamics of Stage 1 carcinoma patients within different fractional orders. The classical integer fractional-order model ( $\alpha = 1.0$ ) shows that the gradual decrease, when the fractional orders ( $\alpha = 0.7 - 0.9$ ) shows the faster reduction in stage 1 of carcinoma population which indicates that the memory effects that captured by fractional calculus gives response to therapy in early disease stages, suggesting that historical health status has an important effect on current treatment outcomes in early breast cancer intervention. Biologically, this finding supports the incorporation of a patient's full treatment history into early-stage risk stratification and personalized therapy planning.

**Table 2**

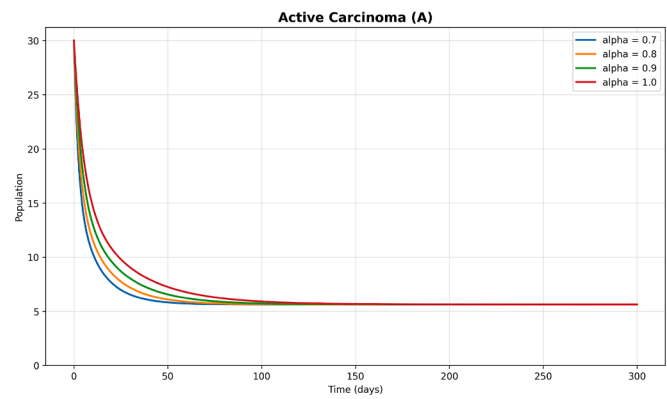
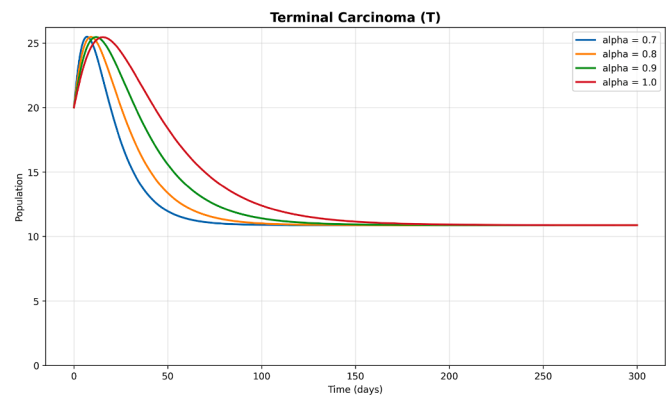
The system's parameters are listed with its value from their source.

| Variable   | Numerical values | Source                    |
|------------|------------------|---------------------------|
| $I_1$      | 20               | Desantis et al. (2015)    |
| $I_2$      | 14               | Abdeljawad et al. (2025b) |
| $A$        | 30               | Desantis et al. (2015)    |
| $T$        | 20               | Desantis et al. (2015)    |
| $R$        | 10               | Desantis et al. (2015)    |
| $C$        | 10               | Abdeljawad et al. (2025b) |
| $L$        | 21               | Estimated                 |
| $b_1$      | 15%              | Desantis et al. (2015)    |
| $b_2$      | 39%              | Desantis et al. (2015)    |
| $b_3$      | 43%              | Desantis et al. (2015)    |
| $b_4$      | 12%              | Desantis et al. (2015)    |
| $\alpha_1$ | 0.046            | Abdeljawad et al. (2025b) |
| $\gamma_1$ | 0.060            | Abdeljawad et al. (2025b) |
| $\gamma_2$ | 0.063            | Abdeljawad et al. (2025b) |
| $\gamma_3$ | 0.035            | Abdeljawad et al. (2025b) |
| $\gamma_4$ | 0.010            | Abdeljawad et al. (2025b) |
| $\gamma_5$ | 0.060            | Estimated                 |
| $\alpha_2$ | 0.056            | Abdeljawad et al. (2025b) |
| $\omega_1$ | 0.024            | Abdeljawad et al. (2025b) |
| $\omega_2$ | 0.030            | Abdeljawad et al. (2025b) |
| $\omega_3$ | 0.036            | Abdeljawad et al. (2025b) |
| $\omega_4$ | 0.062            | Abdeljawad et al. (2025b) |
| $\alpha_3$ | 0.042            | Abdeljawad et al. (2025b) |
| $\tau_5$   | 0.030            | Abdeljawad et al. (2025b) |
| $\tau_3$   | 0.030            | Abdeljawad et al. (2025b) |
| $\tau_4$   | 0.030            | Abdeljawad et al. (2025b) |
| $\mu_3$    | 0.0256           | KSBC (2023)               |
| $\mu_4$    | 0.0256           | KSBC (2023)               |
| $\mu_5$    | 0.0256           | KSBC (2023)               |
| $\beta_2$  | 0.035            | Estimated                 |
| $\beta_3$  | 0.025            | Estimated                 |

**Fig. 21.** Stage 2 Carcinoma evolution with memory effects.

The progression patterns in Fig. 21 shows that the stage of intermediate carcinoma which exhibit a strong dependence on memory. Lower fractional orders ( $\alpha < 1$ ) shows the faster population decay which indicating that disease past history significantly impacts stage 2 effectiveness of treatment. This memory effects gives suggestions that patients past health conditions and treatment responses should be considered under the situation when creating treatment plans for people with intermediate breast cancer stages to get optimal results. Clinically, the pronounced memory-dependent reduction in the intermediate-stage population highlights the importance of aggressive early intervention to capitalize on cumulative therapeutic benefits.

Fig. 22 shows the active carcinoma compartment. The Active carcinoma compartment shows significant from sensitivity to different fractional order variations. The population decreased from the peak and lower  $\alpha$  values for delayed progression which underscore the impact of disease memory effect on advanced carcinoma dynamics. This finding which has clinical significance for chemotherapy treatment and

**Fig. 22.** Active Carcinoma memory-dependent behavior.**Fig. 23.** Terminal Carcinoma fractional-order progression.

its scheduling in stage III breast cancer, where overall treatment effects and historical disease progression patterns heavily affect the current treatment responses. The memory-induced delay in the peak of active carcinoma suggests that historical disease progression data could guide the optimal timing of intensive chemotherapy regimens.

Fig. 23 shows the terminal stage carcinoma dynamics, where memory effects slightly change the disease formulation. The gradual decrease in population with decreasing  $\alpha$  values says that end-stage of cancer progression is strongly affected by previous carcinoma stages. This understanding is crucial for planning palliative care treatment, which showing that a thorough the analyses of patient history that improve our predictions of terminal stage carcinoma progression and improve ways of caring for people at the end of their lives. Even in metastatic disease, the persistent influence of memory effects underscores the relevance of prior treatment success in predicting survival outcomes, reinforcing the need for comprehensive patient history in palliative care.

The recovery patterns from carcinoma in Fig. 24 shows the better recovery rate under fractional-order modeling. Lower  $\alpha$  values are connected to faster growth in recovered population, which showing that memory effects that positively affect the success of long-term treatment. This suggested that patients who complete treatment history and overall recovery response to interventions which gradually impact the final recovery outcome in breast cancer treatment. The accelerated recovery observed with stronger memory effects indicates that cumulative treatment success fosters sustained remission, which may be leveraged to personalize follow-up schedules.

Fig. 25 shows that cardiotoxicity frequency decreases with the lower fractional orders for  $\alpha$ , which showing that memory-effect models predicts reduced the chemotherapy which induced cardiac problems. This results has important for clinical consequences for safety treatment, which suggesting that considering historical treatment sights and over-



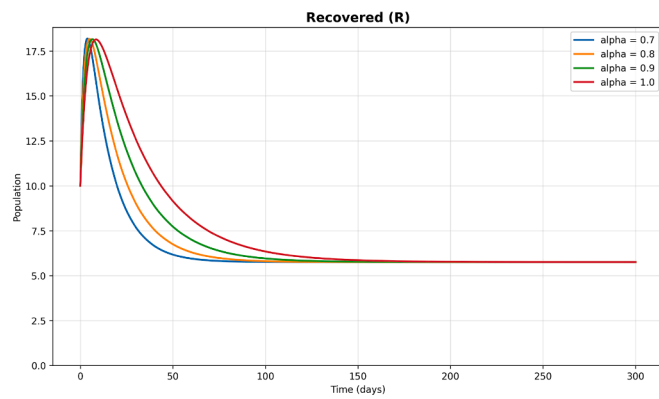


Fig. 24. Recovery compartment fractional dynamics.

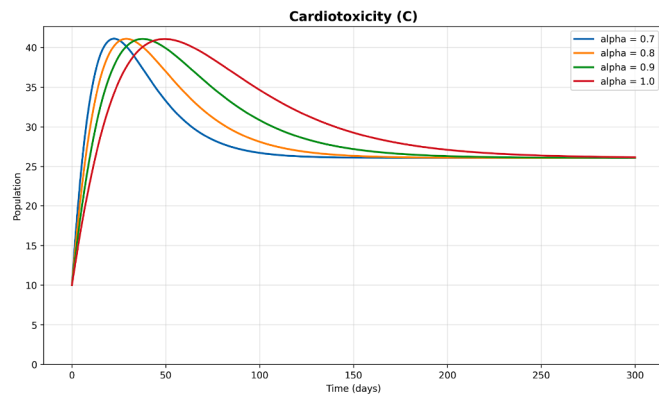


Fig. 25. Cardiotoxicity development under fractional calculus.

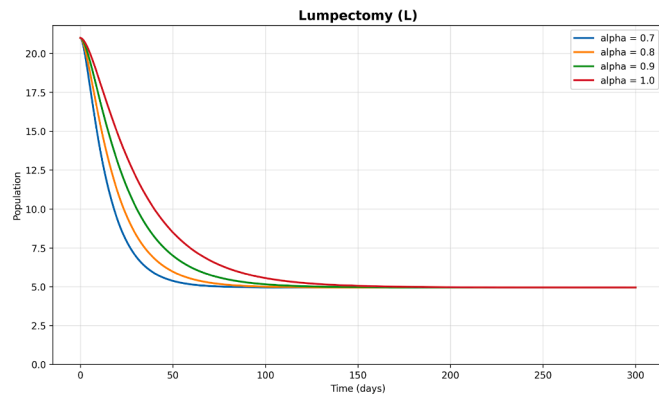


Fig. 26. Lumpectomy surgical outcomes with memory effects.

all effects can help to control chemotherapy regimens to reduce cardiac side effects and preserve anti-cancer effectiveness. Reduced cardiotoxicity incidence at lower fractional orders suggests that memory-aware chemotherapy protocols could improve the therapeutic safety margin, enabling more effective treatment with fewer adverse events.

The Lumpectomy surgical compartment dynamics in Fig. 26 shows that lumpectomy outcomes are memory dependent in fractional orders. It is memory dependent with observed under rapid conclusion with fractional orders. This shows that historical disease progression patterns affects the Lumpectomy surgical recovery paths. The results represent a complete prior to surgery that considers the patient's entire cancer history to forecast and enhance surgical outcomes in breast-conserving procedures. The memory-dependent dynamics of the lumpectomy compartment imply that a patient's complete cancer journey should

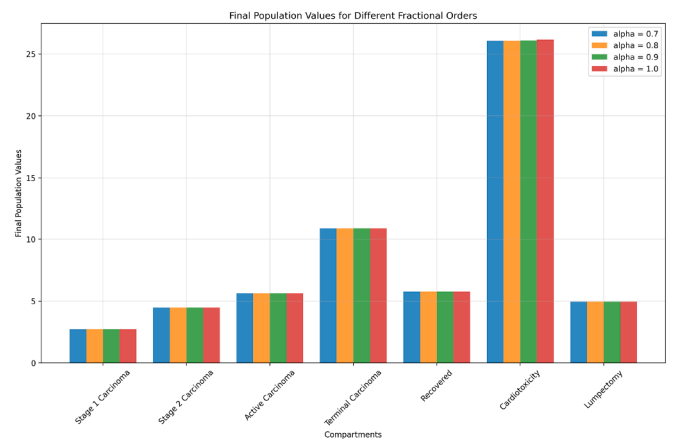


Fig. 27. Final population distribution across fractional orders.

inform preoperative planning and post-surgical recovery expectations.

Fig. 27 provides a complete comparison of steady-state populations for all compartments of carcinoma. The systematic variations shows that fractional-order models gradually predict more favorable outcomes which are reduced carcinoma burdens through over all stages and enhanced strong recovery rates. This body of evidence robustly shows the superiority of fractional calculus in describing biological memory phenomena, which provide more precise predictions for breast cancer progression and treatment results compared to classical integer-order models. Across all compartments, the systematic improvements predicted by fractional-order models reinforce the clinical necessity of incorporating memory effects into breast cancer forecasting tools.

#### Discussion for cancer progression

The fractional-order derivative cancer analysis presented in Figs. 20–27 shows that several critical advancements in breast cancer epidemiology modeling:

- Fractional calculus models with memory effects gives more realistic simulations of breast carcinoma progression for all cancer stages. The consistent improvement in predictions with lower fractional orders provides the importance of historical disease states in determining current treatment outcomes.
- The memory dependent behavior observed in compartments which suggests that individualized treatment strategies accounting for patient's complete medical history could gradually improve the outcomes. This is particularly says in the fast recovery rates and reduced complication profiles which is shown in the fractional-order simulations.
- The observed decrease in cardiotoxicity while preserving effectiveness for therapy in fractional-order models presents promising opportunities for the optimization of chemotherapy regimens, possibly allowing more aggressive treatment with reduced adverse effects.
- These results give mathematical proof for an entire patient assessment that takes into account that diseases have progressed in the past. This will help in clinical decisions about surgery, chemotherapy, and palliative care.

The integration of fractional calculus into breast cancer modeling shows a substantial advancement towards enhanced disease prediction and improved therapeutic strategies in clinical oncology.

#### Conclusion

This research introduced a comprehensive two-stage analytical framework for breast cancer, combining advanced detection method-

ologies with memory-aware disease progression modeling. The detection component, based on the proposed JSPP-Net classifier, achieved remarkable performance with 94% accuracy, perfect sensitivity, and an AUC of 0.997 on a balanced clinical dataset. Comprehensive biomarker analysis identified bcf24, bcf11, and bcf21 as key features, exhibiting clear statistical separation between benign and malignant cases. The Jacobian-based stability regularization embedded in the classifier ensured robust training and interpretable feature importance, making the detection system well-suited for clinical deployment.

For the progression component, a fractional-order compartmental model was formulated using the Caputo-Fabrizio derivative to capture memory-dependent dynamics across seven disease stages, including treatment induced cardiotoxicity and surgical pathways. The model's mathematical rigor was established through fixed-point theory for existence and uniqueness, Ulam-Hyers stability analysis, and equilibrium characterization. Numerical simulations across fractional orders ( $\alpha = 0.7$ – $1.0$ ) demonstrated that stronger memory effects (lower  $\alpha$ ) accelerate early-stage tumor reduction, improve recovery rates, and reduce cardiotoxicity-insights unattainable with classical integer-order models. The proposed Newton interpolation-based numerical scheme offered second-order convergence with minimal memory overhead, ensuring efficient and stable long-term simulations.

Furthermore, the Newton interpolation-based numerical scheme employed to solve the fractional-order system offers distinct advantages over conventional schemes such as the Adams-Bashforth or Euler methods. It achieves second-order convergence while maintaining numerical stability with only two time-step memory, making it computationally efficient and well-suited for the long-term simulation of multi-compartment biological models. This combination of accuracy, stability, and low memory overhead ensures reliable predictions of breast cancer dynamics across all stages and treatment scenarios, reinforcing the practical utility of the proposed framework.

Future work leads to extension of this framework to other cancer types. we plan to extend the proposed fractional-order framework by exploring recently developed  $q$ -analogues of non-singular fractional operators. In particular, the  $q$ -Mittag-Leffler and  $q$ -exponential kernels introduced in Thabet et al. (2024b), the  $q$ -analogs of power-law fractional operators (Thabet et al., 2025a), and the impulsive systems involving  $q$ -analogues of the  $\psi$ -Caputo fractional derivative (Ali et al., 2025) offer enhanced flexibility for modeling complex memory and hereditary effects. Applying these advanced operators to the other cancer progression model may yield even more accurate predictions and provide deeper insights into stage-dependent treatment outcomes.

## CRedit authorship contribution statement

**Vetrivel Muthupandi:** Writing – original draft, Validation, Resources, Methodology, Conceptualization; **Arul Joseph Gnanaprakasam:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Methodology, Investigation, Conceptualization; **Pandit Vivek Kumar Pandey:** Writing – review & editing, Methodology, Validation, Investigation, Visualization.

## Data availability

The dataset that has been used is freely available for research purposes (Wolberg et al., 1993).

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A.

**Definition 7.** (Muthuvel et al., 2024; Nisar, 2024) Let  $\Upsilon : [0, \infty) \rightarrow \mathbb{R}$  and  $\alpha \in (n-1, n)$ ,  $n \in \mathbb{N}$ . The equality represents the left  $\mathfrak{C}\mathfrak{F}$  derivative of the function  $\Upsilon$ , which has an order of  $\alpha$ :

$${}_0^{\mathfrak{C}}D_t^\alpha(\Upsilon(t)) = \frac{1}{\Gamma(n-\alpha)} \int_0^t (t-\mathfrak{s})^{n-\alpha-1} \Upsilon^{(n)}(\mathfrak{s}) d\mathfrak{s}, \quad t > 0. \quad (\text{A.1})$$

**Definition 8.** (Thirthar et al., 2024) The definition of the fractional integral that is associated with the new fractional integral that has a power-law kernel formula:

$${}_0^{\mathfrak{C}}I_t^\alpha(\Upsilon(t)) = \frac{1}{\Gamma(\alpha)} \int_0^t (t-\mathfrak{s})^{\alpha-1} \Upsilon(t) dt. \quad (\text{A.2})$$

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